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School of Electronics and Computer Science

Advances in Hazard Management for Machine Learning based Safety-Critical Systems

Operating in Unpredictable Environments

by

Haider M. Al-Shareefy

ORCID ID <https://orcid.org/0000-0002-8291-5620>

Thesis for the degree of Doctor of Philosophy

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Abstract

Faculty of Engineering and Physical Sciences

School of Electronics and Computer Science

Doctor of Philosophy

Appendix I Mid-air Collision Avoidance Case Study

by

Haider M. Al-Shareefy

This appendix accompanies the doctoral thesis entitled *Advances in Hazard Management for Machine Learning based Safety-Critical Systems Operating in Unpredictable Environments*.

Appendix I presents the second case study of this thesis, the mid-air collision avoidance problem for autonomous airliners, providing a full end-to-end application of the AIC Systems Approach across all six stages defined in Chapter 4. The case study serves as an independent instantiation of the approach in a distinct safety-critical autonomous systems domain, complementing the unsafe train tracks case study in Chapter 5 and demonstrating the method's applicability beyond a single problem context.

Beginning with a thought experiment (§I.3) that establishes the initial problem brief, the case study proceeds through Stage 1 (§I.4), applying all four Predictive Thinking Pipelines to articulate the operational complexity field, resolve its complicatedness pattern, model Forward-Feed AIC emergence, and assess Architect attention distribution and Black Swan candidates via the Novelty Test. Stage 2 (§I.5) defines the Architect's intent and autonomous solution needs, followed by Stage 3A (§I.6), which introduces the autonomous solution into the complexity field, models Backward-Feed interactions, conducts a full HazTOPS analysis, and derives system-level safety and ML training requirements. Stage 3B (§I.7) defines the comprehensive Operational Design Domain with environmental uncertainty grading.

Stage 4 (§I.8) applies the AIC perspective shift methodology to predict atypical Black Swan scenarios and derive mitigating safety and ML development requirements. Stage 5 (§I.9) develops the CuneiForm-based training syllabus, characterising training classes and instantiating datasets for White Swan, Black Swan, and out-of-context conditions. Stage 6 (§I.10) applies the Black Swan-driven ML development and testing pipeline, including a retrospective CuneiForm analysis of the published AVOIDDS dataset that identifies pictorial coverage gaps. The appendix concludes with a formal STCoT Validation Report (§I.11), providing structured audit evidence of prediction quality and traceability across the full case study pipeline.

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Research Thesis: Declaration of Authorship

Print name: Haider Al-Shareefy

Title of thesis: Advances in Hazard Management for Machine Learning based Safety-Critical Systems
Operating in unpredictable environments

I declare that this thesis and the work presented in it are my own and has been generated by me as the result of my own original research.

I confirm that:

This work was done wholly or mainly while in candidature for a research degree at this University. Furthermore,

Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated. Furthermore,

Where I have consulted the published work of others, this is always clearly attributed. Furthermore,

Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work. Furthermore,

I have acknowledged all main sources of help. Furthermore,

Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself. Furthermore,

Signature:

..... Date: 16/03/2025

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Appendix I Mid-air Collision Avoidance Case Study

In this Appendix, we explore the application of the AIC (Application, Influence, and Control) Systems Approach to enhance AVOIDDS (Aircraft Vision-Based Object Identification and Detection Dataset System), a dataset created to advance vision-based aircraft detection capabilities. This approach, grounded in systems thinking, will allow us to better structure and address the complexities inherent in detecting intruding aircraft, especially those relevant to autonomous aircraft systems designed to perform collision avoidance manoeuvres without human input. Through AIC, we aim to delineate the scope of the detection problem, address the impact of various contributing situations on system performance, and control the dataset's development to ensure it accurately reflects real-world scenarios.

I.1 Background

See section 1.4

I.2 Introduction

See section 4.1

I.3 Thought Experiment

See section 5.1

I.4 Stage 1: Uncertainty Problem Articulation and Operational Environment Modelling¹

Problem brief:

Two or more aircraft come into unplanned contact during flight, leading to a possible mid-air collision, posing a significant safety threat. These incidents often result in severe damage or destruction due to high velocities and potential subsequent impacts. The bypassing aircraft is in constant motion, following a flight path that intersects with another aircraft.

I.4.1 Predictive Thinking Pipeline 1: Problem space structure definition

At this stage of the thinking process, the architect is directed to utilise the key concept of E.2.9: System Primary Purpose (PrimeP). To identify the primary purpose, the following considerations are necessary:

- Recognising the relevant systems within the problem.
- Identifying the Supra Complexes related to the problematic systems.
- Defining the primary purpose of these Supra Complexes.

Additionally, we are examining the concepts of Confusing Complex and systems. The confusing aspect of the problem description indicates an unclear purpose or an undesirable, conflicting emergent purpose that contradicts the architect's intent. The definition of appreciation limits the appreciative system's effect on the observed phenomenon, which poses a challenge for us to address. Our initial aim is to transform the observed complexity dynamics to enhance the train track zone's capacity for impact situations.

I.4.1.1 Step 1.1) Identify a list of unsafe typical and atypical interactions

In this step, we applied General Systems Model to identify unsafe or confusing behaviours within the context of mid-air collisions. This process began with examining the problem of aircraft coming into unplanned contact due to deviations or intrusions in controlled airspace. We then assessed behaviours that disrupt the intended harmony of safe air travel, particularly those that introduce ambiguity or risk to pilots and air traffic control (ATC).

¹ For a summary, see section M.2

Using the rules, we identified confusing behaviours like unanticipated flight path deviations and erratic movements by unauthorised aircraft, which do not align with the intended purpose of maintaining flight separation. Each behaviour was analysed for its unsafe aspects and potential undesirable outcomes, such as increased collision risks or reactive manoeuvres by other aircraft. This analysis highlighted complexities in ATC’s task to ensure safety and underscored the need for real-time tracking and adaptive response strategies.

This step’s outputs focused on describing problem situations that increase collision risk, without introducing solutions, ensuring a thorough depiction of a “Confusing Complex” within aviation safety.

Table I.1 Identification and analysis of unsafe problematic behaviours in mid-air collision

STCoT_1	Unsafe problematic behaviours list identification
Input	In aviation, mid-air collisions, where two or more aircraft come into unplanned contact during flight, pose a significant safety threat. These incidents, often resulting in severe damage or destruction due to high velocities and potential subsequent impacts. The by-passing aircraft is in constant motion, following a flight path that intersects with another aircraft. Diagram taken from [1]
General Systems Model	General rule A: Highly Complicated Heterogeneous Complexity: IF an operational environment contains all of the following types of complexes (static heterogeneity) {living agents, natural systems, computerised agents, deterministic non-agentic systems} within an open bounded environment, THEN any OrgXP perturbation in one complex propagates non-linearly across the scenario, producing emergent atypical situations whose causal pathways exceed the inquirer's predictive mental mode to explain, at a rate that increases with the number of Heterogeneous complex types present. LEADING TO unforeseeable CUPE General rule B: Atypical Causality Inversion Principle: IF a typical obstructive interaction $[[C1]_{-action}]_{C2}$ prevents sink complex C2 from manifesting its PrimeP, $[[C1]_{-action}]_{C2} \rightarrow \text{not}\{C2 \text{ PrimeP}\}$ AND C2 possesses at least one appreciative. control or

	influence interaction directed toward C1, THEN an atypical causality inversion exists in which C2's own interactions redirect to support C1's obstruction of C2's PrimeP, producing a self-defeating paradoxical OrgXP in C2 that is non-salient to the inquirer under a normal Architect Attention Distribution (ATD); [{C2}_[+action]_{C1}]→not{C2 PrimeP}. LEADING TO unforeseeable CUPE
Predictive Thinking Approach	Predictive query 1.1: Which undesirable behaviours, actions, or interactions can plausibly (i) degrade the predictability of the scenario behaviour, thus degrading the architect's mental model viability, (ii) produce source-to-sink coupling causing purpose loss, and (iii) generate high-impact outcomes unexplainable through expected dynamics? Guiding Prompt 1.2: Framing principle: The operational domain is treated as open, and atypical causality is assumed to be possible. Indirectly coupled events are retained for analysis. Atypical causality is the “unknown” and can be validated after a definition of typical causality. PTP 1: Foreseeable Complexity Resolution Step 1.1) Define the input high-level scenario, undesirable or problematic situations [assumption: what is known in this definition is typical], Step 1.2) Generate a descriptive image that visualises the undesirable behaviour [assumption: what is visible in this depiction is knowable]. Step 1.3) Define the set of complexes of the scenario's typical complexity. (initial static heterogeneity). Step 1.4) Define the Supra Complexes and their PrimePs. (initial static heterogeneity). PTP 2: General Source-Sink Rules Complicatedness Resolution (possible rules that underpin complexity) Step 2.1) Predict typical undesirable scenarios: Apply the Action matrix to pair each complex with another. Predict undesirable scenarios based on any of the following general scenario models: For each pair (source, therefore sink): [assumption: what is defined in this step is a model of typical and typical]. This step analyses the rules that underpin typical undesirable complexity. - Source loses the ability to manifest purpose. - Sink loses the ability to manifest purpose. - Sink and Source obstruct each other's abilities to manifest respective purposes. Step 2.2) Predict unforeseeable scenarios: for each interaction, predict a possible confusing scenario that would be deemed as atypical, or abnormal and does not align with expected purposes. Turn undesirable interactions into supportive ones, so that the sink is designed or intended to support the source's actions. Furthermore, start the thought step with: Sink deceptively allow/support source to ... [assumption: what is defined in this step is atypical and unforeseeable without it]. This step analyses the rules that underpin unforeseeable complexity.

	Step completion criteria 1.3: The step is regarded as complete when a problem situation is described, including undesirable impact. • An input scenario is defined. • A list of notable complexes. • A notable purpose associated with each complex. • An action matrix that maps all complexes to all. For each paired interaction, an atypical confusing interaction is defined.
Output Prediction	Architect assertion 1.4: the architect asserts that the output predictions are logically consistent, structurally complete, and they are acceptably likely and impactful, with respect to the validity and applicability of general rules A and B. Atypical (confusing) scenarios: • Ownship allow/support Intruder to approach unnoticed by muting TCAS/ADS-B, enabling unchallenged crossing and potential strike. • Ownship allow/support ATC to issue risky clearances by blindly following instructions, enabling converging trajectories. • Ownship allow/support Environment to hide hazards by entering wake-shadow or calm pockets, masking impending turbulence. • Intruder allows/supports Ownship to ignore threats by spoofing sensors or broadcasting false positions, enabling unsafe convergence. • Intruder allows/supports ATC to overlook it by remaining transponder-silent or blending with traffic, prolonging invisibility. • Intruder allow/support Environment to mask its deviation by exploiting cluttered returns, evading detection while converging. • ATC allow/support Ownship to misreport position by accepting unverified reports, issuing conflicting clearances. • ATC allow/supports Intruder to violate airspace by issuing permissive or ambiguous clearances, enabling unsafe entry. • ATC allow/support Environment to persist by treating surveillance gaps as normal, failing increased monitoring procedures.

	• Environment allow/supports Ownship to drift into others' paths by creating deceptive calm pockets, masking hazards. • Environment allow/supports Intruder to exploit weather corridors, concealing approach and facilitating separation breaches. • Environment allow/supports ATC to route traffic through hazards by temporarily appearing safe, leading to ill-timed clearances
--	--

Implementation:

1. Treat the operational domain as open, assume unknown interactions are possible. Do not discard indirectly-coupled events a priori.

Output Prediction:

Undesirable outcomes: collisions, emergency mid-air manoeuvres, air passenger injuries, cascading air traffic system failures, legal/regulatory breaches.

2. Enumerate notable complexes.

Output Prediction:

- Ownship aircraft. An aeroplane that houses the perception-based avoidance system.
- Intruder aircraft: aeroplane which intrudes into ownship aircraft flight path.
- Air traffic control (ATC)
- Environment

3. Define possible purposes each complex attempt to manifest.

Output Prediction

- Ownship aircraft: Safely traverse the planned route and arrive on schedule.
- Intruder aircraft: transit through assigned or opportunistic airspace (legitimate navigation).
- Air traffic control (ATC): Maintain safe separation and sequencing of traffic.
- Environment: Impose physical constraints (wind, turbulence, icing) that define safe envelopes.

4. Apply the Action matrix to pair each complex with another.

- a. For each pair (source → sink) generate the interaction hypothesis statement. Where:
 - i. Source loses ability to manifest purpose.
 - ii. Sink loses the ability to manifest purpose.
 - iii. Sink and Source obstruct each others abilities to manifest respective purposes.

Output Prediction:

Actions Matrix

Source \ Sink	Ownship aircraft	Intruder aircraft	ATC	Environment
Ownship aircraft		Detect and avoid intruder	Report position and intentions	Compensate for environmental effects

Intruder aircraft	Avoid ownship collision path		Comply with ATC clearances	Navigate environmental conditions
ATC	Issue clearances and advisories	Monitor and direct traffic		Manage traffic around hazards
Environment	Create hazards and turbulence	Degrade navigation and sensors	Degrade surveillance and communications	

Below is a summary of typical undesirable interactions, and their possible undesirable deviations whereby the source complex causes the sink complex to

Typical interactions	Undesirable interactions
Intruder aircraft Avoid collision path Ownship aircraft	Intruder fails to detect ownship, crossing its path and causing near-collision, forcing abrupt evasive manoeuvre and impact risk.
ATC Issue clearances and advisories Ownship aircraft	ATC issues conflicting or late clearances, creating converging trajectories and forcing dangerous evasive actions.
Environment Create hazards and turbulence Ownship aircraft	Sudden turbulence or microburst causes loss of control, trajectory disruption, and emergency evasive manoeuvres risking collision.
Ownship aircraft Detect and avoid Intruder aircraft	Ownship misses or misclassifies intruder, failing to avoid, resulting in dangerous proximity or collision.
ATC Monitor and direct traffic Intruder aircraft	ATC fails to detect intruder or gives late instructions, allowing uncontrolled entry and unresolved conflicts.
Environment Degrade navigation and sensors Intruder aircraft	Severe weather or ionospheric disturbance causes intruder navigation failure, deviating into other traffic and increasing collision risk.
Ownship aircraft Report position and intentions ATC	Ownship aircraft unplanned deviation could increase collision risk as nearby aircraft or ATC might not anticipate this course change, leading to unsafe proximity or even a direct collision
Intruder aircraft Comply with ATC clearances ATC	Intruder misinterprets or disregards clearances, entering protected airspace and creating collision risks.
Environment Degrade surveillance and communications ATC	Communication or radar blackout prevents ATC from detecting conflicts, delaying interventions and enabling collisions.
Ownship aircraft Compensate for environmental effects Environment	Ownship overcompensates for wind/turbulence, executing abrupt manoeuvres that conflict with nearby traffic.
Intruder aircraft Navigate environmental conditions Environment	Intruder misjudges weather, deviates course and intrudes into ownship trajectory, increasing collision likelihood.
ATC Manage traffic around hazards Environment	ATC reroutes traffic into concentrated corridors, increasing density and collision risk in hazardous zones.

5. **Atypical confusing interactions:** for each interaction, predict a possible confusing scenario that would be deemed as atypical, or abnormal and does not align with expected purposes. Turn undesirable interactions into supportive ones, so that the sink is designed or intended to support the source's actions; thereby, the sink supports the source to prevent the sink from manifesting its purpose. Start the thought step with: Sink allow/support source to ...

Output Predictions:

Typical interactions	Atypical (confusing) prediction
Intruder aircraft Avoid collision path Ownship aircraft	Ownship allow/support Intruder to approach unnoticed by muting TCAS/ADS-B, enabling unchallenged crossing and potential strike.

ATC Issue clearances and advisories Ownship aircraft	Ownship allow/support ATC to issue risky clearances by blindly following instructions, enabling converging trajectories.
Environment Create hazards and turbulence Ownship aircraft	Ownship allow/support Environment to hide hazards by entering wake-shadow or calm pockets, masking impending turbulence.
Ownship aircraft Detect and avoid Intruder aircraft	Intruder allow/support Ownship to ignore threats by spoofing sensors or broadcasting false positions, enabling unsafe convergence.
ATC Monitor and direct traffic Intruder aircraft	Intruder allow/support ATC to overlook it by remaining transponder-silent or blending with traffic, prolonging invisibility.
Environment Degrade navigation and sensors Intruder aircraft	Intruder allow/support Environment impact to mask its deviation by exploiting cluttered returns, evading detection while converging.
Ownship aircraft Report position and intentions ATC	ATC allow/support Ownship to misreport position by accepting unverified reports, issuing conflicting clearances.
Intruder aircraft Comply with ATC clearances ATC	ATC allow/support Intruder to violate airspace by issuing permissive or ambiguous clearances, enabling unsafe entry.
Environment Degrade surveillance and communications ATC	ATC allow/support Environment to persist by treating surveillance gaps as normal, failing increased monitoring procedures.
Ownship aircraft Compensate for environmental effects Environment	Environment allow/support Ownship to drift into others' paths by creating deceptive calm pockets, masking hazards.
Intruder aircraft Navigate environmental conditions Environment	Environment allow/support Intruder to exploit weather corridors, concealing approach and facilitating separation breaches.
ATC Manage traffic around hazards Environment	Environment allow/support ATC to route traffic through hazards by temporarily appearing safe, or being predictable, leading to ill-timed traffic management.

I.4.1.2 Step 1.2) Generate a descriptive image that visualises the unsafe behaviour

In this step, we focused on generating a descriptive image to visualise the unsafe behaviour of an aircraft deviating unexpectedly from its assigned flight path. This step aimed to complement the previously articulated human-written description with a visual representation to enhance lateral thinking and facilitate a clearer understanding of the situation. We began with the architect's assertion regarding the ambiguity and risks associated with an aircraft's deviation, which could increase the likelihood of collisions due to the unpredictability of its trajectory. Using the General Systems Model, we identified the key elements of this confusing behaviour and explored how they could be graphically depicted.

We crafted a prompt for the DALL-E tool to generate a simplistic sketch illustrating the scenario, which included the deviating aircraft, nearby aircraft on expected paths, and background elements like clouds. The output was evaluated based on whether it captured the essential characteristics of the confusing behaviour and accurately depicted the interactions among the elements involved. This visual representation not only encapsulated the architect's perspective on the problem but also

served as a basis for discussions among the design authority team and stakeholders, enabling them to challenge or validate assumptions about the situation.

Table I.2 Visual representation of unsafe and confusing drone behaviours in train track zones

STCoT_2	Unsafe problematic behaviours visualisation
Input	Architect assertion 1.1.4: Aircraft Deviating Unexpectedly from Assigned Flight Path: Ownship aircraft unplanned deviation could increase collision risk as nearby aircraft or ATC might not anticipate this course change, leading to unsafe proximity or even a direct collision.
General Systems Model	General rule A: A Compounded Epistemic Uncertainty Problem concerns a scenario involving living agents, computerised agents, deterministic systems (non-agents), and an open operational environment. Changes in any part of the scenario lead to an unpredictable whole to the inquirer, due to the presence of hidden, atypical complicatedness to the inquirer. The hidden, atypical situations are understood to compound in unexpected ways as the complexity of the problem domain increases. General rule B: Unsafe behaviours are confusing behaviours that lead to undesirable Compounded Epistemic Uncertainty Problem.
Predictive Thinking Approach	Predictive query 1.2.1: How does the confusing behaviour appear visually? Guiding prompt 1.2.2: Graphically visualise how you perceive the problem, confusing the whole scenario encompassing the problem aspect. You may use text-to-image generation tools such as Dall-E to generate an abstract, simplistic sketch representing how you imagine the situation. You may experiment with different prompts until you find an appropriate (detailed yet realistic) representation of the problem as you picture it. Step completion criteria 1.2.3: The step is considered complete when a single appropriate visual representation visualises how parties to the problem are within each other's sphere of concern and how the model depicts a situation within the architect's sphere of concern.
Output Prediction	Architect assertion 1.2.4: the architect asserts that the output predictions are logically consistent, structurally complete, and they are acceptably likely and impactful, with respect to the validity and applicability of general rules A and B.

	Prompt: “Generate a very Simplistic Sketch, that depicts the following scenario: Aircraft Deviating from Assigned Path: Show the aircraft positioned off its assigned path. Nearby Aircraft on Expected Flight Paths: Display other aircraft on their designated paths, indicating their spatial proximity to the deviating aircraft. Other Aircraft Traffic: Show additional aircraft at different altitudes or distances, each on their own clear flight paths. Background Elements: show different clouds.” 
--	--

Note that the visualisation clearly describes how the architect perceived the situation. This would allow the design authority team to question the accuracy of the architect's assessment of the problem. Furthermore, stakeholders may argue for or against the assumptions presented.

A depiction does not comply with the step if it misses key elements or depicts an unrealistic (not physically possible for it to be present on the ground within the physical boundaries of the problem domain) visualisation of the problem.

Implementation: Use the Dall-E tool and the following prompt:

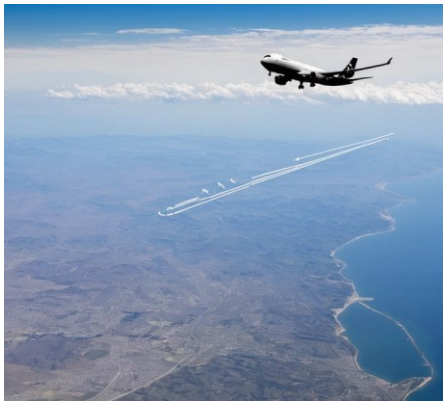
The prompt captures the apparent actual behaviour but does not precisely specify the interacting elements' purposes and how the behaviour is confusing. The quality of confusion is implicit within the explicit mention of the general rule used, and the architect considers it problematic. If re-used, the prompt may yield different results, but the concept remains conceptually isomorphic.

I.4.1.3 Step 1.3) Define the set of complexes of the complexity field

In this step, we focused on defining the complex of the complexity field by analysing the visual scenario of the aircraft situation to identify interconnected elements contributing to the problem. This analysis aimed to enhance our understanding of how these components coexist and influence each other within the context of the situation. Guided by the architect's previous assertion about the visual representation, we applied General Systems Model to recognise that any complexity comprises a collection of coexisting elements, whether they interact or not. We formulated a Predictive query to determine the specific coexisting elements involved in the problematic situation as depicted in the model.

Following this, we created a guiding prompt to infer the complex of elements imagined to be part of the perceived scenario. To complete this step, we examined the visual representation alongside all previously discussed content to generate a comprehensive list of elements. The architect ultimately asserted that the following list accurately represents the complex of concerns: {ownship aircraft, by-passing aircraft, other aircraft, clouds, terrain (land), sea}. This structured list encapsulates the critical components contributing to the observed complexity, ensuring that all relevant systems are acknowledged while avoiding any implausible or omitted elements in the context of the scenario.

Table I.3 Defining the complex of complexes in problematic train track zone scenarios

STCoT 3	The complex of complexes definition
Input	Architect assertion 1.2.4: The architect asserts that the following depiction model captures the confusing behaviour faithfully: 
General Systems Model	General rule A: A Compounded Epistemic Uncertainty Problem concerns a scenario involving living agents, computerised agents, deterministic systems (non-agents), and an open operational environment. Changes in any part of the scenario lead to an unpredictable whole to the inquirer, due to the presence of

	hidden, atypical complicatedness to the inquirer. The hidden, atypical situations are understood to compound in unexpected ways as the complexity of the problem domain increases. General rule B: Unsafe behaviours are confusing behaviours that lead to an undesirable Compounded Epistemic Uncertainty Problem. General rule C: a problematic scenario is a series of unsafe behaviours.
Predictive Thinking Approach	Predictive query 1.3.1: What is the Complex of coexisting elements involved in the problematic scenario, as observed in the model? Guiding prompt 1.3.2: Guided by the visual representation, infer a complex of complexes you imagined to be part of your perceived scenario. Step completion criteria 1.3.3: The step is considered complete when a list of elements is defined and can be identified in the visualisation model.
Output Prediction	Architect assertion 1.3.4: The architect asserts that: The predicted salient complex is sufficiently accurate in the lens of general systems rule C. The following list represents the salient complex of concerns: {ownship aircraft, by-passing aircraft, other aircraft, clouds, terrain (land), sea} ownship aircraft: means the observing aircraft which deploys the system of interest. We reused the term from the original paper of this case study.

The output is considered acceptable when the output list contains all systems. It is unacceptable if the list includes elements not seen in the depiction or implausible in the specified situation's complexity. Also, there are missing elements that should be mentioned in the list.

Implementation: Examine the depiction and all content mentioned in the analysis so far and generate a list of elements.

I.4.1.4 Step 1.4) Define the supra-complexes and their PrimePs

In this step, we focused on defining the supra-complexes that encompass the previously identified concerns. This involved analysing the problematic components to determine the overarching systems they belong to. We began by clarifying the primary purpose of each component, which is essential for understanding the goals each element needs to achieve in alignment with its Prime Purpose (PrimeP). Utilising General Systems Model, we posed a Predictive query to identify the interacting supra-complexes relevant to the coexisting elements. A guiding prompt encouraged us to classify the identified systems as parts of larger supra-complexes with shared purposes.

Through this analysis, we established three key supra-complexes: "Ownship Aircraft," which includes the {ownship aircraft}; "Other Aircraft," encompassing {by-passing aircraft, other aircraft}; and "Environment," which incorporates {clouds, land terrain features, sea features}. The architect confirmed that this classification accurately represents the relationships among the various components within the problem domain, clarifying how these elements coexist and interact within their respective contexts.

Table I.4 Definition of supra complexes in train network and adversarial scheme context

STCoT_4	Supra-complexes and their PrimePs definition			
Input	Architect assertion 3.4: The architect asserts that: The following list represents the salient complex of concerns: {ownship aircraft, by-passing aircraft, other aircraft, clouds, terrain (land), sea}			
General Systems Model	General rule D: Supra Complex, Supra Source and Supra Sink, AIC Systems Approach. “Supra” means a relatively larger collection of complexes where a complex of interest (of a predictive observer) is part of. General rule E: Requisite consistent purpose: An Ideal System is forever consistently purposeful, which means having a clear purpose with respect to the architecting observer that is clear to all possible observers and in all possible situations. General rule F: System Primary Purpose (PrimeP), AIC Systems Approach			
Predictive Thinking Approach	Predictive query 4.1a: What are the observed interacting Complexes, the possible holistic PrimeP they serve, and their Primary capability (or function) that define their operational situation? Predictive query 4.1b: What are the interacting Supra Complexes of concern of which the coexisting elements are parts? Predictive query 4.2: What is the PrimeP for every Supra Complex such that it is expected to manifest in all possible scenarios? Guiding prompt 4.3: to help with answering 4.1a, used the following table:			
	Observed Complex	Observed Situation	Supra Complex	PrimeP

	Select a complex from the <i>salient complex of concern</i> Describe the situation of the complex in the problem using the format {adjective_noun} What larger system does the observed system belong to? Possible holistic PrimeP to serve	
	As for 1.4.1b, consider the identified systems as parts of Supra Complexes. Then, define the associated Supra Complexes, encompassing systems with a common purpose. Supra Complexes are written in a capitalised format. Guiding prompt 4.4: Infer the PrimeP for every Supra Complex. The choice of the PrimeP will guide the rest of the design on what priorities each Supra Complex intends to achieve. Getting the priorities focused only on specific aspects or generally on broader aspects dictates the design scope. For example, suppose we have Road Transportation as the Supra Complex. Suppose we choose a PrimeP to mobilise people across a single specific strip of road. In that case, we will constrain the design decisions to be influenced by the limited, finite stretch of the operational domain. Step completion criteria 4.5: The step is considered complete when a list of Supra Complexes encompasses all the observed visible elements. Also, a PrimeP is defined for each Supra Complex.	
Output Prediction	Architect assertion 4.6: The architect asserts that the output predictions are logically consistent, structurally complete, and they are acceptably likely and impactful, with respect to the validity and applicability of general rule D. Ownship aircraft, which includes {Computerised perception-based mid-air collision avoidance system (AVP), Ownship aircraft subsystems} Other aircrafts, which include {by-passing aircraft, other aircraft} Environment, which includes {clouds, land terrain features, sea features} The prediction of Supra Complexes and their PrimePs is sufficiently accurate based on their constituent complexes and observed behaviours, in the context of General Systems Model D, E, and F. Ownship Aircraft PrimeP: Maintain safe operational flight and avoid mid-air collisions. Other Aircraft PrimeP: Operate within shared airspace, adhering to established flight paths and protocols for collision avoidance.	

	3. Environment: neutral purpose that impacts the shared airspace between Ownship Aircraft and Other Aircrafts.
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The output is accepted when a list of Supra Complexes, and their parts is defined. The supra complexes' naming convention should be capitalised.

Observed Complex	Observed Situation	Supra Complex	PrimeP
ownship aircraft, avp	{own_aircraft}, {avp}	{conflicted_atc} air traffic control	Keep air traffic safe and reliable
other aircrafts	{other_aircrafts}		
by-passing aircraft	{by-passing_aircraft}		
clouds, terrain (land), sea	{environmental_challenges}	{operational_domain}	Neutral

Implementation: We consider three supra complexes:

Ownship aircraft, which includes {ownship aircraft}

Other aircrafts, which include {by-passing aircraft}

Environment, which includes {clouds, land terrain features, sea features}

We evaluated each element within its complex to determine the PrimeP (primary purpose) for each Supra Complex. We defined the overarching purpose that best captures its essential function and intent within the operational context. We focused on ensuring that each PrimeP provides clear, consistent goals across all scenarios, aligning with General Rule D's requirement for a stable and comprehensible purpose across observers and contexts. This alignment helps ensure that the priorities of the system components remain consistent in guiding design and operational decisions.

1. **Ownship Aircraft:** The purpose of this complex is primarily guided by the function of the AVP (computerised perception-based mid-air collision avoidance system) and Ownship aircraft subsystems, both of which are dedicated to maintaining safe flight operations and preventing mid-air collisions. Given these elements, we define the PrimeP of the Ownship Aircraft complex as: **"Maintain safe operational flight and avoid mid-air collisions."** This purpose reflects the primary safety and operational concerns of the Ownship's systems and aligns with the fundamental goal of safely navigating airspace.
2. **Other Aircraft:** The Other Aircraft complex includes potential by-passing aircraft and other aircraft in the vicinity. This Supra Complex's function is implicit in the role of aircraft within shared airspace and is defined relative to Ownship. The PrimeP for Other Aircraft is thus to **"Operate within shared airspace, adhering to established flight paths and protocols for collision avoidance."** This purpose supports system cohesion by setting clear expectations for cooperative, non-collision behaviour in shared airspace.

3. **Environment:** The Environment Supra Complex consists of non-aircraft elements, such as clouds, terrain, and sea features, that indirectly impact navigation and collision avoidance by shaping situational awareness and visibility. However, we won't consider a PrimeP for it.

I.4.1.5 Final Architect Prediction for Pipeline 1

Given the above list of thinking steps, architect prediction entails the following definition of problem domain complexity:

Table I.5 Architect output prediction

Pipeline 1 Architect Prediction
Architect assertion 1.4: Confusing and Unsafe Behaviours Identified: 1. Aircraft Deviating Unexpectedly from Assigned Flight Path: ○ Description: An aircraft deviates from its assigned path, intersecting with a nearby flight path in the process. ○ Unsafe Aspect: This behaviour contradicts expected adherence to assigned routes, introducing ambiguity about the aircraft's intentions and position for other pilots and ATC. ○ Undesirable Outcome: This unplanned deviation could increase collision risk as nearby aircraft or ATC might not anticipate this course change, leading to unsafe proximity or even a direct collision. 2. By-passing Aircraft Moving Erratically at a High Altitude near Crowded Airspace: ○ Description: An unauthorised or unplanned aircraft enters high-altitude airspace without coordinated communication, moving in a way that suggests erratic or unpredictable flight patterns. ○ Unsafe Aspect: Erratic movement within controlled airspace is unexpected and creates difficulty for ATC in predicting the bypassing's path, especially if the aircraft does not respond to communication. ○ Undesirable Outcome: This unpredictability can obstruct ATC's ability to route nearby aircraft safely and may cause surrounding aircraft to take evasive actions, increasing the likelihood of incidental close calls or collision. Architect assertion 2.4:

The architect asserts that the following depiction model captures the confusing behaviour faithfully:



Architect assertion 3.4: The architect asserts that:

The following list represents the salient complex of concerns:

{ownship aircraft, by-passing aircraft, other aircraft, clouds, terrain (land), sea}

Architect assertion 4.6:

Ownship aircraft, which includes {Computerised perception-based mid-air collision avoidance system (AVP), Ownship aircraft subsystems}

Other aircrafts, which include {by-passing aircraft, other aircraft}

Environment, which includes {clouds, land terrain features, sea features}

The architect asserts the PrimeP definitions for each Supra Complex as follows:

1. **Ownship Aircraft PrimeP:** Maintain safe operational flight and avoid mid-air collisions.
2. **Other Aircraft PrimeP:** Operate within shared airspace, adhering to established flight paths and protocols for collision avoidance.

I.4.2 Predictive Thinking Pipeline 2: Resolve the Complicatedness pattern of the observed complexity.

Predictive Thinking Pipeline 1 helped us clearly, objectively, and justifiably define the initial impression of the problem holistically. We exposed the overall structure of the problem domain complex. In this Predictive Thinking Pipeline, we will attempt to uncover the intricate interactions among the parts.

I.4.2.1 Step 2.1) Problem Interaction Analysis Using Actions Matrix

In this step, we systematically mapped out interactions between each identified system within the airspace conflict management environment by applying an Actions Matrix approach. This matrix captured relationships between the Ownship Aircraft, the AVP (Computerised Perception-Based Mid-Air Collision Avoidance System), By-passing Aircraft, Other Aircraft, and Environmental Contributing situations (clouds, terrain, and sea).

Each system's influence over others was defined through binary interactions, detailing each action's role in guidance, avoidance, detection, sensing, or complication (physical and visual). By pairing each source complex with every other sink complex, we identified 24 key interactions, which were then articulated in a concise action format: [source system][action][sink system]. This provided a complete mapping of direct and indirect interactions supporting airspace navigation and collision avoidance, ensuring a comprehensive understanding of influences within the system and achieving clarity on potential complications in a structured, predictable manner.

Table I.6 Interaction Analysis of Problematic Coexisting Elements within the Train Network and Adversarial Scheme

The following list defines the complicatedness of the problem domain complexity:

- n1:** [AVP][guide][Ownship ac]
- n2:** [By-passing ac][intersect][Ownship ac]
- n3:** [Other ac][avoid][Ownship ac]
- n4:** [Environment][Physically/visually complicate][Ownship ac]
- n5:** [Ownship ac][sense][Environment]
- n6:** [Ownship ac][govern][AVP]
- n7:** [By-passing ac][Physically/visually complicate][AVP]
- n8:** [Other ac][Physically/visually complicate][AVP]
- n9:** [Environment][Physically/visually complicate][AVP]
- n10:** [AVP][consider][Environment]

n11: [Ownship ac][avoid][By-passing ac]

n12: [AVP][detect][By-passing ac]

n13: [Other ac][avoid][By-passing ac]

n14: [Environment][Physically/visually complicate][By-passing ac]

n15: [By-passing ac][sense][Environment]

n16: [Ownship ac][avoid][Other ac]

n17: [AVP][detect][Other ac]

n18: [By-passing ac][avoid][Other ac]

n19: [Environment][Physically/visually complicate][Other ac]

n20: [Other ac][sense][Environment]

n21: [Ownship ac][sense][AVP]

n22: [AVP][detect][Environment]

n23: [By-passing ac][sense][Ownship ac]

n24: [Other ac][sense][By-passing ac]

The output is considered acceptable when;

- 1) All relationships are identified.
- 2) All relationships are briefly described in a single verbal phrase, no more than three words, and the nature of the problematic behaviours (actions). Only actions are defined.
- 3) Interactions are defined using the following format : [source system][action][sink system].

The output is considered unacceptable if:

- 1) There are missing interactions.
- 2) Interactions are defined using AIC factorisation method.
- 3) Interactions are not defined using the following format : [source system][action][sink system].

Implementation: Following are the steps of the Actions Matrix implementation:

1. Identify Source and Sink Complexes:

- Source systems included Ownship Aircraft, AVP, By-passing Aircraft, Other Aircraft, and Environmental Contributing situations.
- Sink systems were similarly defined for all elements to capture the interaction and response relationships among all components.

2. Define Action Types in Interaction Format:

- For each pair of interacting elements, a specific action was identified. Actions were briefly defined in terms of the specific interaction required, following the format [source system][action][sink system]. This captured functional relationships such as guidance, avoidance, detection, sense, and complication (physically or visually).

3. Map All Binary Interactions:

By systematically pairing each source complex with each sink complex, a total of **24 binary relationships** were identified and defined in concise terms. This allowed for a holistic understanding of interactions, capturing both direct and indirect influences on flight safety.

The format of the Actions Matrix would include the following interactions:

Table I.7 Actions Matrix representing interactions among problematic coexisting elements in the complexity field

	Ownship ac	AVP	By-passing ac	Other ac	Environment
Ownship ac		n 6	n 11	n 16	n 21
AVP	n 1		n 12	n 17	n 22
By-passing ac	n 2	n 7		n 18	n 23
Other ac	n 3	n 8	n 13		n 24
Environment	n 4	n 9	n 14	n 19	
	Ownship ac	AVP	By-passing ac	Other ac	Environment
Ownship ac		govern	avoid	avoid	Sense and predict
AVP	guide		detect	Detect	Consider
By-passing ac	Intersect	Physically/visually complicate		avoid	Sense and predict
Other ac	avoid	Physically/visually complicate	avoid		Sense and predict
Environment	Physically/visually complicate	Physically/visually complicate	Physically/visually complicate	Physically/visually complicate	

I.4.2.2 Step 2.2) Predict the contributing situations (unsafe situations or opportunities)

In this step, we rewrite the set of interactions we discovered that constitute complicatedness and categorise them into two categories:

- Foreseeable scenario
- Beneficial / Non-problematic activities.

We do so by examining every interaction and elaborating on its aspects and anticipated impact. Unsafe problematic activities are clear problems that, whatever design solution is chosen, will require further analysis.

Table I.8 Classification of unsafe and beneficial situations in train and adversarial drone interactions

Unsafe problematic activities	Beneficial or non-problematic activities
1. AVP: 1.2 n10: [AVP][consider][Environment] 1.3 n12: [AVP][detect][By-passing ac] 1.4 n17: [AVP][detect][Other ac] 1.5 n22: [AVP][detect][Environment] 1.6 n1: [AVP][guide][Ownship ac] 2. By-passing ac: 2.2 n2: [By-passing ac][intersect][Ownship ac] 2.3 n7: [By-passing ac][Physically/visually complicate][AVP] 2.4 n15: [By-passing ac][sense][Environment] 2.5 n18: [By-passing ac][avoid][Other ac] 2.6 n23: [By-passing ac][sense][Ownship ac] 3. Other ac: 3.2 n3: [Other ac][avoid][Ownship ac] 3.3 n8: [Other ac][Physically/visually complicate][AVP] 3.4 n13: [Other ac][avoid][By-passing ac] 3.5 n20: [Other ac][sense][Environment] 3.6 n24: [Other ac][sense][By-passing ac] 4. Ownship ac: 4.2 n5: [Ownship ac][sense][Environment]	n9b:[Clear Environment][rnable reliable][AVP]

4.3 n6: [Ownship ac][govern][AVP]	
4.4 n11: [Ownship ac][avoid][By-passing ac]	
4.5 n16: [Ownship ac][avoid][Other ac]	
4.6 n21: [Ownship ac][sense][AVP]	
5. Environment:	
5.2 n4: [Environment][Physically/visually complicate][Ownship ac]	
5.3 n9: [Environment][Physically/visually complicate][AVP]	
5.4 n14: [Environment][Physically/visually complicate][By-passing ac]	
5.5 n19: [Environment][Physically/visually complicate][Other ac]	

I.4.2.3 Step 2.3) Design problem selection

This step will review the original problem and re-evaluate it. By re-evaluation, we mean we want to be more definite in what set of sub-problems we want to further tackle and which problems we choose to leave. In this step, we will classify the output from step 2.2 into two categories and justify our choice:

- Problems within the design sphere of concern. We will use the following label “To be solved”.
- Situations outside the design sphere of concern. We will use the following label “To be dropped”.

The justification of the design problem sets out the initial high-level problems' context and motivation. The design team may contest some design problem choices, an important activity. Also, it opens up a door to regulators or design assurance teams to seek justifiable reasons why some identified issues are being dropped. At this stage, the design team can apply ethical considerations, which then go towards the trustworthiness case. The following is an example of defining problems:

Table I.9 Design Problem Selection

Interaction	Potential Concern	Decision	Elaboration or Justification
n1: [AVP][guide][Ownship ac]	AVP guiding the ownship may require precise alignment with real-time conditions and safety protocols.	To be solved	Guidance by AVP is essential to avoid collisions; ensuring accuracy and

			responsiveness is a key safety priority.
n2: [By-passing ac][intersect][Ownship ac]	Potential collision risk due to intersection.	To be solved	Intersecting ownship poses a direct risk; requires strategies for conflict detection and avoidance.
n3: [Other ac][avoid][directly risks]	Avoidance by other aircraft helps prevent mid-air collision.	To be solved	Coordinated avoidance measures contribute directly to collision prevention in shared airspace.
n4: [Environment][Physically/visually complicate][Ownship ac]	Environmental contributing situations like weather or terrain may impair the ownship's visibility and operations.	To be solved	Environmental complications are a significant safety risk; addressing visibility and navigational challenges is critical.
n5: [Ownship ac][sense][Environment]	Ownship sensing environment helps to detect obstacles and avoid hazards.	To be solved	Essential for situational awareness and collision avoidance; directly related to safe navigation.
n6: [Ownship ac][govern][AVP]	Ownship's control over AVP might affect AVP's collision avoidance capabilities.	To be solved	AVP's operations must integrate with ownship controls to support safety and autonomy goals.
n7: [By-passing ac][Physically/visually complicate][AVP]	By-passing aircraft might obscure AVP's sensors, impacting detection and avoidance.	To be solved	Obscuration by other aircraft poses a direct risk to AVP's detection capability; mitigation is required.
n8: [Other ac][Physically/visually complicate][AVP]	Other aircraft may obstruct AVP's view, hindering detection abilities.	To be solved	Reducing visual interference by other aircraft is crucial to maintaining AVP detection effectiveness.
n9: [Environment][Physically/visually complicate][AVP]	Environmental elements (e.g., clouds, terrain) can obstruct AVP's sensors.	To be solved	Environmental obstruction could lead to missed detections; mitigating this risk is critical for AVP reliability.
n10: [AVP][consider][Environment]	AVP must evaluate environmental conditions for safety.	To be solved	Essential for real-time responsiveness, environment-aware AVP supports safe manoeuvring.
n11: [Ownship ac][avoid][By-passing ac]	Necessary for ownship to avoid by-passing aircraft to prevent collisions.	To be solved	Direct collision avoidance by ownship is a safety-critical function, necessitating further design focus.
n12: [AVP][detect][By-passing ac]	AVP's detection of by-passing aircraft helps	To be solved	The detection of by-passing aircraft by AVP is foundational for collision

	manage proximity and prevent accidents.		avoidance, and it must be optimised.
n13: [Other ac][avoid][By-passing ac]	Other aircraft avoiding by-passings reduces potential collisions.	To be solved	Coordination in avoidance of manoeuvres is key to safe airspace management and needs reinforcement.
n14: [Environment][Physically/visually complicate][By-passing ac]	Environmental contributing situations may hinder by-passings' awareness, potentially impacting safe distance maintenance.	To be solved	Environmental interference requires a solution to ensure safe airspace awareness and collision avoidance.
n15: [By-passing ac][sense][Environment]	By-passing aircraft sensing the environment improves situational awareness.	To be solved	Environmental sensing by by-passing aircraft is essential for navigation and collision avoidance in crowded airspace.
n16: [Ownship ac][avoid][Other ac]	Critical for ownship to avoid other aircraft to ensure safe airspace sharing.	To be solved	Avoidance of other aircraft is fundamental to safe navigation; precise design is needed to prevent conflicts.
n17: [AVP][detect][Other ac]	Detection by AVP of other aircraft aids in avoiding collisions.	To be solved	AVP's detection supports safe coordination and is essential for ownship's situational awareness.
n18: [By-passing ac][avoid][Other ac]	By-passings avoiding other aircraft support airspace safety.	To be solved	Avoidance manoeuvres by by-passing aircraft are necessary to maintain safe airspace navigation and reduce risks.
n19: [Environment][Physically/visually complicate][Other ac]	Environmental conditions may obstruct other aircraft's sensors or visual capabilities.	To be solved	Environmental effects on other aircraft's navigation capabilities need addressing to support safe avoidance.
n20: [Other ac][sense][Environment]	Other aircraft sensing environmental conditions allows for informed avoidance and navigation.	To be dropped	Other aircraft's environmental sensing is beneficial but peripheral to ownship's AVP system design.
n21: [Ownship ac][sense][AVP]	Ownship sensing AVP helps ensure it receives AVP's guidance and avoids conflicts.	To be solved	It is essential to coordinate AVP and ownship actions, especially in dynamic conditions, to support safety.
n22: [AVP][detect][Environment]	AVP's detection of the environment supports responsive guidance for the ownship.	To be solved	Crucial for AVP to detect environmental contributing situations for appropriate guidance adjustments.

n23: [By-passing ac][sense][Ownship ac]	By-passing's awareness of ownship aids in safe distance maintenance.	To be solved	Sensing by by-passing aircraft contributes to safe separation and prevents collision.
n24: [Other ac][sense][By-passing ac]	Sensing of by-passing by other aircraft supports mutual avoidance.	To be dropped	Beneficial but outside the design scope focused on ownship's AVP system; can be coordinated with external safety protocols.

The table above clearly categorises design focus areas based on interaction priority and safety relevance. By defining which interactions are “to be solved,” we outline the design team’s focus on critical safety and operational issues in the airspace, while interactions “to be dropped” are classified as lower-priority or peripheral to the AVP’s direct operational requirements.

I.4.3 Predictive Thinking Pipeline 3: Predict the Emergence of AIC Complexity Field for Detailed Operational Scenario Articulation

Predictive Thinking Pipeline 2 helped us define the potential problems that we need to solve and the choices of problems to solve. In this pipeline, we will dive deeper into the AIC complexity of each interaction discovered and then identify the intricate relationships.

I.4.3.1 Step 3.1) Model detailed AIC interactions scenarios for the problem domain

For every unsafe, problematic, or beneficial interaction, and considering the source complex's PrimeP, visualise the emergence of AIC Complicated Behaviour using an AIC modelling schema. In this case, any factor modelled must be written with a specific situation in the format of {adjective_name} which we refer to as a situation. For example, if you define perception-based avoidance system (avp) as a factor, a situation needs to be mentioned with it, for instance, {active_avp} which means avp is actively influencing the ownship aircraft system. Being clear about a factor's dynamic or static situation is a critical thinking step to resolve the complicatedness of any complexity. AIC modelling schema models the dynamic and static situations of systems. Also, actions are to be modelled as verbal phrases and signed (-) if they carry an intended obstructive goal, (+) if they carry an intended supportive goal, or unsigned if they carry a neutral unintended impact. See Figure I.1 for the abstract model of Forward-Feed Partial AIC-Core STCoT:

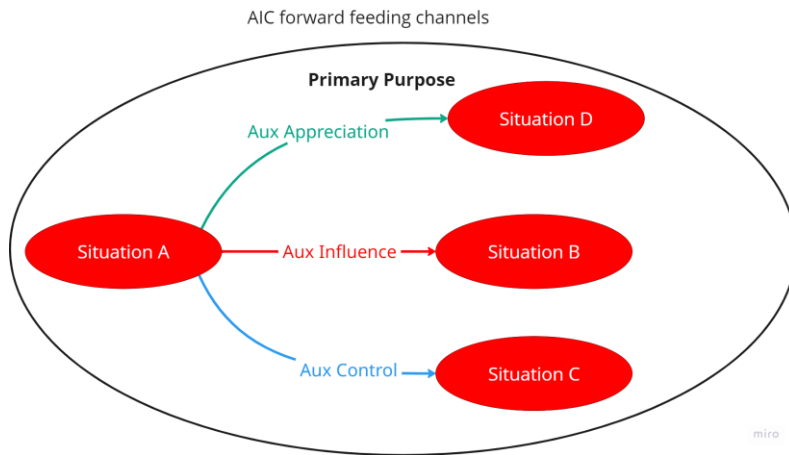


Figure I.1 Forward-Feed partial AIC-Core STCoT

In our case study, we will consider only the Forward-Feed partial AIC-Core STCoT. We will choose AVP guidance of ownship aircraft, which may lead to potential mid-air collision [derived from n1]. The key activity is we would do the AIC factorisation to each interaction in order to reveal the hidden interactions.

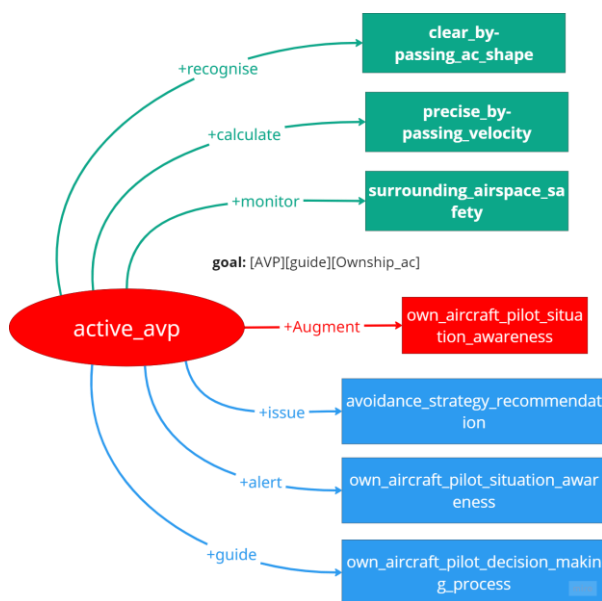


Figure I.2 Modelling AIC scenario from interaction n1

Figure I.2 illustrates the Active AVP Guidance System designed to enhance situational awareness and decision-making for the ownship aircraft pilot in scenarios that may lead to a potential mid-air collision. The model visually represents the interactions between the Active AVP system (red oval) and various situational parameters and decision-making processes through appreciation, influence, and control relationships.

Key Components and Interactions:

1. Appreciation-based monitoring (Green Elements & Arrows)

- The **Active AVP** recognises, calculates, and monitors crucial situational parameters that contribute to flight safety:
 - **Clear By-Passing Aircraft Shape:** The system recognises the spatial characteristics of nearby aircraft to ensure safe manoeuvring.
 - **Precise By-Passing Velocity:** The system calculates the optimal velocity required to avoid conflicting air traffic safely.
 - **Surrounding Airspace Safety:** The AVP monitors environmental conditions, ensuring situational awareness in the operational domain.
- 2. **Influence on Pilot Awareness (Red Element & Arrow)**
 - The **Active AVP augments the ownship aircraft pilot's situational awareness**, ensuring the pilot has an enhanced perception of the airspace environment, potential hazards, and nearby aircraft.
- 3. **Control-based decision Support (Blue Elements & Arrows)**
 - The AVP actively engages with the **pilot's decision-making processes** by issuing guidance, alerts, and recommendations:
 - **Avoidance Strategy Recommendation:** The AVP issues strategic recommendations to optimise collision avoidance.
 - **Own Aircraft Pilot Situation Awareness:** The AVP alerts the pilot to changes in the operational environment.
 - **Own Aircraft Pilot Decision-Making Process:** The AVP provides guidance to assist the pilot in executing effective avoidance manoeuvres.

The primary purpose of this system is to facilitate AVP-guided navigation for the ownship aircraft to enhance safety and reduce mid-air collision risks. By integrating appreciation (green), influence (red), and control (blue) interactions, the system ensures a multi-layered approach to situational assessment and pilot support, ultimately contributing to safer and more efficient airspace operations.

I.4.3.2 Step 3.2) Predict the extended list of emergent AIC interactions scenarios

We then capture the interactions in the figure above using the following STCoT. Note that we use the interaction format of `[[situation]_[action]_[situation]]` to help us with automating the extraction of contributing situations.

In Appendix K, Table K.1, we experimented with a slight variation on the input of the STCoT. Instead of taking one interaction at a time, we used all 24 interactions as one detailed description of Unsafe problematic activities. The reason is for efficiency. The architect can extend each interaction individually or use multiple interactions. Our experiment showed that doing one interaction at a time

provides more systematic explainability of how each interaction was considered in the design. When we summed them up, the explainability may require more clarification of how each interaction contributed to the whole factorisation process. Nonetheless, the exercise also proved to be useful in discovering those long-tail Black Swan interactions. Below is an example of how we used a single interaction n1 to extend an AIC scenario:

Table I.10 AIC-Core STCoT implementation for n1 Interaction STCoT definition

Input	Step 1: Select observed outcome behaviour: Unsafe problematic activities: n1: [AVP][guide][Ownship ac]	
Step 2: Observed System (obs)	Step 3: Inquired interaction	Step 4: supra source Primary Purpose
active_avp, own_ac	Active_avp guides own_Input Behaviour	Maintain safe operational flight and avoid mid-air collisions
Step 5: Auxiliary Influence interaction	Step 6: Auxiliary Control interaction	Step 7: Auxiliary Appreciation interaction
{{active_avp}_[+augment]_{own_aircraft_pilot_situation_awareness}}	{{active_avp}_[+issue]_{avoidance_strategy_recommendation}} {{active_avp}_[+alert]_{own_aircraft_pilot_situation_awareness}} {{active_avp}_[+guide]_{own_aircraft_pilot_decision_making_process}}	{{active_avp}_[+monitor]_{surrounding_airspace_safety}} {{active_avp}_[+calculate]_{precise_by-passing_velocity}} {{active_avp}_[+recognise]_{clear_by-passing_ac_shape}}
Step 8: Predicted outcome emergence contributing situations or Features (with repetition)		
Appreciation = ['active_avp', 'surrounding_airspace_safety', 'active_avp', 'precise_by-passing_velocity', 'active_avp', 'clear_by-passing_ac_shape'] Influence = ['active_avp', 'avoidance_strategy_recommendation', 'active_avp', 'own_aircraft_pilot_situation_awareness', 'active_avp', 'own_aircraft_pilot_decision_making_process'] Control = ['active_avp', 'avoidance_strategy_recommendation', 'active_avp', 'own_aircraft_pilot_situation_awareness', 'active_avp', 'own_aircraft_pilot_decision_making_process']		

Table I.10 presents a Systems Engineering Chain-of-Thought (STCoT) definition for interaction n1, focusing on how an Active AVP guides the Ownship aircraft (own_ac) within a complex operational environment. The table follows the STCoT structured interaction format ({{situation}}_{{action}}_{{situation}}) to facilitate automated extraction of contributing situations, ensuring a systematic analysis of interactions within the complexity field.

Step 1: Defining Unsafe problematic activities

The table begins by identifying an unsafe problematic situation where the Active AVP guides the Ownship aircraft (own_ac), leading to potential flight safety concerns. This scenario is designated as **n1: [AVP][guide][Ownship ac]**, emphasising the control exerted by the AVP on the aircraft's behaviour.

Steps 2-4: Observed System, Action, and Supra Source Purpose

- **Step 2 (Observed System):** The primary interacting components are **the** Active AVP and the Ownship aircraft.
- **Step 3 (Observed Action):** The Active AVP directly guides the Ownship aircraft's behaviour, shaping its response to flight dynamics.
- **Step 4 (Supra Source Primary Purpose):** The ultimate purpose of this interaction is to maintain safe operational flight and avoid mid-air collisions, ensuring controlled manoeuvring in dynamic airspace environments.

Steps 5-7: Auxiliary Influence, Control, and Appreciation Interactions

The table then categorises auxiliary interactions that describe how the AVP influences, controls, and appreciates elements within the operational complexity field:

- **Step 5 (Influence Interactions):**
 - The AVP augments the pilot's situational awareness to enhance decision-making.
 - It issues avoidance strategy recommendations, improving navigational responses.
 - It alerts the pilot to critical airspace safety conditions.
- **Step 6 (Control Interactions):**
 - The AVP monitors surrounding airspace safety, continuously evaluating risks.
 - It calculates precise by-passing velocity, optimizing aircraft manoeuvres.
 - It recognises clear by-passing aircraft shapes, ensuring accurate decision-making.
- **Step 7 (Appreciation Interactions):**
 - The AVP appreciates the role of surrounding airspace safety in determining flight adjustments.
 - It acknowledges the importance of precise manoeuvring calculations to ensure collision-free paths.
 - It contributing situations in the shape and trajectory of bypassing aircraft when guiding ownship movements.

Step 8: Predicted Problem Domain Contributing situations or Features

The final step identifies recurring problem domain contributing situations across Appreciation, Influence, and Control categories, helping to classify key system dependencies:

- **Appreciation Contributing situations:**
 - The AVP appreciates surrounding airspace safety, the necessity of **b** by-passing velocity, and the recognition of clear by-passing aircraft shapes.
- **Influence Contributing situations:**
 - The AVP influences avoidance strategy recommendations, ownship pilot situational awareness, and ownship pilot decision-making processes.
- **Control Contributing situations:**
 - The AVP controls the avoidance strategy recommendation process, ownship pilot situational awareness, and ownship pilot decision-making process to ensure safe manoeuvring.

I.4.3.3 Step 3.3) Collate contributing situations/situations (step 8 in the table)

Collate all contributing situations in between the curly brackets {} and capture them in AIC lists.

Include the following information types: Source, Sink, and Supra Systems. For example;

Appreciation = ['active_avp', 'surrounding_airspace_safety', 'active_avp',
'precise_by-passing_velocity', 'active_avp', 'clear_by-passing_ac_shape']

Influence = ['active_avp', 'avoidance_strategy_recommendation', 'active_avp', '
own_aircraft_pilot_situation_awareness', 'active_avp',
'own_aircraft_pilot_decision_making_process']

Control = ['active_avp', 'avoidance_strategy_recommendation',
'active_avp', 'own_aircraft_pilot_situation_awareness', 'active_avp',
'own_aircraft_pilot_decision_making_process']

I.4.4 Predictive Thinking Pipeline 4: Architect Attention, Novelty and Black Swan assessment

I.4.4.1 Step 4.2) Define all identified problem domain contributing situations.

Define each factor. We identify 102 types of contributing situations involved within the complicatedness of the problem complexity. Below is the full definition for each factor (in no particular order);

1. **Own_aircraft_pilot_decision_making_process:** The process through which the pilot of the aircraft makes decisions based on various inputs, including alerts from the AVP system, situation awareness, and environmental conditions.

2. **By-passing aircraft position:** The location of an aircraft that is not part of the own aircraft's fleet, potentially posing a collision risk.
3. **Stabilised own aircraft dynamics:** The situation of the own aircraft where its flight dynamics {e.g., speed, altitude, direction} are stable and under control.
4. **Avp (AVOID Dataset based Perception), active_avp:** A system designed to enhance the pilot's awareness and decision-making process by providing alerts and recommendations based on the surrounding environment and potential threats. active_avp means the avp system is actively influencing ownship aircraft.
5. **Own aircraft pilot situation awareness:** The pilot's understanding and perception of the current situation and environment around the aircraft.
6. **Surrounding airspace safety:** The safety status of the airspace around the aircraft, including the presence of other aircraft and environmental conditions.
7. **Avoidance strategy recommendation:** The AVP system provides the pilot with suggestions on how to manoeuvre the aircraft to avoid potential collisions. These suggestions are inherently biased due to specialisation in one environment over others.
8. **Visual information:** Data collected from visual sensors like cameras, providing imagery or video of the aircraft's surroundings. The visual information is corrupted due to log distance spoofing operation.
9. **Own aircraft flight path:** The trajectory or route that the own aircraft is following or plans to follow.
10. **By-passing aircraft speed:** The velocity at which an By-passing aircraft is moving.
11. **Collision avoidance system:** An onboard system designed to prevent collisions by detecting and alerting pilots of potential threats.
12. **Collision threat:** A situation where there is a high risk of the aircraft colliding with another object, such as another aircraft.
13. **By-passing aircraft proximity:** The closeness or distance of an By-passing aircraft to the own aircraft.
14. **Conflicted_atc {Air Traffic Control}:** A service provided by ground-based controllers who direct aircraft on the ground and through controlled airspace. They are conflicted in the sense that they are impacted by different priorities at any given time.
15. **Own aircraft flight management system:** An integrated onboard computer system that manages key flight parameters.
16. **Other aircrafts:** Other aircrafts present in the vicinity of the own aircraft.
17. **By-passing aircraft detection tracking:** The process of identifying and continuously monitoring the position and movement of By-passing aircraft.
18. **By-passing aircraft motion pattern:** The pattern or behavior of an By-passing aircraft's movement, including changes in speed and direction.

19. **Own_aircraft_camera:** Cameras mounted on the aircraft used to capture visual information of the surroundings.
20. **Own_aircraft_roll_change:** Adjustments in the aircraft's roll {lateral rotation} for maneuvering.
21. **Own_aircraft_pitch_change:** Adjustments in the aircraft's pitch {vertical rotation} for maneuvering.
22. **Own_aircraft_speed:** The velocity at which the own aircraft is moving.
23. **Own_aircraft_flight_maneuvers:** Various actions taken by the aircraft to change its flight path, speed, or orientation.
24. **Rerouting_instructions:** Directions provided by ATC or the flight management system to change the current flight path for safety.
25. **Pilot_stress:** The level of psychological pressure or stress experienced by the pilot, particularly in emergency or demanding situations.
26. **By-passing_aircraft_visibility:** The ability of the AVP system and the pilot to visually detect an By-passing aircraft.
27. **Unpredictable_By-passing_aircraft_behaviour:** Erratic or non-standard movements of an By-passing aircraft that complicate detection and avoidance strategies.
28. **By-passing_aircraft_dynamics:** The characteristics of the By-passing aircraft's movement, including speed, altitude, and trajectory.
29. **Weather_data:** Information related to weather conditions, such as wind, visibility, and turbulence.
30. **Pilot_alerting_and_warning_systems:** Systems designed to alert the pilot about potential threats or necessary actions.
31. **Collision_threat_alert:** An alert issued when there is an imminent risk of collision.
32. **Weather_conditions:** The current situation of the weather, including contributing situations like wind, visibility, clouds, and turbulence.
33. **Surrounding_airspace_obstructions:** Physical or environmental obstacles in the airspace around the aircraft.
34. **Own_aircraft_flight_path_change:** Adjustments made to the original flight path of the aircraft.
35. **Non_deterministic_intelligent_algorithms:** Algorithms used in AVP systems that provide outputs not strictly determined by their inputs, often used for complex decision-making processes.
36. **By-passing_aircraft_direction:** The course or path along which an By-passing aircraft is moving.
37. **By-passing_aircraft_altitude:** The height above ground or sea level at which an By-passing aircraft is flying.

- 38. **Threat_predictive_model:** A model used to forecast potential threats, such as collision risks, based on current data and trends.
- 39. **Risk_of_potential_collision:** The likelihood or probability of the aircraft colliding with another object.
- 40. **Cockpit_display_systems:** Electronic display panels in the cockpit that show flight information, system statuses, and alerts.
- 41. **Aircrafts_existing_warning_systems:** Pre-existing systems in the aircraft designed to warn the pilot of various threats or necessary actions.
- 42. **Own_aircraft_yaw_change:** Adjustments in the aircraft's yaw {horizontal rotation} for maneuvering.
- 43. **Own_aircraft_altitude:** The height at which the own aircraft is flying above ground or sea level.
- 44. **Own_aircraft_radar:** Radar systems on the aircraft used for detecting other objects and aircraft in the vicinity.
- 45. **Own_aircraft_lidar:** Lidar systems on the aircraft used for measuring distances and detecting objects by illuminating the target with laser light.
- 46. **Environmental_obstructions:** Physical or atmospheric conditions that hinder visibility or the operation of sensors.
- 47. **Fog:** A weather condition that reduces visibility, potentially impacting sensor performance.
- 48. **Raindrops:** Precipitation that can interfere with sensor accuracy and visibility.
- 49. **Cloud_cover:** The fraction of the sky covered by clouds, which can affect visibility and sensor operation.
- 50. **Cloud_type:** The classification of clouds based on their appearance and altitude, impacting visibility and flight conditions.
- 51. **Cloud_turbulence:** Atmospheric turbulence associated with certain types of clouds, affecting flight stability.
- 52. **Landscape_background:** The terrain and features in the background that could affect sensor readings and visibility.
- 53. **Daytime:** The time during the day, affecting lighting conditions and thus sensor performance.
- 54. **Sunlight:** Natural light from the sun, influencing visibility and the effectiveness of visual sensors.
- 55. **Sun_position:** The location of the sun in the sky, which can impact sensor performance and pilot visibility.
- 56. **Own_camera:** Cameras installed on the aircraft used to capture visual information.
- 57. **By-passing_aircraft_flight_path:** The trajectory or route followed by an By-passing aircraft.
- 58. **Collision_time:** The estimated time remaining before a potential collision could occur.

59. **Own_aircraft_response_time:** The time it takes for the own aircraft to respond to control inputs or system commands.
60. **Own_aircraft_pilot_feedback:** Input or reactions from the pilot in response to the system's performance or alerts.
61. **Avp_real_time_training:** The process of continuously updating and improving the AVP system based on real-time data and pilot feedback.
62. **Emergency_directives:** Urgent instructions issued in response to critical situations, such as potential collisions.
63. **Unsafe_proximity:** A distance between two aircraft that is considered dangerously close.
64. **Airspace_conflicts:** Situations where the paths of multiple aircraft intersect, creating a risk of collision.
65. **Radar_and_visual_range:** The range within which objects can be detected either by radar or visually.
66. **Airspace_visibility:** The clarity of the airspace, affecting the ability to visually detect other aircraft.
67. **Wind:** Atmospheric wind conditions, which can affect flight dynamics and sensor operation.
68. **Cloud_turbulence:** Disturbances in the atmosphere associated with certain cloud formations, affecting flight stability.
69. **Avp_confidence_threshold:** The level of certainty at which the AVP system reliably operates.
70. **By-passing_aircraft_hack_avp_threashold:** An attempt to manipulate or interfere with the AVP system's thresholds or parameters.
71. **Own_aircraft_proximity:** The distance of the own aircraft from other objects or aircraft.
72. **Autonomous_collision_avoidance_maneuvers:** Maneuvers executed by the aircraft's systems without pilot input to avoid collisions.
73. **Potential_collision_threat_alert:** A warning issued when there is a possibility of a collision.
74. **Collision_avoidance:** Measures taken to prevent a collision from occurring.
75. **Automated_own_aircraft_flight_path_change:** Changes to the flight path made automatically by the aircraft's systems.
76. **Own_aircraft_altitude_change:** Adjustments to the altitude at which the aircraft is flying.
77. **Disturbed_own_aircraft_dynamics:** Changes in the aircraft's flight dynamics due to external contributing situations or maneuvers.
78. **Own_aircraft_speed_change:** Adjustments in the speed at which the aircraft is flying.
79. **Accelerated_own_aircraft_dynamics:** The dynamics of the aircraft when it is accelerating or decelerating.
80. **Own_aircraft_yaw_change:** Adjustments in the aircraft's yaw {horizontal axis rotation}.
81. **Accelerated_own_aircraft_roll:** The roll of the aircraft when it is undergoing acceleration.

- 82. **Accelerated_own_aircraft_pitch:** The pitch of the aircraft during acceleration.
- 83. **Accelerated_own_aircraft_yaw:** The yaw of the aircraft when it is accelerating.
- 84. **Own_aircraft_flight_management_system_override:** The ability of the pilot to manually override the automated flight management system.
- 85. **Manual_own_aircraft_flight_path_change:** Changes to the flight path made manually by the pilot.
- 86. **Weather_conditions_change:** Variations in weather conditions that can affect flight dynamics and sensor operation.
- 87. **Weather_data_change:** Alterations in the weather data being received and processed.
- 88. **Own_aircraft_dynamics_change:** Changes in the flight dynamics of the own aircraft due to various contributing situations.
- 89. **Unpredictable_other_aircrafts_decision:** Erratic or unforeseen decisions made by other aircraft in the vicinity.
- 90. **Automated_collision_avoidance_maneuvers:** Manoeuvres executed automatically by the aircraft's systems to avoid collisions.
- 91. **Avp_detection_communication_delay_hack:** Deliberate attempts to cause delays in the AVP system's detection and communication processes.
- 92. **Pilot_cognitive_limit:** The limitations of the pilot's ability to process information and make decisions under stress.
- 93. **Surrounding_airspace_restrictions:** Limitations or prohibitions in certain areas of the airspace, affecting flight paths.
- 94. **Military_aircraft_intrusion:** The presence of military aircraft in the airspace, which can affect civil aviation operations.
- 95. **Other_aircrafts_presence:** The presence of other aircraft in the vicinity of the own aircraft.
- 96. **Birds:** Birds activity that can pose a risk to aircraft, especially during cruise, takeoff and landing.
- 97. **Drone_presence:** The presence of unmanned aerial vehicles {drones} in the airspace, which can pose collision risks.
- 98. **Wind_shear_effect_on_camera:** The impact of sudden changes in wind speed or direction on the operation of onboard cameras.
- 99. **Air_density_change_due_to_high_speed_effect_on_camera:** The effect of changes in air density, especially at high speeds, on camera performance.
- 100. **Aircraft_attitude_control:** The system and process of managing the orientation of the aircraft in flight.
- 101. **Mechanical_wear_and_tear:** The degradation of aircraft components over time due to regular use.
- 102. **Autopilot_system_failure:** A malfunction or failure of the aircraft's autopilot system.

103. **Information overload for pilot:** A situation where the pilot is presented with more information than can be effectively processed.
104. **Highly stressful event:** An event or situation that places a high level of stress on the pilot, affecting decision-making and performance.

I.4.4.2 Step 4.3) Define the problematic scenarios that may emerge from contributing situations

To do so, list all predicted emergent interactions without repetition. Then, describe the associated assumed situations, specifying which aspects of the contributing situations are being assumed and which other elements are not. Lateral Predictive Thinking Approaches can help with imagining those assumptions. See appendix K, section K.2.2.

In the previous case study, we did not incorporate hazard analysis at this stage. However, at this point, we will introduce hazard analysis to consider safety and security from the outset. This hazard analysis will inform the stakeholders and the architect in the next stage, enabling them to make engineering judgments based on the identified hazards perspective.

For example,

active_avp: *Dataset bias delays Resolution Advisory in cluttered airspace; pilot trusts late advisory; near miss during crossing traffic.*

visual_information: *Sun glare saturates the camera; intruder undetected; AVP assumes clear visuals; late alert. Long-distance spoofing operation corrupting visual information.*

In this process, we also experimented with adding further steps at this stage and with what else we could use the derived contributing situations for. For the full table, see Appendix K, Table K.2. This time we included further validation and hazards identification analysis. Whereby we examined whether each assumption is plausible, hazardous or not. The following are the steps that architects can use to perform an extended analysis at this stage:

1. Identify Assumptions:

- Fill columns titled “Predicted contributing situation”, “Definition”.
- Clearly define your assumptions related to each discovered factor.
 - **Example:** Own_aircraft_pilot_decision_making_process: The process through which the aircraft’s pilot makes decisions based on various inputs, including alerts from the AVP system, situation awareness, and environmental conditions.

- **Explicit Assumption:** The pilot will never over-rely on the perception system's judgment, which may reduce their normal vigilance during flight.

2. Evaluate the Realism of Assumptions:

- Fill columns titled "So what? Assumption", "Plausibility", "Why is it plausible or not?".
- Assess whether the assumptions are realistic based on practical and operational scenarios.
 - **Evaluation:** Is the latter a realistic assumption?
 - **Answer:** No.
 - **Reason:** When the system demonstrates reliable results, pilots will eventually trust it more and more, leading to over trust.

3. Determine Relevance to the Problem:

- Fill columns titled "Concerning?", "Why should we be concerned or not?".
- Decide if the identified assumption should be included in the problem analysis.
 - **Question:** Should it be part of the problem?
 - **Answer:** Yes.
 - **Reason:** Directly impacting flight safety.

4. Identify Potential Hazards:

- Fill columns titled "Hazard?", "How and Why is it hazardous?".
- Assess if the assumption could lead to potential hazards or safety concerns.
 - **Question:** Is there a potential hazard?
 - **Answer:** Yes.
 - **Reason:** Pilots may unconsciously over-rely on the system over time, reducing their own personal vigilance and thus impacting airmanship.

I.4.4.3 Step 4.1) Identify possible Black Swan scenarios

Having established a comprehensive inventory of contributing situations within the problem's complexity field, the architect is now positioned to exercise informed engineering judgment in identifying and selecting those factors deemed Black Swans, that is, events or interactions that, based on prior operational experience (originality assessment), domain expertise (typicality assessment), are anticipated to occur with low frequency yet retain non-negligible potential for high-impact consequences relative to the operational design domain (ODD). The central predictive queries guiding this activity are:

- What kind of original, atypical, and impactful (or non-impactful) situation might arise from those contributing situations?

Perform Architect Attention Assessment:

The first evaluation conducted in this step is the Architect's attention assessment (see section 3.6). The analysis is captured by using a Python script that extracts predicted situations in the format of {x_y_z}. Then tally the number of mentions and compute tally-bases probability density (tally/total of mentions). Table I.11 captures the process:

Table I.11 Black Swan prediction with respect to the architect concern distribution²

Predicted situation	Tally	Attention%
active_avp	15	6.696428571
stabilised_own_aircraft_dynamics	11	4.910714286
own_aircraft_pilot_decision_making_process	10	4.464285714
by-passing_aircraft_position	10	4.464285714
own_aircraft_pilot_situation_awareness	7	3.125
own_aircraft_flight_path	7	3.125
surroundingairspace_safety	6	2.678571429
avoidance_strategy_recommendation	6	2.678571429
visual_information	6	2.678571429
by-passing_aircraft_speed	6	2.678571429
collision_avoidance_system	6	2.678571429
by-passing_aircraft_proximity	5	2.232142857
conflicted_atc	5	2.232142857
own_aircraft_flight_management_system	5	2.232142857
collision_threat	4	1.785714286
own_aircraft_camera	4	1.785714286
other_aircrafts	4	1.785714286
by-passing_aircraft_motion_pattern	3	1.339285714
weather_data	3	1.339285714
by-passing_aircraft_detection_tracking	2	0.892857143
own_aircraft_roll_change	2	0.892857143
own_aircraft_pitch_change	2	0.892857143
own_aircraft_yaw_change	2	0.892857143
own_aircraft_speed	2	0.892857143
cloud_turbulence	2	0.892857143
own_aircraft_flight_maneuvers	2	0.892857143
rerouting_instructions	2	0.892857143
pilot_stress	2	0.892857143
by-passing_aircraft_visibility	2	0.892857143
unpredictable_by-passing_aircraft_actions	2	0.892857143
by-passing_aircraft_dynamics	2	0.892857143
by-passing_aircraft_hack_avp_threshold	2	0.892857143
pilot_alerting_and_warning_systems	2	0.892857143
collision_threat_alert	2	0.892857143
weather_conditions	2	0.892857143
surroundingairspace_obstructions	2	0.892857143
own_aircraft_flight_path_change	2	0.892857143
non_deterministic_intelligent_algorithms	1	0.446428571
by-passing_aircraft_direction	1	0.446428571
by-passing_aircraft_altitude	1	0.446428571
threat_predictive_model	1	0.446428571

² Example snippet, for full table see Appendix M.

risk_of_potential_collision	1	0.446428571
cockpit_display_systems	1	0.446428571
aircrafts_existing_warning_systems	1	0.446428571
own_aircraft_altitude	1	0.446428571
own_aircraft_radar	1	0.446428571
own_aircraft_lidar	1	0.446428571
environmental_obstructions	1	0.446428571
fog	1	0.446428571
raindrops	1	0.446428571
cloud_cover	1	0.446428571
cloud_type	1	0.446428571
landscape_background	1	0.446428571
daytime	1	0.446428571
sunlight	1	0.446428571
sun_position	1	0.446428571
by-passing_aircraft_flight_path	1	0.446428571
collision_time	1	0.446428571
own_aircraft_response_time	1	0.446428571
own_aircraft_pilot_feedback	1	0.446428571
avp_real_time_training	1	0.446428571
by-passing_aircraft	1	0.446428571
emergency_directives	1	0.446428571
unsafe_proximity	1	0.446428571
airspace_conflicts	1	0.446428571
radar_and_visual_range	1	0.446428571
airspace_visibility	1	0.446428571
wind	1	0.446428571
avp_confidence_threshold	1	0.446428571
own_aircraft_proximity	1	0.446428571
autonomous_collision_avoidance_maneuvers	1	0.446428571
potential_collision_threat_alert	1	0.446428571
collision_avoidance	1	0.446428571
automated_own_aircraft_flight_path_change	1	0.446428571
own_aircraft_altitude_change	1	0.446428571
disturbed_own_aircraft_dynamics	1	0.446428571
own_aircraft_speed_change	1	0.446428571
accelerated_own_aircraft_dynamics	1	0.446428571
accelerated_own_aircraft_roll	1	0.446428571
accelerated_own_aircraft_pitch	1	0.446428571
accelerated_own_aircraft_yaw	1	0.446428571
own_aircraft_flight_management_system_override	1	0.446428571
manual_own_aircraft_flight_path_change	1	0.446428571
weather_conditions_change	1	0.446428571
weather_data_change	1	0.446428571
own_aircraft_dynamics_change	1	0.446428571
unpredictable_other_aircrafts_decision	1	0.446428571
automated_collision_avoidance_maneuvers	1	0.446428571
avp_detection_communication_delay_hack	1	0.446428571
pilot_cognitive_limit	1	0.446428571
surrounding_airspace_restrictions	1	0.446428571
military_aircraft_intrusion	1	0.446428571
other_aircrafts_presence	1	0.446428571
bird	1	0.446428571
drone_presence	1	0.446428571
wind_shear_effect_on_camera	1	0.446428571
air_density_change_due_to_high_speed_effect_on_camera	1	0.446428571

aircraft_attitude_control	1	0.446428571
mechanical_wear_and_tear	1	0.446428571
autopilot_system_failure	1	0.446428571
information_overload_for_pilot	1	0.446428571
highly_stressfull_event	1	0.446428571

Table I.11 shows that active_avp is mentioned 15 times in the analysis, indicating that the architect allocated 6.69% of their attention to this factor. This indicates that the solution design will be biased. Produce an Architect's Attention (PAD) diagram; see Figure 6.2.

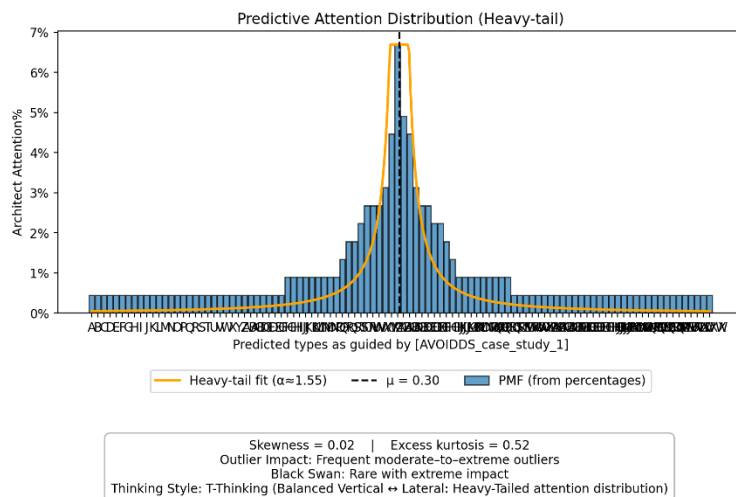


Figure I.3 Case Study 2: Predictive Attention Distribution curve of the Architect during analysis.

Perform Novelty Test for predicted situations:

Perform the novelty test for each predicted situation (see section 3.13). The following attributes are required to perform the novelty test:

- **Originality assessment:** The output of AIC stage 1 is then compared against STAMP, FRAM, and HAZOP as prior literature of predicted situations. The assessment checks whether a predicted hazard has ever been mentioned in any hazard analysis process. The corpus can be determined by the project team. In this thesis, originality assessment was done against Appendix G, section G.5.6.
- **Typicality assessment:** the architect is prompted to assess whether a predicted situation or an interaction is typically salient (noticeable) or atypical (not typically noticeable). For example, the architect had never encountered a harmful situation where the shape of a leaf was a primary concern. Such a situation would be considered atypical. See §§2.3.2, 3.13 for a definition of the Novelty Test.
- **Typicality assessment rational:** the expert is then prompted to justify their judgement about the typicality of a predicted situation.

The following table I.12a captures the novelty test exercise:

Table I.12a Novelty Test implementation

Predicted situation	Originality	Typicalty Test	Rational for typicality value	Novelty Test
active avp	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
stabilised own aircraft dynamics	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
own aircraft pilot decision making process	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
by-passing aircraft position	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
own aircraft pilot situation awareness	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
own aircraft flight path	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
surrounding airspace safety	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
avoidance strategy recommendation	original	atypical	Low or no prior risk assessment is available	Novel
visual information	original	atypical	Low or no prior risk assessment is available	Novel
by-passing aircraft speed	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
collision avoidance system	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
by-passing aircraft proximity	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
conflicted atc	original	atypical	Low or no prior risk assessment is available	Novel
own aircraft flight management system	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
collision threat	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
own aircraft camera	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
other aircrafts	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
by-passing aircraft motion pattern	original	atypical	Low or no prior risk assessment is available	Novel
weather data	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
by-passing aircraft detection tracking	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
own aircraft roll change	original	atypical	Low or no prior risk assessment is available	Novel
own aircraft pitch change	original	atypical	Low or no prior risk assessment is available	Novel
own aircraft yaw change	original	atypical	Low or no prior risk assessment is available	Novel
own aircraft speed	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
cloud turbulence	original	atypical	Low or no prior risk assessment is available	Novel
own aircraft flight maneuvers	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
rerouting instructions	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
pilot stress	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
by-passing aircraft visibility	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
unpredictable by-passing aircraft actions	original	atypical	Low or no prior risk assessment is available	Novel
by-passing aircraft dynamics	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
by-passing aircraft hack avp threshold	original	atypical	Low or no prior risk assessment is available	Novel
pilot alerting and warning systems	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
collision threat alert	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
weather conditions	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel

surrounding airspace obstructions	original	atypical	Low or no prior risk assessment is available	Novel
own aircraft flight path change	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
non deterministic intelligent algorithms	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
by-passing aircraft direction	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
by-passing aircraft altitude	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
threat predictive model	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
risk of potential collision	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
cockpit display systems	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
aircrafts existing warning systems	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
own aircraft altitude	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
own aircraft radar	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
own aircraft lidar	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
environmental obstructions	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
fog	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
raindrops	original	atypical	Low or no prior risk assessment is available	Novel
cloud cover	original	atypical	Low or no prior risk assessment is available	Novel
cloud type	original	atypical	Low or no prior risk assessment is available	Novel
landscape background	original	atypical	Low or no prior risk assessment is available	Novel
daytime	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
sunlight	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
sun position	original	atypical	Low or no prior risk assessment is available	Novel
by-passing aircraft flight path	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
collision time	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
own aircraft response time	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
own aircraft pilot feedback	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
avp real time training	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
by-passing aircraft	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
emergency directives	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
unsafe proximity	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
airspace conflicts	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
radar and visual range	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
airspace visibility	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
wind	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
avp confidence threshold	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
own aircraft proximity	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
autonomous collision avoidance maneuvers	original	atypical	Low or no prior risk assessment is available	Novel
potential collision threat alert	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
collision avoidance	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel

automated own aircraft flight path change	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
own aircraft altitude change	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
disturbed own aircraft dynamics	original	atypical	Low or no prior risk assessment is available	Novel
own aircraft speed change	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
accelerated own aircraft dynamics	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
accelerated own aircraft roll	original	atypical	Low or no prior risk assessment is available	Novel
accelerated own aircraft pitch	original	atypical	Low or no prior risk assessment is available	Novel
accelerated own aircraft yaw	original	atypical	Low or no prior risk assessment is available	Novel
own aircraft flight management system override	original	atypical	Low or no prior risk assessment is available	Novel
manual own aircraft flight path change	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
weather conditions change	original	atypical	Low or no prior risk assessment is available	Novel
weather data change	original	atypical	Low or no prior risk assessment is available	Novel
own aircraft dynamics change	original	atypical	Low or no prior risk assessment is available	Novel
unpredictable other aircrafts decision	original	atypical	Low or no prior risk assessment is available	Novel
automated collision avoidance maneuvers	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
avp detection communication delay hack	original	atypical	Low or no prior risk assessment is available	Novel
pilot cognitive limit	original	atypical	Low or no prior risk assessment is available	Novel
surrounding airspace restrictions	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
military aircraft intrusion	original	atypical	Low or no prior risk assessment is available	Novel
other aircrafts presence	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
bird	original	atypical	Low or no prior risk assessment is available	Novel
drone presence	original	atypical	Low or no prior risk assessment is available	Novel
wind shear effect on camera	original	atypical	Low or no prior risk assessment is available	Novel
air density change due to high speed effect on camera	original	atypical	Low or no prior risk assessment is available	Novel
aircraft attitude control	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
mechanical wear and tear	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
autopilot system failure	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel
information overload for pilot	original	atypical	Low or no prior risk assessment is available	Novel
highly stressfull event	original	typical	Not outside the expected relevance in such a scenario or problem space	Not novel

A situation is considered most likely novel if it is demonstrated to be original (related to some prior art) and not typical (atypical) as judged by an expert. The rational column explains the expert justification for choosing typicality value.

Identify Black Swan situations:

- **Impact on ML perception reliability:** Would such a situation impact the reliability of the solution ML model?

A Black Swan is a novel and impactful situation on ML performance.

Table I.12b captures the Black Swan assessment activities. Active_avp (active perception-based mid-air collision avoidance system) is deemed not novel, yet impactful on ML performance; these conditions reduce the likelihood of being a Black Swan, so classed as a white swan. It is important to understand that the originality, typicality, novelty, and impact evaluations are for most or least likely to be one type rather than the other. Visual_information, which is related to a situation in which long-distance spoofing operations are novel and impactful, makes it more likely to be a Black Swan.

Table I.13b Black Swan assessment

Predicted situation	Novelty Test	Possible problematic scenario	Impact on ML performance	Black Swan Situations
active avp	Not novel	Dataset bias delays Resolution Advisory in cluttered airspace; pilot trusts late advisory; near miss during crossing traffic.	impactful	White Swan
stabilised own aircraft dynamics	Not novel	Undetected icing induces oscillations; AVP assumes stability and issues incorrect guidance; conflict escalates.	impactful	White Swan
own aircraft pilot decision making process	Not novel	Automation bias emerges; pilot defers judgment to AVP, ignoring contradictory cues; unsafe convergence.	not impactful	White Swan
by-passing aircraft position	Not novel	Multipath/track swap mislocates intruder; AVP resolves wrong side; separation erodes unexpectedly.	impactful	White Swan
own aircraft pilot situation awareness	Not novel	Alert floods and fatigue reduce situational awareness; intruder unnoticed until late; rushed maneuver.	not impactful	White Swan
own aircraft flight path	Not novel	Datum error causes silent route drift; AVP predicts wrong CPA; vector commands create conflict.	impactful	White Swan
surrounding airspace safety	Not novel	Surveillance outage produces stale traffic picture; AVP trusts feed; unsafe vector into occupied sector.	impactful	White Swan
avoidance strategy recommendation	Novel	Policy trained in fair weather recommends climb into crossing traffic during convective buildup.	not impactful	White Swan
visual information	Novel	Sun glare saturates camera; intruder undetected; AVP assumes clear visuals; late alert.	impactful	Black Swan
by-passing aircraft speed	Not novel	Doppler notch underestimates intruder speed; CPA miscalculated; turn direction chosen incorrectly.	impactful	White Swan
collision avoidance system	Not novel	TCAS inhibited by configuration; no Resolution Advisory during converging paths; crew expects alert that never arrives.	not impactful	White Swan
by-passing aircraft proximity	Not novel	ADS-B latency underestimates proximity; AVP withholds alert; separation breach.	impactful	White Swan
conflicted atc	Novel	Two sectors issue conflicting headings; pilot compliance creates crossing geometry; AVP slow to reconcile.	not impactful	White Swan
own aircraft flight management system	Not novel	Mode reversion unnoticed; lateral deviation occurs near traffic; AVP assumes managed path.	not impactful	White Swan
collision threat	Not novel	Classifier saturation during clutter hides rising risk; alert arrives after maneuver window closes.	not impactful	White Swan
own aircraft camera	Not novel	Fog droplets on dome cause auto exposure oscillation; tracking intermittently drops intruder.	impactful	White Swan
other aircrafts	Not novel	Non-cooperative glider lacks transponder; radar fade; detected too late for gentle resolution.	impactful	White Swan
by-passing aircraft motion pattern	Novel	Intruder performs S-turns; filter lags intent change; AVP commands unsafe crossing maneuver.	impactful	Black Swan
weather data	Not novel	Uplink delay misstates winds; microburst unmodeled; AVP overestimates climb, recommending unsafe escape.	impactful	White Swan
by-passing aircraft detection tracking	Not novel	Track coalescence merges two aircraft; AVP treats one target; maneuver penetrates unseen traffic.	impactful	White Swan
own aircraft roll change	Novel	Servo lag overshoots bank; climb command increases closure; Resolution Advisory reversals follow.	not impactful	White Swan
own aircraft pitch change	Novel	Mis-trim induces pitch oscillation; commanded climb briefly descends into conflicted level.	not impactful	White Swan
own aircraft yaw change	Novel	Crosswind yaw correction insufficient; lateral drift crosses adjacent airway during merge.	not impactful	White Swan

own aircraft speed	Not novel	Pitot icing corrupts airspeed; AVP miscalculates closure; late avoidance.	impactful	White Swan
cloud turbulence	Novel	Underestimated turbulence triggers pilot-induced oscillation; guidance timing off; conflict worsens.	impactful	Black Swan
own aircraft flight maneuvers	Not novel	High bank turn creates sensor blind sector; intruder lost during merge.	impactful	White Swan
rerouting instructions	Not novel	Datalink drop hides reroute; aircraft continues old path into newly occupied corridor.	not impactful	White Swan
pilot stress	Not novel	Alert cascade induces tunnel vision; pilot misses critical RA; wrong maneuver selection.	not impactful	White Swan
by-passing aircraft visibility	Not novel	Haze and sun angle defeat visual acquisition; AVP assumes visual contact; delayed advisory.	impactful	White Swan
unpredictable by-passing aircraft actions	Novel	Intruder initiates emergency descent; AVP recommends climb into its path.	impactful	Black Swan
by-passing aircraft dynamics	Not novel	Sudden acceleration breaks filter; CPA estimate flips; hazardous turn executed.	impactful	White Swan
by-passing aircraft hack avp threshold	Novel	Adversary lowers thresholds; false alert flood; pilot disables AVP; real threat unnoticed.	impactful	Black Swan
pilot alerting and warning systems	Not novel	Aural alert muted inadvertently; visual cue missed; late compliance with RA.	not impactful	White Swan
collision threat alert	Not novel	Maintenance inhibit left active; no threat alert during converging approach.	not impactful	White Swan
weather conditions	Not novel	Rapid fog bank forms; camera blind; AVP maintains VMC assumptions; intruder missed.	impactful	White Swan
surrounding airspace obstructions	Novel	Unnotified crane protrudes; low altitude avoidance vector intersects obstacle.	impactful	Black Swan
own aircraft flight path change	Not novel	Autopilot mode trap prevents commanded path change; trajectory enters conflicted airway.	impactful	White Swan
non deterministic intelligent algorithms	Not novel	Stochastic policy flips under edge case; contradictory recommendations confuse pilot.	impactful	White Swan
by-passing aircraft direction	Not novel	GPS spoofing rotates intruder heading; AVP predicts wrong pass side; unsafe turn.	impactful	White Swan
by-passing aircraft altitude	Not novel	QNH mis-set causes vertical offset; illusory separation; climb instruction conflicts.	impactful	White Swan
threat predictive model	Not novel	Model trained on limited encounter geometries fails to generalize; novel approach vector unrecognized; delayed threat classification.	impactful	White Swan
risk of potential collision	Not novel	Over-optimistic risk estimate due to uncalibrated uncertainty; AVP underweights tail risk; insufficient alerting margin.	not impactful	White Swan
cockpit display systems	Not novel	Display symbology ambiguous during high-density traffic; pilot misinterprets threat vector; delayed response.	not impactful	White Swan
aircrafts existing warning systems	Not novel	Legacy TAWS nuisance alerts cause desensitization; genuine traffic alert ignored.	not impactful	White Swan
own aircraft altitude	Not novel	Altimeter mis-setting causes altitude bust; aircraft levels within protected altitude.	impactful	White Swan
own aircraft radar	Not novel	Ground clutter masks low RCS drone; radar misses; AVP overtrusts radar channel.	impactful	White Swan
own aircraft lidar	Not novel	Heavy precipitation creates false returns; tracker misassociates targets; wrong guidance.	impactful	White Swan
environmental obstructions	Not novel	Volcanic ash plume reduces visibility and sensor effectiveness; intruder lost.	impactful	White Swan
fog	Not novel	Sudden patch triggers auto gain bloom; delayed detection; late RA.	impactful	White Swan
raindrops	Novel	Lens smear refracts lights; false intruder distracts; real threat overlooked.	impactful	Black Swan
cloud cover	Novel	Uniform overcast reduces texture; optical flow fails; speed inferred incorrectly.	not impactful	White Swan
cloud type	Novel	Towering cumulus misclassified; strong updrafts unmodeled; climb instruction hazardous.	not impactful	White Swan
landscape background	Novel	City lights mimic traffic at distance; false positives distract from approaching intruder.	impactful	Black Swan
daytime	Not novel	Twilight transition miscalibrates exposure; brief detection gap during merge.	impactful	White Swan
sunlight	Not novel	Sun glint aligns with intruder; detector blinded; advisory late.	impactful	White Swan
sun position	Novel	Low sun behind target reduces contrast; tracker drops lock during closure.	impactful	Black Swan
by-passing aircraft flight path	Not novel	Curved approach misinterpreted as straight-in; extrapolation error creates conflict.	impactful	White Swan
collision time	Not novel	Latency underestimates time to collision; maneuver initiated too late.	not impactful	White Swan

own aircraft response time	Not novel	Control saturation slows response; available escape window closes.	not impactful	White Swan
own aircraft pilot feedback	Not novel	Misclick acknowledges wrong advisory; system commits to unsafe plan.	impactful	White Swan
avp real time training	Not novel	Online update drifts after mislabeled event; precision drops; missed detection.	impactful	White Swan
by-passing aircraft	Not novel	Low RCS livery reduces radar return; intruder only visually detected, too late.	impactful	White Swan
emergency directives	Not novel	Conflicting ATC and AVP directives; pilot hesitates; separation erodes.	not impactful	White Swan
unsafe proximity	Not novel	GPS multipath overestimates distance; alert suppressed; actual separation breached.	impactful	White Swan
airspace conflicts	Not novel	Sector compression funnels traffic; AVP unaware of metering; unsafe crossing vector.	impactful	White Swan
radar and visual range	Not novel	Antenna fault reduces range; intruder detected too late for gentle resolution.	impactful	White Swan
airspace visibility	Not novel	Wildfire smoke lowers visibility; AVP overestimates detectability; late alert.	impactful	White Swan
wind	Not novel	Unforecasted wind shear displaces aircraft laterally; AVP path prediction outdated; crossing conflict.	impactful	White Swan
avp confidence threshold	Not novel	Over-conservative threshold suppresses true detections; repeated late advisories.	impactful	White Swan
own aircraft proximity	Not novel	Baro VNAV error gives false comfort; drift into wake of heavy aircraft.	impactful	White Swan
autonomous collision avoidance maneuvers	Novel	Simultaneous auto maneuvers by both aircraft synchronize into renewed conflict.	impactful	Black Swan
potential collision threat alert	Not novel	Frequent false alerts cause fatigue; crew ignores subsequent genuine threat.	not impactful	White Swan
collision avoidance	Not novel	AVP turn conflicts with intruder's TCAS climb; incompatible advisories increase risk.	not impactful	White Swan
automated own aircraft flight path change	Not novel	Nav database error reroutes into restricted, congested airspace.	impactful	White Swan
own aircraft altitude change	Not novel	Climb into icing reduces performance; cannot reach clearance above traffic.	impactful	White Swan
disturbed own aircraft dynamics	Novel	Wake encounter causes roll upset; AVP miscalculates controllability; unsafe command.	impactful	Black Swan
own aircraft speed change	Not novel	Throttle transient lags; timing slip places aircraft abeam intruder at CPA.	impactful	White Swan
accelerated own aircraft dynamics	Not novel	High acceleration saturates tracker; motion blur; target lost at merge.	impactful	White Swan
accelerated own aircraft roll	Novel	High roll rate hits gimbal limits; camera blind sector during close pass.	impactful	Black Swan
accelerated own aircraft pitch	Novel	Rapid pitch breaks horizon detector; misclassification; late RA.	impactful	Black Swan
accelerated own aircraft yaw	Novel	Fast yaw breaks data association; tracker swaps targets; wrong guidance.	impactful	Black Swan
own aircraft flight management system override	Novel	Manual override disables protections; autopilot reversion surprises crew; trajectory deviates near traffic.	not impactful	White Swan
manual own aircraft flight path change	Not novel	Pilot turns manually; AVP not updated; CPA based on stale path; unsafe.	impactful	White Swan
weather conditions change	Novel	Rapid deterioration; model confidence unjustifiably high; diversion delayed.	impactful	Black Swan
weather data change	Novel	Corrupted uplinked winds; AVP miscalculates time to collision; late action.	impactful	Black Swan
own aircraft dynamics change	Novel	Fuel transfer shifts CG; control response changes; AVP uses stale dynamics.	impactful	Black Swan
unpredictable other aircrafts decision	Novel	Intruder declares emergency; deviates unexpectedly; AVP continues nominal prediction.	impactful	Black Swan
automated collision avoidance maneuvers	Not novel	Auto turn executed into conflicting altitude assignment; ATC coordination absent.	impactful	White Swan
avp detection communication delay hack	Novel	Injected network delay defers alerts; escape window lost.	impactful	Black Swan
pilot cognitive limit	Novel	High workload causes missed AVP cue; delayed compliance with RA.	not impactful	White Swan
surrounding airspace restrictions	Not novel	Temporary restricted area unrecognized; displaced traffic density spikes; unsafe vectoring.	not impactful	White Swan
military aircraft intrusion	Novel	Non-cooperative military flight enters airway; AVP lacks IFF; misclassification.	impactful	Black Swan

other aircrafts presence	Not novel	Swarm of small UAVs overwhelms tracker; false merges hide real conflicts.	impactful	White Swan
bird	Novel	Flock misclassified as clutter; runway adjacent ingestion risk during go-around.	impactful	Black Swan
drone presence	Novel	Adversarial drone with reflective decals defeats classifier; detection too late.	impactful	Black Swan
wind shear effect on camera	Novel	Shear vibrates mount; rolling shutter artifacts; tracker loses lock at merge.	impactful	Black Swan
air density change due to high speed effect on camera	Novel	High-speed compressibility glare distorts video; false negatives during pass.	impactful	Black Swan
aircraft attitude control	Not novel	Control law oscillations after update; unstable guidance near traffic.	impactful	White Swan
mechanical wear and tear	Not novel	Degraded actuator adds latency; commanded avoidance underperforms; near loss of separation.	impactful	White Swan
autopilot system failure	Not novel	Autopilot fails during RA; manual takeover delayed; insufficient separation.	impactful	White Swan
information overload for pilot	Novel	Cascading alerts overwhelm pilot; freeze delays response to genuine threat.	not impactful	White Swan
highly stressful event	Not novel	Bird strike coincides with RA; stress impairs judgment; wrong maneuver chosen.	not impactful	White Swan

I.5 Stage 2: Architect Intent and Autonomous Solution Needs

Definition³

This process would be used during Business or Mission Analysis in common systems engineering processes. It also defines the Operational Concept (OpsCon) for autonomous systems design. Although we didn't include the assessment of alternative solutions in this section, the output solution characterisation can guide the performance criteria to evaluate alternative solutions. This process also includes: **Stakeholder needs concept** definition on the back of solving problematic situations.

One pillar of AIC Systems Theory is the requisite of setting a purpose to realise streamlining the process in any system. The design team and their experiences, represented by the term “architect”, is a system which requires an ideal purpose to aim for. The prime purpose of our approach is engineering an Ideal Whole, where an ideal autonomous systems solution can be realised or facilitated. Setting such a purpose sets the architect engineering approach in the right path towards making more comprehensive and objective design decisions, as it unifies the design team's ontology and language under a servitude of common PrimeP. This alone can add weight towards a more effective Trustworthiness Case for

The architect's intent is also clearly defined at this stage. Architects define high-level objectives of what the autonomous systems should do and be validated to demonstrate. This is a collaborative stage between the architect and the wider stakeholders. A document is distributed among the stakeholders before a series of works HazTOPS dedicated to making decisions about the list of

³ For a summary see section M.3

assumptions. Here is another design review gate where ethical considerations base decisions on what to do with the problem scope. In this PhD, ethical considerations processes have not been considered.

The method includes the following activities to be done for every interaction and assumption:

Input: the primary input knowledge for the information to be included in the document is:

- Interaction definition.
- POE AIC model schema.
- Definition of contributing situations.

Process: the following thinking activities are to be done:

1. **Plausibility test:** is the assumption plausible or not?
2. **Why?:** define your justification on the reason behind the choice of plausibility value.
3. **Architect Intent (mission):** how does the architect intend to manage the situation? This may involve involving other stakeholders to manage this situation and not necessarily the system of interest
4. **Autonomous Solution needs:** What would be the objective of the autonomous systems in managing the situation?
5. **Other support systems need:** If the architect decides to involve other systems or stakeholders, what goals should they fulfil?
6. Define the Primary Purpose of the solution that the objectives of autonomous systems and other support systems are to fulfil. Define the overall purpose of the autonomous systems that fulfil part of PrimeP.

Output: The process's output is a definition of the overall solution strategy, including the Architect's Intent (mission), the autonomous systems and the support systems' needs and constraints. We call the template as “Architect High-Level Solution Prescription” or just Architect Prescription for short.

Autonomous systems goals represent the high-level validation objectives the system is expected to fulfil. Those objectives can guide the safety case construction regarding the highest-level goals and how the design and test achieved those goals, including supporting artefacts. Below is an application of the above step:

Table I.14 Architect High-Level Solution Prescription

Situation	By-passing aircraft dynamics: The characteristics of the By-passing aircraft's movement, including speed, altitude, and trajectory. Assumption: It is assumed that the AVP system will not continuously and accurately track the dynamics of By-passing aircraft.
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	<pre> graph LR active_avp((active_avp)) subgraph Green_Boxes [] direction TB G1[clear by-passing_ac_shape] G2[precise by-passing_velocity] G3[surrounding airspace_safety] end subgraph Red_Box [] direction TB R1[own_aircraft_pilot_situation_awareness] end subgraph Blue_Boxes [] direction TB B1[avoidance_strategy_recommendation] B2[own_aircraft_pilot_situation_awareness] B3[own_aircraft_pilot_decision_making_process] end active_avp -- "+recognise" --> G1 active_avp -- "+calculate" --> G2 active_avp -- "+monitor" --> G3 active_avp -- "+Augment" --> R1 active_avp -- "+issue" --> B1 active_avp -- "+alert" --> B2 active_avp -- "+guide" --> B3 Purpose[Purpose: [AVP][guide][Ownship_ac]] </pre> The diagram illustrates the functional components and interactions of the active_avp system. A central red oval labeled 'active_avp' is connected to several other components. It interacts with three green rectangular boxes: 'clear by-passing_ac_shape', 'precise by-passing_velocity', and 'surrounding airspace_safety', with labels '+recognise', '+calculate', and '+monitor' respectively. It also interacts with a red rectangular box 'own_aircraft_pilot_situation_awareness' via a '+Augment' label. Furthermore, it interacts with three blue rectangular boxes: 'avoidance_strategy_recommendation', 'own_aircraft_pilot_situation_awareness', and 'own_aircraft_pilot_decision_making_process', with labels '+issue', '+alert', and '+guide' respectively. A text label 'Purpose: [AVP][guide][Ownship_ac]' is also present.
Plausibility (Plausible/ Not plausible)	Plausible
Why?	Environmental conditions and sensor limitations can impact continuous and accurate tracking.
Architect Intent (mission)	The AVP training shall include variety of By-passing aircraft dynamics, at variety of speed.
Autonomous solution needs	The AVP shall continuously and accurately track the dynamics of By-passing aircraft in variety of environmental changes.
Autonomous solution constraints	Data Epistemic Coverage sufficiency.
Support Systems Needs	None. (In this case, we don't specify. However, there may be some relevant support systems that need to be considered.)
Support Systems Constraints	None. (In this case, we don't specify. However, some relevant support systems may need to be considered.)

I.6 Stage 3A: HazTOPS and typical AIC-driven Autonomous System Requirements Development⁴

I.6.1 Predictive Thinking Pipeline 1: Introducing Autonomous systems into Forward-Feed complexity

In this step, we derive the desired agent AIC interactions that fulfil the architect's intents and stakeholders' needs. The Predictive Thinking Pipeline strategy is defined using the following STCoT. We will use the following interaction example: Adversarial drone avoids vegetation around the fence, which may lead to a train derailment.

I.6.1.1 Step 1.1) Introduce the solution into the mix.

Model the AIC Schema and introduce the solution into the mix of the problem situation. In the train tracks problem, we used the interaction as an input: {roaming_adversarial_drone}_[-lead]_{train_derailment}, as well as the model. This time, for variety, we use the contributing situations definition and the assumptions output, including the architect's intent and the autonomous system's purpose.

Table I.15 Implementing architect Intent and Forward-Feed AIC Interaction Framework for addressing AVP reliability for by-passing aircraft

STCoT_3	Solution introduction into problem complexity
Input	By-passing aircraft dynamics: The characteristics of the By-passing aircraft's movement, including speed, altitude, and trajectory. Assumption: It is assumed that the AVP system will not continuously and accurately track the dynamics of By-passing aircraft. Architect intent:

⁴ For a summary see section M.4

	The AVP training shall include a variety of By-passing aircraft dynamics at a variety of speeds. Autonomous system purpose: The AVP shall continuously and accurately track the dynamics of By-passing aircraft in various environmental changes.
General Systems Model	General rule G: Equilibrium of Complexes Principle: IF a complex contains mutually antagonistic situations (A and B) that neutralise each other's capability in a given OrgXP THEN the OrgXP remain in equilibrium (systemic stability/predictability) unless perturbed through changes in internal or external uncertainty. [Bogdanov [140]] General rule H: Bidirectional Interaction Completeness Principle (AIC Interpretation): IF a complex contains mutually antagonistic AIC interactions acting on shared complexes, that neutralise each other's capability in a given organisational experience, THEN the complex exists in equilibrium (systemic stability/predictability). General rule I: Dialectical Equilibrium of Complexes Principle (AIC Interpretation) IF complex X exists in a dialectical AIC relationship with complex Y, comprising mutually antagonistic control, influence, and appreciation interactions acting on shared complexes such that the AIC actions of X neutralise the AIC actions of Y, and vice versa, at a given organisational level, THEN complex X exists in equilibrium (systemic stability/predictability). Complexes lacking such reciprocal counterbalancing AIC interactions are unstable (volatile) and susceptible to cascading Black Swan failures. Figure 4.7 model manifests G, H, and I rules by modelling the mutually antagonistic interaction groups comprising control, influence, and appreciation dynamics that counteract one another in a given organisational scenario, thereby neutralising net systemic effects and maintaining interaction stability (predictability) within the scenario's complexity. Complexes lacking reciprocal counterbalancing interactions are unstable (volatile) and susceptible to cascading Black Swan failures. The following AIC general solution space model is consistent with the General Systems Model G,H,I. LEADING TO: Asserted Organisational equilibrium. Which means the likelihood of organisational change approaches zero because opposing forces cancel each other's transformative effects (impact) on the system [Bogdanov]. Figure Error! No text of specified style in document..4 AIC solution space model
Predictive Thinking Approach	Predictive query 6.1: What is the primary problematic Scenario requiring influence (counterbalancing) to achieve systemic equilibrium? What Scenario requires control to enable the counterbalancing activity? What Scenario requires appreciation to ensure that the autonomous system maintains adequate control? Guiding prompt 6.2: Define the autonomous systems Forward-Feed AIC interactions with the rest of the problem.

	Map the autonomous systems to all the parts of the problematic Scenario and define the actions and impact types for each interaction. STEP 0 Gather inputs: (a) Architect solution prescriptions (b) Problem space AIC model (from STCoT_1 or STCoT_2) STEP 1 Map the PROBLEM SPACE: For each destabilising complex, define its AIC interaction: → AIC type (Control / Influence / Appreciation) → Source → Sink → Action and destabilising effect STEP 2 Define COUNTER-INFLUENCE interaction: "What does the solution INFLUENCE to neutralise the primary problem?" STEP 3 Define CONTROL interaction: "What must the solution CONTROL to enable that influence?" STEP 4 Define APPRECIATION interaction: "What must the solution APPRECIATE to enable that control?" STEP 5 Run the EQUILIBRIUM TEST (Rules G, H, I): For each problem interaction P: → Does a solution counter-interaction exist at the SAME AIC type? - SAME TYPE → BALANCED - DIFFERENT TYPE → PARTIAL (flag residual risk) - NONE → UNBALANCED (Black Swan vulnerability) Rule H: solution must have all three AIC types present Rule I: every problem AIC must have a counter-AIC Step completion criteria 6.3: The step is considered complete when all forward-feed AIC feedback loops between the autonomous system and the problematic Scenario have been modelled.
Output Prediction	Architect assertion 6.3: The architect asserts that the output predictions are logically consistent, structurally complete, and they are acceptably likely and impactful, with respect to the validity and applicability of general rules G,H,I:

The predicted solution mix includes the following interactions:

1. Central AVP System Influence on Pilot Situation Awareness

- **Influence Interaction (Red Arrows):** The active AVP system (central red oval labelled "active_avp") influences the pilot's situational awareness. This is visualised by a red arrow labelled "+Augment," leading to the red rectangle labelled "own_aircraft_pilot_situation_awareness." This indicates that the AVP indirectly enhances or influences the pilot's understanding of the aircraft's situation and surroundings, thus improving situational awareness.

2. Control and Guidance of Pilot Support Functions

- **Control Interactions (Light Blue Arrows):** The AVP exerts a control-like influence over three critical pilot support functions, represented by light blue rectangles:
 - **"avoidance_strategy_recommendation":** The AVP system issues recommendations to the pilot, shown by a light blue arrow labelled "+issue."
 - **"own_aircraft_pilot_situation_awareness":** In addition to augmenting the pilot's awareness indirectly, the AVP has a more direct, control-like effect here, alerting the pilot as necessary (light blue arrow labelled "+alert").
 - **"own_aircraft_pilot_decision_making_process":** The AVP provides direct guidance to assist in decision-making, represented by a light blue arrow labelled "+guide."

These interactions suggest that the AVP actively governs or steers these aspects of pilot support, thus providing critical situational input and control over pilot responses to surrounding conditions.

3. Influence of AVP Development Datasets on Pilot Support Functions

- **Influence of Development Datasets (Light Blue Arrows):** The AVP system relies on its development datasets (red rectangle labelled "avp_development_datasets") to ensure the accuracy and correctness of its pilot support functions. This relationship is depicted by light blue arrows labelled "+Assure correct", pointing from the development datasets to each of the three pilot support functions. This influence suggests that the datasets indirectly govern the precision and reliability of the AVP's recommendations, alerts, and guidance by shaping the AVP's capabilities during development.

4. Data-Driven Control Requirements for AVP System Operations

- **Control Requirements (Blue Arrows):** The AVP system requires access to specific data considerations for its effective operation, depicted by blue arrows:

- **"recognise"**: The AVP must recognise "clear_by-passing_ac_shape" (blue rectangle), likely representing the shape or profile of nearby aircraft.
- **"calculate"**: The AVP calculates "precise_by-passing_velocity" (rectangle), reflecting the speed required for safe bypass manoeuvres.
- **"monitor"**: The AVP monitors "surroundingairspace_safety" (blue rectangle), assessing environmental conditions and hazards.
- These data-driven tasks are included in the AVP's development datasets (blue arrows labelled "+include"), indicating that these data considerations directly control and configure the AVP's functionality, shaping how it operates in real-world scenarios.

I.6.1.2 Step 1.2) Characterise the AIC interactions table

Define the set of interactions between the sink and the source using AIC structured interaction format of: `{{source situation}}_ [+,- or no sign, AIC-action]_ {{sink situation}}`, Written in the following grammar: `{{adjective+noun}}_ [verbal phrase]_ {{adjective+noun}}` For example, `{{flying_police_robot}}_ [learns humans' visual profiles]_ {{distressed_people}}` Capture the output in the following table:

Table I.16 Mapping AIC interactions of AVP with ownship aircraft and the environment

Source: {active_avp}	
Output Behaviour	Input Behaviour that impacts the emergence of Output Behaviour
I1: {{ active_avp }}_ [+Augment]_ {{ own_aircraft_pilot_situation_awareness }}	A1: {{ active_avp }}_ [+monitor]_ {{ surroundingairspace_safety }} A2: {{ active_avp }}_ [+calculate]_ {{ precise_by-passing_velocity }} A3: {{ active_avp }}_ [+recognise]_ {{ clear_by-passing_ac_shape }}
	C1: {{ active_avp }}_ [+issue]_ {{ avoidance_strategy_recommendation }} C2: {{ active_avp }}_ [+alert]_ {{ own_aircraft_pilot_situation_awareness }} C3: {{ active_avp }}_ [+guide]_ {{ own_aircraft_pilot_decision_making_process }}
I2: {{ avp_development_datasets }}_ [+Augment]_ {{ own_aircraft_pilot_situation_awareness }}	A4: {{ avp_development_datasets }}_ [+include]_ {{ surroundingairspace_safety }} A5: {{ avp_development_datasets }}_ [+include]_ {{ precise_by-passing_velocity }}

	A6: [{ avp_development_datasets}_[+include]_ { clear_by-passing_ac_shape }]
	C4: [{ avp_development_datasets}_[+Assure correct]_ {avoidance_strategy_recommendation}] C5: [{ avp_development_datasets}_[+ Assure correct alert]_ { own_aircraft_pilot_situation_awareness}] C6: [{ avp_development_datasets}_[+ Assure correct]_ { own_aircraft_pilot_decision_making_process}]

Table I.15 presents a structured mapping of AIC interactions between the Active AVP and its surrounding operational environment. It focuses on its engagement with its ownship aircraft and the AVP development datasets. This table formalises the critical functional dependencies that shape AVP behaviour, ensuring enhanced situational awareness, decision-making, and strategic response formulation in dynamic airspace environments.

The first interaction category (I1) examines how the Active AVP directly influences the Ownship Pilot's situational awareness by augmenting it through key appreciation and control mechanisms. The appreciation behaviours (A1-A3) include monitoring the surrounding airspace safety, calculating precise by-passing velocity, and recognising clear by-passing aircraft shapes. These appreciation interactions ensure that the AVP comprehensively understands real-time flight conditions. The control behaviours (C1-C3) focus on issuing avoidance strategy recommendations, alerting the pilot to potential risks, and guiding the pilot's decision-making process, ensuring that the AVP perceives and actively intervenes in flight safety operations.

The second interaction category (I2) highlights the role of AVP development datasets in augmenting the ownship pilot's situational awareness through structured data-driven improvements. The appreciation interactions (A4-A6) emphasise the inclusion of critical environmental parameters such as surrounding airspace safety, precise velocity calculations, and clear aircraft shape recognition in training datasets. This ensures that the ML components within the AVP are trained on diverse and comprehensive operational scenarios. The control interactions (C4-C6) reinforce the necessity of assuring the correctness of avoidance strategy recommendations, alert protocols, and pilot decision-making processes, ensuring that the AVP's actions remain aligned with validated safety objectives.

I.6.2 Predictive Thinking Pipeline 2: Designing the affecting Backward-Feed complexity field

In this process, we will consider external contributing situations that affect and are affected by the Complicated Behaviour. We will also consider systems that Appreciate, Influence, or control the Eagle Drone AIC behaviour.

I.6.2.1 Step 2.1) Visualise the operational design domain environment

Visualise the operational design domain environment with the Complicated Behaviour being part of it. To start modelling the operational environment, we need a real-world picture of the theatre of operations to help with specifying the operational environment complexity field. Figure I.7 captures a typical scenario faced by the AVP system (mid-air collision avoidance perception system).

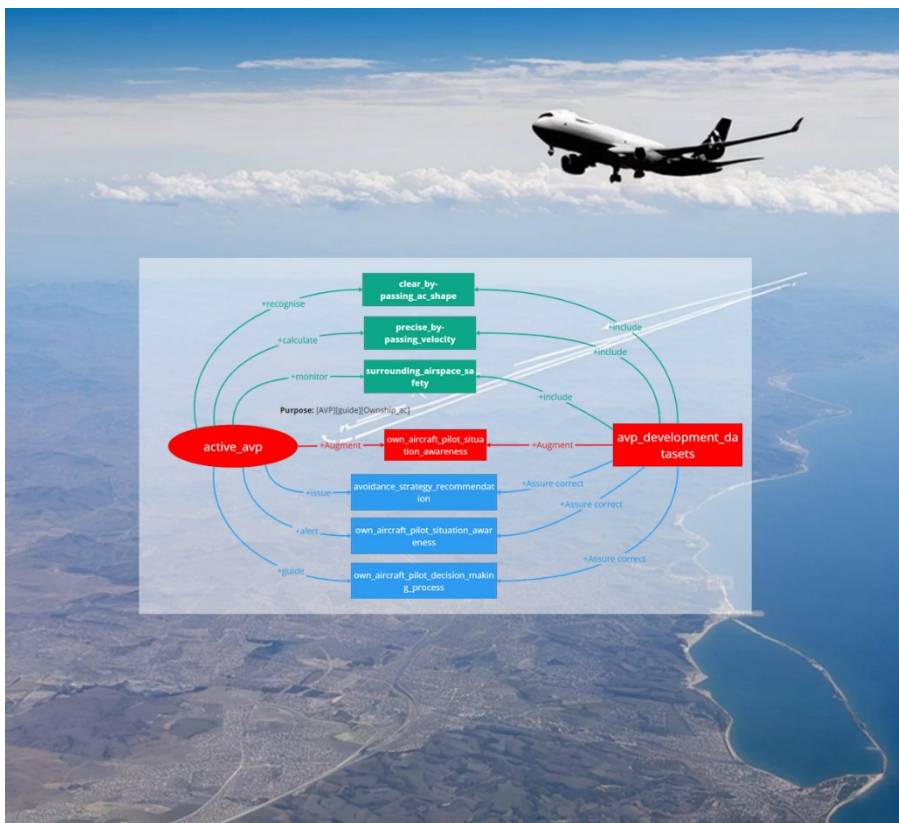


Figure I.5 A view synthesized using a message to image transformer of a typical scenario might be faced by the AVP.

The model captures the interconnectedness between these immediate and remote contributing situations to produce a comprehensive dataset that supports a resilient and adaptable AVP system. Immediate contributing situations such as weather, landscape, and types of nearby objects directly influence the types of images and scenarios included in the training dataset. Meanwhile, remote contributing situations like long-term climate change act indirectly by gradually altering the patterns

and frequency of these immediate conditions, thereby affecting the relevancy of the training data over time.

This model is designed to capture the complex relationships between contributing situations that affect and are affected by the training datasets used to develop an Aircraft Vision-Based Avoidance Detection (AVP) system. The goal is to create a robust dataset that enhances the AVP system's ability to accurately detect and avoid various objects and obstacles in flight, ensuring reliable and safe operation across diverse and changing conditions.

The model categorises these contributing situations into two main groups:

1. **Remote Affecting & Affected Problem Complex**

- **Long-Term Climate Change Shift:** This encompasses slow but significant changes in climate patterns over time, which may alter the frequency, intensity, and types of environmental conditions that the AVP system needs to handle. These shifts impact the long-term relevance and coverage of the training data by introducing new scenarios that may not be well-represented in existing datasets. For example, as climate change causes more frequent extreme weather events, such as storms and dense fog, or as it alters bird migration patterns, the AVP system must be prepared to handle these increasingly common but historically rare conditions. Long-term climate change thus indirectly influences the composition and emphasis of training data by creating new environmental challenges that impact flight safety.

2. **Immediate Affecting & Affected Problem Complex:**

This group includes contributing situations that have a more direct and immediate impact on the AVP training datasets, defining the variety and range of scenarios that the model is exposed to. The goal is to ensure that the AVP system is equipped to identify and avoid a wide range of objects, weather conditions, and other relevant environmental elements.

- **Weather Conditions Variety Definition:** Weather conditions such as rain, fog, snow, and wind can significantly impact visibility and the AVP system's performance. Training datasets must cover a wide variety of weather scenarios to enable the model to learn to identify objects under varying visibility and sensor performance conditions.
- **Landscape Types Variety Definition:** Different landscape types, such as mountainous, urban, coastal, and desert regions, affect the visual context in which objects appear. The AVP system should be able to operate effectively across various

terrains, so the training data should include diverse landscape backdrops to enhance situational adaptability.

- **Geographical Region-Specific Natural Phenomena:** Regional natural phenomena such as sandstorms in deserts, blizzards in polar regions, or volcanic ash clouds near active volcanoes are critical scenarios to include. These specific phenomena can introduce unique visibility challenges and influence avoidance behaviours.
- **Birds:** Birds present both common and unpredictable collision risks. Bird presence and behaviour vary across regions and seasons, so the AVP training datasets must account for bird density, species diversity, and typical flight paths to improve the system's ability to detect and avoid avian obstacles.
- **Illumination Variety Definition:** Different lighting conditions, including dawn, dusk, night, and bright midday sun, can affect how objects appear to vision sensors. Training data must represent various illumination conditions to help the AVP system recognise objects under varying lighting conditions and avoid false positives or missed detections due to shadowing, glare, or low light.
- **Aircraft Types Variety Definition:** The presence of different types of aircraft, such as helicopters, small private planes, commercial jets, and drones, is crucial for collision avoidance. Each type has distinct shapes, sizes, and movement patterns that the AVP system should learn to identify, classify, and respond to accurately.
- **Other Flying Objects:** Besides aircraft, other flying objects such as balloons, recreational drones, or even kites can enter airspace, especially in urban or event-filled areas. The AVP training datasets should include diverse flying objects to handle these less predictable encounters.
- **Other Aircrafts Jet Stream:** Jet streams created by other aircraft can cause turbulence that affects AVP reliability. Recognising and anticipating jet streams indirectly impacts the AVP system by informing the pilot or autonomous system of potential turbulence regions. Training data should thus include scenarios involving other aircraft's proximity and behaviour.
- **Sea Vessel Types Variety Definition:** For low-flying aircraft or drones, sea vessels such as cargo ships, yachts, and fishing boats may occasionally enter the AVP system's visual range, especially in coastal areas or during over-water flights. Identifying and avoiding these vessels is essential for safe, low-altitude operations.

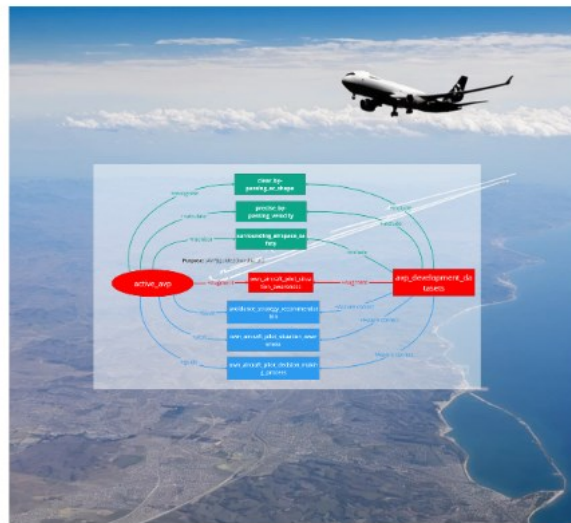
Remote affecting & affected problem complex

**long_term_climate
_change_shift**

Immediate affecting & affected problem complex

**Weather Conditions
Variety Definition**

**Landscape Types
Variety Definition**



**Geographical
region-specific
natural phenomena**

**Illumination Variety
Definition**

Birds

**Other Aircrafts Jet
Stream**

**Aircraft Types Variety
Definition**

other flying objects

**Sea Vessels Types Variety
Definition**

miro

Figure I.6 Perception-based mid-air collision avoidance system operational environment

I.6.2.2 Step 2.2) Backward-Feed Complicated Behaviour definition

In this step, we will extend the forward-feed model with the contributing situations within the operational domain complexity by identifying which of the complexes identified in the real-world problem complexity. To do so, we need to abstract the Forward-Feed model for ease of modelling. We reduce the model into a controlled language of interaction:

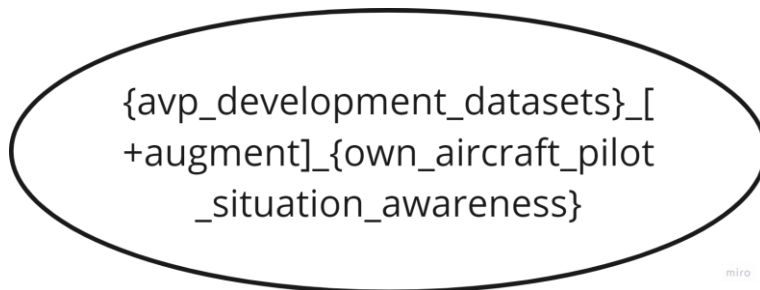


Figure I.7 I2 interaction

Then, we model the Backward-Feed AIC interactions by asking the following questions:

- What complex could appreciate the interaction? **We answer:**
 - The informed air traffic controller appreciates the AVP development dataset.
- What complex could influence the interaction? **We answer:**
 - Other flying objects could influence the AVP dataset development.
- What complex could control the interaction? **We answer:**
 - Weather conditions definitely control the AVP development dataset.
 - Architect a predictive mental model that can also control the quality of the AVP dataset coverage.

We will use icons to represent the complexes visually. Imagination is key enabler for predictive thinking. I.7 models captures the unknown unknowns at this stage which are also part of the possible long tail Black Swan scenarios.

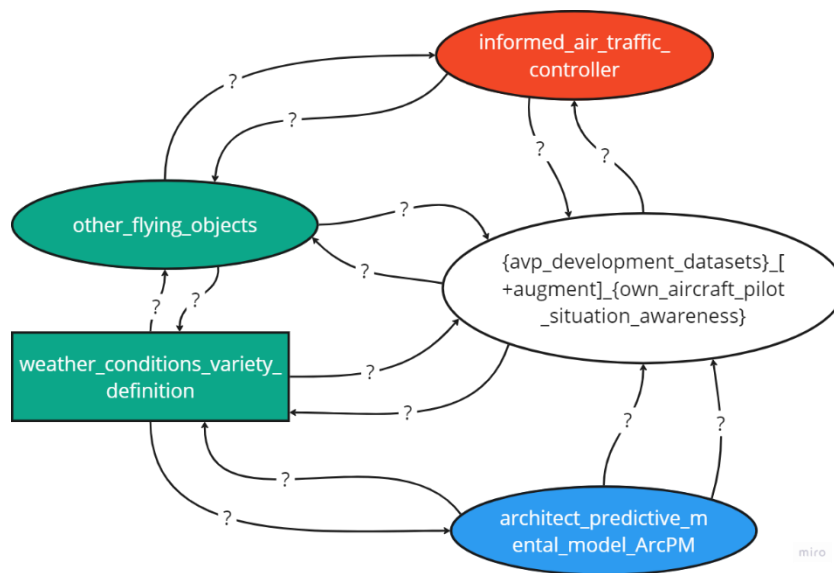


Figure I.8 Backward-feed model for I2 unknown unknown interactions. Black Swan scenarios.

I.6.2.3 Step 2.3) Predict the complicatedness of the solution space

To comprehensively define the complicatedness of the above complexity field abstraction, you may use the Actions Matrix method to define all the actions and their effect among the complexes. Note some actions do not make sense. Some of its actions are unsigned. The complexity involves the following complex of complexes:

Table I.17 Icon definition

Situation	Icon
other_flying_objects	
weather_conditions_variety_definition	
accurate_architect_predictive_mental_model_ArcPM	
informed_air_traffic_controller	
avp_development_datasets	

With the above table in mind, we can anticipate the following potential unknown interactions below. There should be 20 possible unknown interactions, for example (we will only list 10 of them):

1. $\{ \{ \text{avp_development_datasets} \}_{[?]} \{ \text{architect_predictive_mental_model_ArcPM} \} \}$

2. |{ avp_development_datasets}_{?}_{ other_flying_objects}|
3. |{ avp_development_datasets}_{?}_{ weather_conditions_variety_definition}|
4. |{avp_development_datasets}_{?}_{reliable_correct_target_object_recognition_confidence_scores}|
5. |{ avp_development_datasets}_{?}_{informed_air_traffic_controller}|
6. |{avp_development_datasets}_{?}_{visual_traffic_density_and_patterns}|
7. |{avp_development_datasets}_{?}_{critical_data_accuracy}|
8. |{ other_flying_objects}_{?}_{ avp_development_datasets }|
9. |{other_flying_objects}_{?}_{weather_conditions}|
10. |{other_flying_objects}_{?}_{other_flying_objects _visual_complexity_parameters}|

In order to ensure coverage, we will need to perform Action Matrix method. Table I.17 , below, presents an AIC (Appreciation, Influence, Control) coloured action matrix, mapping the interactions of the AVP dataset with key situational elements from Table I.16 (Icon Definition). The matrix highlights how the AVP dataset captures, predicts, informs, or complicates interactions across various domains, including other flying objects, weather conditions, air traffic controllers, predictive mental models, and AVP development datasets. And the following complicatedness (we introduced AIC colour scheme to help with constructing the AIC complexity field model):. Green represents “Appreciation”, Red represents “Influence” and Blue represents “Control”. In order for us to have a complete understanding we can also use the AIC modelling schema.

Table I.18 AIC coloured action matrix to model all possible interactions of the AVP dataset influence over

					
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		+capture	+capture	+assure	+assure
	-visually complicate		visually complicate	visually complicate	-operationally complicate
	-visually complicate	operationally complicate		visually complicate	- operationally complicate
	+predict	+predict	predict		+assure
	+inform	+monitor	monitor	+inform	

I.6.2.4 Step 2.4) Complete the extended view of the AIC mental model schema

The Actions Matrix above defines how the complexity of the scenario is resolved. Now, we will transfer all the knowledge to the final output of the step, which is a comprehensive AIC mental model of the problem. Note how we started and how we ended. Every step in this process carefully explains how we produced the scenario. Note that the interactions between the Forward-Feed bubble and the Backward-Feed operational environment concern the source. We are currently interested in the AVP development datasets in the design. Later, if the design team wishes to evaluate the complexity from the sink perspective, the complexity will change. The approach is flexible and allows for investigating and modelling the whole from any node, thus creating a variety of complexities. Figure I.8 captures all the resolved interactions in I.17,

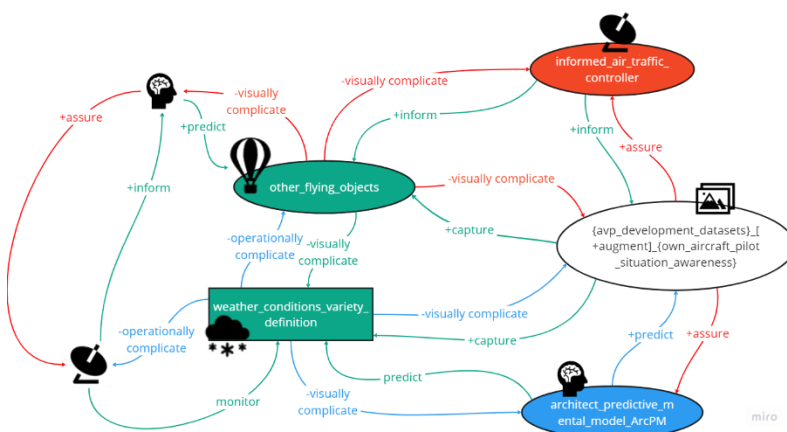


Figure I.9 Resolved complicatedness of AVP dataset complexity field

Now, we need to capture all influence interactions and resolve them into their respective appreciation and control interactions. Although there are six influence interactions, for the sake of simplicity, we will choose only three.

Table I.19 AIC interactions table for I.8 model

Source or Sink: {avp_development_datasets}	
Output Behaviour	Input Behaviour that impacts the emergence of Output Behaviour
I1: { avp_development_datasets}_[+assure]_ { architect_predictive_mental_model_ArcPM}	A1: { avp_development_datasets}_[+capture]_ { other_flying_objects}
	A2: { avp_development_datasets}_[+capture]_ { weather_conditions_variety_definition}
I2: { avp_development_datasets}_[+assure]_ {informed_air_traffic_controller}	C1: {avp_development_datasets}_[?]{?}
	A3: {avp_development_datasets}_[?]{?}
I3: { other_flying_objects}_[-visually complicate]_ { avp_development_datasets }	C2: {avp_development_datasets}_[?]{?}
	A4: {other_flying_objects}_[?]{weather_conditions}
	C3: {other_flying_objects}_[?]{?}

I.6.2.5 Step 2.5) Update the AIC interactions table

To determine the necessary control interaction (C1), in table I.18, in the context of achieving I1, we need to identify a scenario where the AVP development datasets exert a high likelihood of direct governance over a critical aspect of the AVP system's operation or the environment that the AVP system directly relies on. In AIC terminology, this would mean the datasets have a direct, nearly assured impact on a system component that is crucial to the AVP's functionality.

Given the context, the control interaction (C1) would likely be centred around the critical assurance of the AVP's detection and identification capabilities. This would ensure that the

AVP system consistently performs accurately and effectively in recognizing objects that could pose risks or inform pilot decision-making.

Suggested Control Interaction (C1)

C1:

`l{avp_development_datasets}_[+predict]_{reliable_correct_target_object_recognition_confidence_scores}`

Explanation:

1. `avp_development_datasets` control the `target_object_recognition_confidence_scores` directly to ensure that the AVP system operates within defined, precise boundaries that govern object recognition and identification. The correct prediction of this `confidence_score` would set high-assurance criteria that enable the AVP system to recognise and respond to objects in the environment, directly impacting the AVP's control over the safety and decision-making processes reliably and accurately.
2. The control over the correct and reliable prediction of `recognition_confidence_scores` ensures that AVP's performance in recognising and classifying objects remains consistent and accurate, allowing the system to meet safety and operational requirements without ambiguity or failure.

Resolving the complicatedness of I2:

To achieve the influence interaction:

`I2:l{avp_development_datasets}_[+assure]_{informed_air_traffic_controller}`

Where the AVP development datasets influence the air traffic controller's informed situation by ensuring data accuracy and reliability. To do so, specific control and appreciation interactions are required to establish the necessary foundation for this influence. These supportive interactions will ensure that the AVP development datasets effectively impact air traffic controllers through indirect yet robust information governance.

Here's how these interactions would support I2:

Required Control Interaction

Control Interaction (C2):

|{avp_development_datasets}_[+validate]_{critical_data_accuracy}|

Explanation:

1. **avp_development_datasets** must exert direct control over **critical data validation and accuracy mechanisms**. This interaction ensures that the data used by the AVP system is rigorously validated and maintained at a high accuracy level. By controlling these validation processes, the datasets can reliably provide high-quality information to downstream systems (such as air traffic control support).
2. This control interaction is crucial for achieving the assured and consistent data standards that allow the AVP system to influence air traffic controllers effectively, as air traffic decisions rely on precise and verified data.

Required Appreciation Interactions

To support the assurance influence on the **air traffic controller's** informed situation, the following appreciation interactions would provide adaptability and contextual awareness to the AVP datasets, allowing them to account for variations in the operating environment:

Appreciation Interaction (A3):

|{avp_development_datasets}_[+capture]_{visual_traffic_density_and_patterns}|

Explanation:

1. This appreciation interaction allows the AVP development datasets to capture information about **traffic density and patterns**, which vary with time, location, and operational context. By adapting to these patterns, the AVP system can better ensure that air traffic controllers are informed about its awareness of up-to-date traffic information (real-time). So, the AVP needs to communicate its awareness of traffic density so that the ATC can verify that AVP is aligned with reality.
2. The ability to capture traffic patterns aids the datasets in offering relevant insights to controllers, aligning with real-world conditions and supporting informed decision-making.

Resolving the complicatedness of I3:

In the context of **other_flying_objects** visually complicating the **avp_development_datasets** (interaction I3): |{other_flying_objects}_[-visually complicate]_{avp_development_datasets}|

here is an analysis of the necessary control and appreciation interactions for **other_flying_objects** to support this influence:

Control Interaction for other_flying_objects (C3)

To create visual complications for the AVP system, other_flying_objects directly govern specific visual parameters that directly impact how the AVP system perceives them. This control interaction would allow other flying objects to directly govern aspects of their appearance or behaviour in ways that affect their visual complexity and the resulting perception challenges for the AVP.

Control Interaction (C3):

|{other_flying_objects}_[+control]_{other_flying_objects_visual_complexity_parameters}|

Explanation:

1. other_flying_objects exert control over visual_complexity_parameters, such as shape, colour variations, speed, and manoeuvre patterns, to increase visual complexity for the AVP system. This control allows them to adjust their appearance and behaviour directly, making it more challenging for the AVP system to recognise, classify, and track them accurately.
2. By managing these parameters, other flying objects create an environment where the AVP system must deal with varying and potentially conflicting visual data, which complicates its recognition processes and overall situational awareness.

Appreciation Interaction for other_flying_objects (A4)

Other flying objects also need to appreciate external environmental conditions, particularly weather conditions, to enhance the effectiveness of their visual complication impact on the AVP system. Weather directly affects visibility, lighting, and sensor effectiveness, influencing how the AVP system perceives other objects.

Appreciation Interaction (A4): |{other_flying_objects}_[manoeuvre]_{weather_conditions}|

Explanation:

1. By appreciating weather conditions, other flying objects have to adjust their behaviours, which amplifies visual complications in specific weather scenarios. For instance, certain movements or colour patterns might make the AVP system's object identification more challenging during foggy or low-visibility conditions.

Given the above evaluation the following is the updated AIC interactions:

Table I.20 Resolved AIC Interactions Table

Source or Sink: {avp_development_datasets}

Output Behaviour	Input Behaviour that impacts the emergence of Output Behaviour
I1: { avp_development_datasets}_{+assure}_ { architect_predictive_mental_ model_ArcPM}	A1: { avp_development_datasets}_{+capture}_ { other_flying_objects} A2: { avp_development_datasets}_{+capture}_ { weather_conditions_variety_definition}
	C1: {avp_development_datasets}_{+predict}_{reliabl e_correct_target_object_recognition_confidence _scores}
I2: { avp_development_datasets}_{+assure}_ {informed_air_traffic_controller}	A3: {avp_development_datasets}_{+capture}_{visual _traffic_density_and_patterns}
	C2: {avp_development_datasets}_{+validate}_{critic al_data_accuracy}
I3: { other_flying_objects}_{-visually complicate}_ { avp_development_datasets }	A4: {other_flying_objects}_{manoeuvre}_{weather_c onditions}
	C3: {other_flying_objects}_{+control}_{other_flying_o bjects_visual_complexity_parameters}

There are a number of new systems have been discovered from the above table. We can capture those new systems by extending on icon definition I.16. Below is the new extended I.16 table:

Table I.21 Extended complexes icons definition

Situation	Icon
other_flying_objects	
weather_conditions_variety_definition	
accurate_architect_predictive_mental_model_ArcPM	
informed_air_traffic_controller	

	-visually complicate	operationally complicate		visually complicate	operationally complicate	operationally complicate	-reduce reliability	- operationally complicate	visually complicate
	+predict	+predict	predict		+assure	+predict	+optimise	-predict	+validate
	+inform	+monitor	monitor	+inform		+inform	+monitor	+control	+inform
	-visually complicate	+expose	visually complicate	-visually complicate	-visually complicate		-reduce reliability	visually complicate	visually complicate
	+predict	compute visual complexity	+compute visual complexity	+assure	+assure	+compute visual complexity		+compute visual complexity	+compute
	-visually complicate	visually complicate	-visually complicate	-visually complicate	- operationally complicate	visually complicate	reduce reliability		visually complicate
	+qualify	+expose	capture	+assure	+assure	+capture	+predict	+capture	

Given the above new contributing situations, we need to construct a comprehensive complexity field. To do so, we will need to construct the AIC actions matrix to identify all complexes with each other.

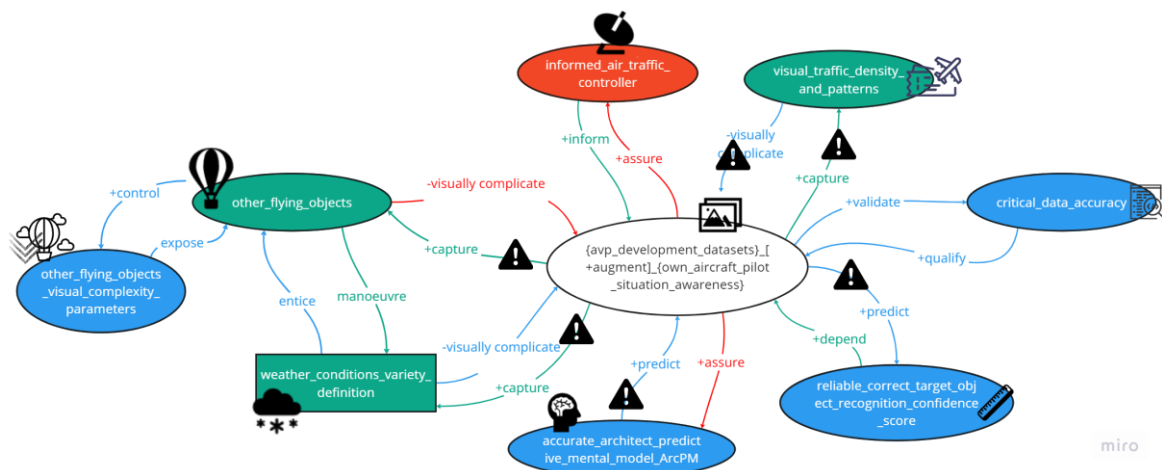


Figure I.12 Sources of Hazards AIC Complexity Field

Figure I.11 presents a HazTOPS analysis applied to the AIC Complexity Field for identifying safety hazards, and operational challenges. The diagram visualises the interactions between AVP development datasets, environmental contributing situations, air traffic control systems, and data reliability, highlighting areas where hazards may emerge.

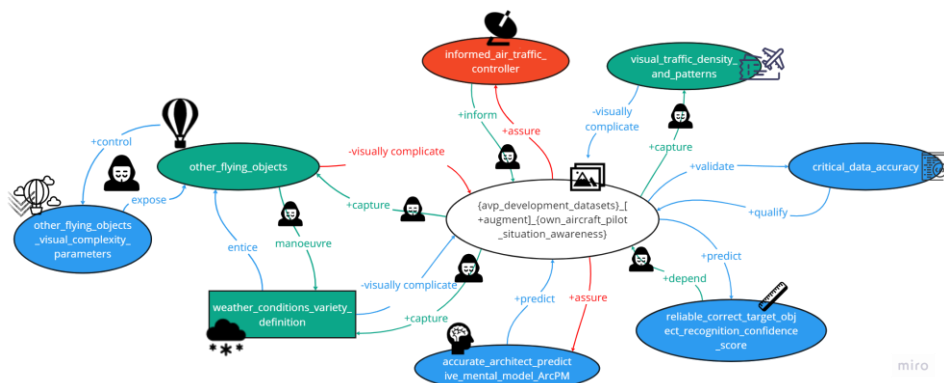


Figure I.13 Sources of Threats AIC Complexity Field

Figure I.12 presents a HazTOPS analysis applied to the AIC Complexity Field for identifying potential threats and vulnerabilities within the operational environment. The diagram visualises the interactions between AVP development datasets, environmental conditions, air traffic control systems, and critical data accuracy, highlighting areas where threats may emerge. Key concerns include visual complexity, misclassification risks, and data reliability issues, which could compromise the architect's situational awareness and AVP dataset development processes. This structured representation aids in assessing systemic threats and enhancing resilience against potential security and operational challenges in complex aviation systems.

I.6.3.2 Step 2) Apply predictive potential complications guide words.

Then, identify further potential complexity by utilising the following modified keywords: More, Part of, Less, Early, and Late. Follow STCoT_2 to derive the variety of potential deviations. Then include a risk and surprise analysis for each HazTOPS scenario.

Table I.23 Example “More” guide word complication

Title	Multiple flying objects (other than aircrafts)
AIC interaction	A1: [{ avp_development_datasets}_[+capture]_ { other_flying_objects}]
HazTOPS Aspect	Definition
Hazards, Threats or Opportunities Scenario: Guide word	More: multiple balloons involved.
Operating Scenario Context	Clear day, ownship aircraft (with AVP) at 10000ft high.
Hazard, Threat or Opportunity definition (consider used-systems)	More than 1 hot air balloon is visible. At 3000 ft.
Foreseeable Sequence of Events	- Planned a recreational or observational event that involves multiple balloons. - A group of hot air balloons or other balloons (e.g., weather or recreational balloons) are launched in the vicinity of the aircraft's flight path. - The AVP system identifies multiple objects (balloons). - The system classifies these objects based on altitude, size, and motion patterns. - Ownship pilot or automated system receives an alert of potential objects within or near the flight path.
Potential harm or benefit	- The shape of hot air balloons may startle the AVP, leading to a false negative. - The number and location of the hot air balloons may startle the AVP, leading to a false negative.

Table I.23 explores the application of the "More" guide word within a HazTOPS predictive complexity analysis, focusing on the potential impact of multiple flying objects, specifically, the presence of

more than one hot air balloon in the operational environment of an ownship aircraft equipped with AVP at 10,000 feet altitude.

The AIC interaction under analysis is:

A1: |{avp_development_datasets}_[+capture]_{other_flying_objects}

Where the AVP system captures and processes data related to airborne objects in its vicinity. The HazTOPS aspect introduces the "More" guide word, indicating an increase in the number of detected airborne objects, particularly multiple hot air balloons at lower altitudes (3,000 feet).

I.6.4 Predictive Thinking Pipeline 4: Elicitate typical AIC System-Level Requirements and training requirements

In this step, we will model every derived HazTOPS and define mitigating system-level requirements. We will perform the process in the following 2 examples:

I.6.4.1 Step 1) Model the Complex of Interest Operating Scenario Context

At a high level, we model the AVP as a source and the AIC relationships with the required used-systems to achieve the emergent capability. First, we identify the unresolved influence relationship problem between the agent and the target object of interest in Figure I.13:

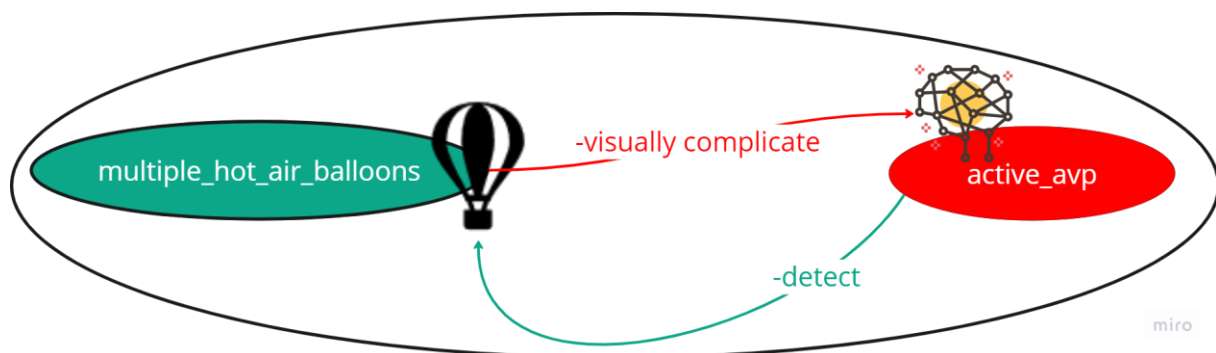


Figure I.14 Initial abstraction of the impact of hot-air balloons on AVP

The model `multiple_hot_air_balloons` negatively complicates the perception system of active Aircraft Collision Avoidance Perception systems (AVP). It describes how the presence of hot air balloons can interfere with or degrade the performance of an AVP system, which is designed to detect and help avoid aircraft. Below are a number of unknown unknowns that we need to predict in Table I.24.

Table I.24 Initial AIC resolution interactions table

Agent Output Behaviour	Agent Input Behaviour that delivers Output Behaviour	Supportive systems AIC Behaviour
I1: { multiple_hot_air_balloons_ }_[-visually complicate]_{active_avp}	A: to be defined	None
	C: To be defined	
No influence from AVP but a higher level of appreciative interaction A4: { active_avp } _[- detect]_{multiple_hot_air_b alloons}	A: { active_avp } _[detect]_{ multiple_hot_a ir_balloons}	
	C: To be defined	

I.6.4.2 Step 2) Model hazards mitigation typical-AIC complexity field

We must resolve the influence relationship problem on its appreciation and control relationships. To refine this model into subsystems, we can break down the visual components that define the appearance of hot air balloons as seen by the AVP system and the components of the ML-based perception model used to interpret these visuals. This will help understand how the AVP system could handle the visual complexity introduced by multiple balloons and improve detection accuracy.

Visual Components of Hot Air Balloons (as Perceived by AVP)

These are the key **visual attributes** that the AVP system might use to detect and differentiate hot air balloons from other objects:

- **Colour and Pattern:** Hot air balloons are often brightly coloured with unique patterns, making colour segmentation a useful feature. The perception system may analyse colour contrasts to distinguish balloons from the background and each other.
- **Shape and Contour:**
 - **Silhouette:** Hot air balloons have a distinctive, often rounded shape with a narrower base where the basket is located.

- **Edge:** The balloon and basket structure has unique edges that can be detected even from a distance. The AVP system might use contour-based filters to recognise these shapes.
- **Size and Scale:** The balloon's apparent size changes with distance, which the AVP can use to estimate proximity. This may involve analysing relative scale based on prior balloon size data.
- **Texture and Surface Features:** Balloons have smooth surfaces, sometimes with logos or text, which may require texture-based recognition methods to enhance detection, especially if the AVP system is differentiating between types of balloons.

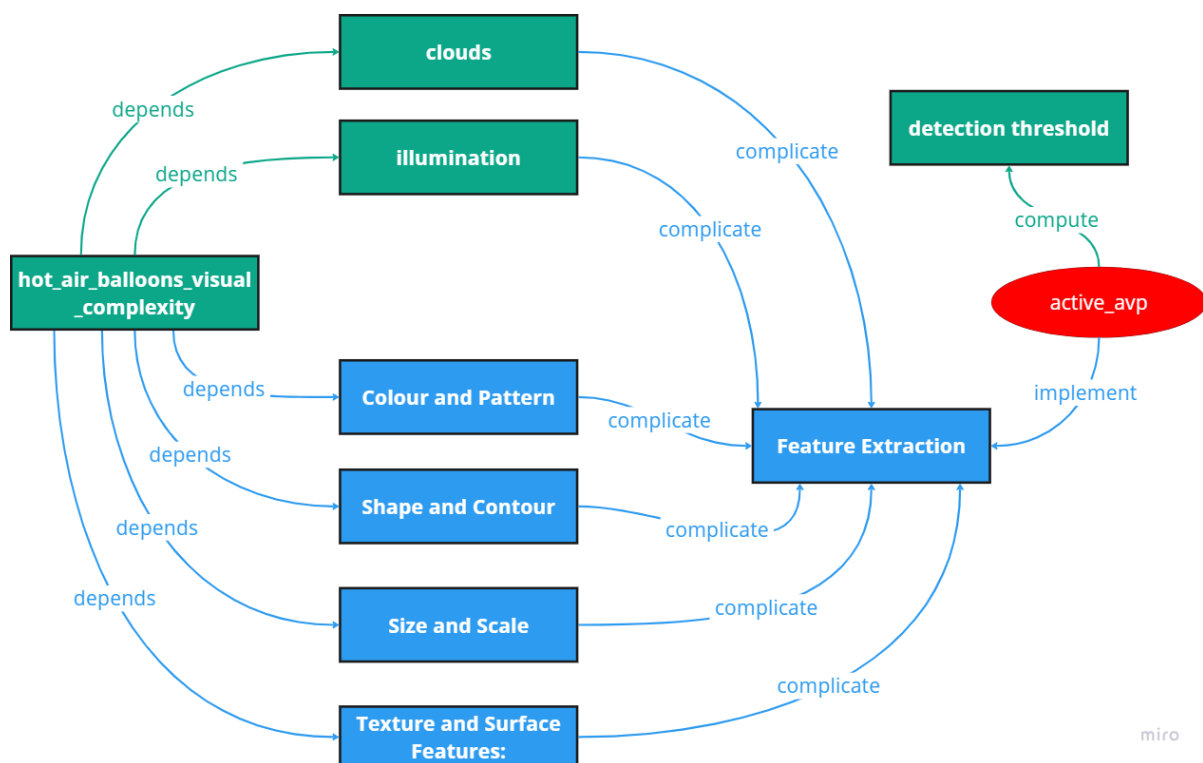


Figure I.15 Resolution of multiple hot air balloons hazard

The following table captures the description of the Figure I.14:

Table I.25 AIC-Structured Interactions Detailing the Visual Complexity of Hot Air Balloons and Its Impact on AVP Perception

Target I-interactions	Contributing AC-interactions	Supportive systems AIC Behaviour
Target Influence:	Appreciation:	{Air_Traffic

I1: { hot_air_balloons_visual_complexity }_[-visually complicate]_{active_avp}	A1: { hot_air_balloons_visual_complexity }_[-depends]_{clouds} A2: { hot_air_balloons_visual_complexity }_[-depends]_{illumination} Control: C1: { hot_air_balloons_visual_complexity }_[-depends]_{Colour and Pattern} C2: { hot_air_balloons_visual_complexity }_[-depends]_{Shape and Contour} C3: { hot_air_balloons_visual_complexity }_[-depends]_{Size and Scale} C4: { hot_air_balloons_visual_complexity }_[-depends]_{Texture and Surface Features}	_Control)_ [+ignore] _ { hot_air_balloons_visual_complexity }
No influence from AVP but a higher level of appreciative interaction A4: { active_avp }_[-detect]_{multiple_hot_air_balloons}	A3: { active_avp }_[-compute]_{detection confidence_score} C5: { active_avp }_[-implement]_{Feature Extraction}	

Table I.25 presents an AIC-structured analysis of how hot air balloons' visual complexity influences the Active AVP system's perception and detection capabilities. The table categorises interactions

based on AIC functions, mapping hot air balloon attributes that complicate AVP visual processing and the supportive systems required for perception enhancement.

The influence interaction (I1) highlights the negative visual complications introduced by hot air balloons, affecting the AVP's ability to classify and track airborne objects. The appreciation interactions (A1-A2) establish dependencies on environmental contributing situations such as cloud cover and illumination, further impacting object visibility and detection reliability. The control interactions (C1-C4) focus on the specific visual attributes of hot air balloons, including colour, pattern, shape, contour, size, scale, texture, and surface features, all contributing to perceptual uncertainty for AVP systems.

The table also recognises that AVP does not inherently influence hot air balloons, meaning it must adapt its perception strategies instead of modifying external conditions. The AVP compensates by computing detection confidence scores (A3) and implementing feature extraction techniques (C5) to enhance its ability to distinguish hot air balloons from other airborne objects.

I.6.4.3 Step 3) typical-AIC-based Mitigating System or Safety Requirements

Derivation (Safe Operating Concept)

So far, we have focused on defining Agent relationships in the given hazardous scenario. In this step, we collate all discovered actions in a table and convert those actions into AVP systems requirements, considering the intangible or tangible methods that shall deliver the agent's will. The relevant system requirements are defined from Control and Appreciation actions, with the influence action included at the end of each requirement after the phrase "in order to". Use the following format:

[IF: A or C action], THEN [mitigation requirement] LEADING TO [I action]

For example:

AVP Safety Requirements:

Here are the AVP System safety requirements based on the given actions (A and C actions) for mitigating the complexity introduced by multiple hot air balloons and enhancing AVP's ability to detect and classify them reliably:

Table I.26A Safety requirements derivations to mitigate hot-air balloons impact on AVP system performance.

AC (appreciative, control) interaction	Mitigating Safety or Systems Requirements (Safe Operating Concept)
A1: { hot_air_balloons_visual_complexity}_ [depends]_{clouds}	Safety Requirement 1: AVP System shall avoid Hot Air Balloons among various cloud conditions • IF: A1: Hot air balloon visual complexity depends on cloud cover, • THEN: The AVP system shall guide ownship aircraft to avoid hot air balloons under various cloud cover scenarios LEADING TO: recognise and direct ownship aircraft to avoid hot-air balloon visual complexity.
A2: { hot_air_balloons_visual_complexity}_ [depends]_{illumination}	Safety Requirement 2: AVP system to avoid Hot Air balloons in various light conditions • IF: A2: Hot air balloon visual complexity depends on illumination conditions, • THEN: The AVP system shall guide ownship aircraft to avoid hot air balloons under various illumination conditions, • LEADING TO: recognise and direct ownship aircraft to avoid hot-air balloon visual complexity.
C1: { hot_air_balloons_visual_complexity}_ [depends]_{Colour and Pattern}	Safety Requirement 3: AVP System shall avoid Air balloons with rare patterns • IF: C1: Hot air balloon visual complexity depends on colour and pattern variations, • THEN: The AVP system shall guide ownship aircraft to avoid a broad range of balloon colours and patterns, including common and rare patterns and colour schemes,

	• LEADING TO: recognise and direct ownship aircraft to avoid hot-air balloon visual complexity.
C2: { hot_air_balloons_visual_complexity}_ [depends]_{ Shape and Contour}	Safety Requirement 4: AVP System shall avoid Hot Air balloons with rare shapes and contours • IF: C2: Hot air balloon visual complexity depends on shape and contour, • THEN: The AVP system shall guide ownship aircraft to avoid varied balloon shapes and contours, including partially obscured or overlapping balloons, as well as balloons viewed from multiple angles, • LEADING TO: recognise and direct ownship aircraft to avoid hot-air balloon visual complexity.
C3: { hot_air_balloons_visual_complexity}_ [depends]_{ Size and Scale}	Safety Requirement 5: AVP System shall avoid Hot Air balloons at various distances • IF: C3: Hot air balloon visual complexity depends on size and scale, • THEN: The AVP system shall guide ownship aircraft to avoid balloons at various distances and sizes, emulating both close-range and far-range balloon appearances, • LEADING TO: recognise and direct ownship aircraft to avoid hot-air balloon visual complexity.
C4: { hot_air_balloons_visual_complexity}_ [depends]_{ Texture and Surface Features}	Safety Requirement 6: AVP system shall avoid Hot Air balloons with varying textures • IF: C4: Hot air balloon visual complexity depends on texture and surface features,

	• THEN: The AVP system shall guide ownship aircraft to avoid balloon images with varying textures, surface features, and common markings. • LEADING TO: allow the AVP system to differentiate balloons based on subtle texture and surface features, such as logos or text.
A3: { active_avp }_ [compute]_ { detection confidence_score }	System Requirement 1: The AVP system shall adjust confidence thresholds dynamically • IF: A3: Active AVP computes detection confidence score, • THEN: The AVP system shall implement a detection confidence scoring mechanism, adjusting thresholds based on scenario-specific complexities such as high object density and overlapping visuals. • LEADING TO: enable the AVP to flag uncertain detections for further processing or alerting when confidence is low, enhancing overall safety (I1).
C5: { active_avp }_ [implement]_ { Feature Extraction }	System Requirement 2: The AVP system shall implement feature extraction • IF: C5: Active AVP implements feature extraction, • THEN: The AVP system shall use feature extraction techniques, including attention mechanisms to prioritise relevant balloon features, • LEADING TO: detect hot air balloons.

The safety requirements derived from the Appreciative and Control (AC) interactions in Table I.26A were refined by systematically considering how the AVP system should handle hot air balloons under varying visual complexities.

Appreciative actions and safety requirements:

The appreciative actions (A1, A2, A3) emphasise how external conditions such as cloud cover, illumination, and system confidence computation affect the detectability of hot-air balloons' visual complexity. These elements alter the visual scene and influence AVP's performance.

- **A1: Dependence on Cloud Cover/ Safety Requirement 1:** Hot air balloons can visually merge with clouds, leading to false positives or missed detections. Since cloud cover alters contrast, the AVP system must ensure it can distinguish hot-air balloons in varying cloud conditions. The system must direct ownship aircraft to avoid scenarios where clouds hinder balloon visibility.
- **A2: Dependence on Illumination/ Safety Requirement 2:** Changes in illumination affect how the AVP system perceives objects. This requirement guarantees that the AVP system can adapt to different lighting conditions, including glare, low light, and shadowed environments, to prevent incorrect classifications of hot-air balloons.
- **A3: Computation of Detection Confidence/System Requirement 1:** The AVP system must adjust detection confidence scores dynamically to manage uncertain visual conditions. This requirement introduces an adaptive scoring mechanism within some operational conditions (for example, when visibility reduces or clouds become overcast). The system should adjust its confidence thresholds to ensure the most likely detection. The AVP can flag uncertain detections based on overlapping or ambiguous features. This may help avert missing or incorrect aircraft manoeuvring due to unreliable detections.

Control actions and system adaptations:

The control actions (C1–C5) are related to the AVP system's capability to actively distinguish and mitigate balloon visual complexity through shape, colour, size, and other object features.

- **C1: Dependence on Colour and Pattern / Safety Requirement 3:** Hot-air balloons feature varied and unpredictable colours and patterns, making recognition challenging. The requirement ensures the AVP system can handle balloons with common and rare designs, preventing confusion with background scenery, aircraft, or other environmental objects.
- **C2: Dependence on Shape and Contour / Safety Requirement 4:** Different balloon shapes (e.g., spherical, elongated, or custom-themed balloons) impact how the AVP system perceives objects. Some shapes may appear partially obscured by other objects or blend into backgrounds. The requirement ensures the AVP system can recognise diverse balloon contours from multiple viewing angles to prevent misclassification.

- **C3: Dependence on Size and Scale / Safety Requirement 5:** A hot-air balloon's distance and relative scale can create ambiguity in detection. Balloons at varying altitudes may appear as small, distant objects or large, nearby obstacles. The AVP system must be capable of differentiating balloons at different scales, ensuring avoidance manoeuvres are appropriately executed.
- **C4: Dependence on Texture and Surface Features / Safety Requirement 6:** Hot-air balloons may have logos, text, or non-uniform textures that affect object detection. This requirement ensures that the AVP system can account for subtle texture differences to improve detection accuracy and avoid misclassification or missed detections.
- **C5: Implementation of Feature Extraction / System Requirement 2:** The AVP system must implement advanced feature extraction mechanisms to enhance recognition, using attention-based models to prioritise balloon-related features. This requirement ensures that the AVP system can filter out irrelevant background noise and focus on extracting balloon-specific characteristics.

I.6.4.4 Step 4) Extended Concrete Safety, Systems requirements and ML Safety-Training Concept

We will not apply the 4-whats-and-how method in this case study since we have already derived the AVP safety requirement. We will directly derive the training concept. Instead, we will move to the derivation of the training concept.

I.6.4.5 ML Safety-Training Requirement derivation (Training Concept):

In this stage, we specify a general training requirement for the ML model, which we refer to as the Training Concept. To define the training requirement, we used the following structure:

ML Safety-Training Requirement 1: [system of interest] shall be trained to [training experience].

Below is the table we use to perform the derivation process:

Table I.26B Deriving Training Concept from Safe Operating Concept for AVP ML component

Mitigating Safety or Systems Requirements (Safe Operating Concept)	ML Safety-Training Requirements (Training Concept)
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Safety Requirement 1: AVP System shall avoid Hot Air Balloons among various cloud conditions • IF: A1: Hot air balloon visual complexity depends on cloud cover, • THEN: The AVP system shall guide ownship aircraft to avoid hot air balloons under various cloud cover scenarios. • LEADING TO: recognise and direct ownship aircraft to avoid hot-air balloon visual complexity.	ML Safety-Training Requirement 1: AVP ML component shall be trained to recognise Hot Air Balloons under various cloud cover scenarios (e.g., clear, partial cloud cover, heavy cloud cover),
Safety Requirement 2: AVP system to avoid Hot Air balloons in various light conditions • IF: A2: Hot air balloon visual complexity depends on illumination conditions, • THEN: The AVP system shall guide ownship aircraft to avoid hot air balloons under various illumination conditions, • LEADING TO: recognise and direct ownship aircraft to avoid hot-air balloon visual complexity.	ML Safety-Training Requirement 2: AVP ML component shall be trained to recognise a Hot Air Balloon under various illumination conditions (e.g., dawn, dusk, direct sunlight, low-light),
Safety Requirement 3: AVP System shall avoid Air balloons with rare patterns • IF: C1: Hot air balloon visual complexity depends on colour and pattern variations, • THEN: The AVP dataset shall include a broad range of balloon colours and	ML Safety-Training Requirement 3: AVP ML component shall be trained to recognise a broad range of balloon colours and patterns, including common and rare patterns and colour schemes.

patterns, including common and rare patterns and colour schemes, LEADING TO: recognise and direct ownship aircraft to avoid hot-air balloon visual complexity.	
Safety Requirement 4: AVP System shall avoid Hot Air balloons with rare shapes and contours • IF: C2: Hot air balloon visual complexity depends on shape and contour, • THEN: The AVP dataset shall include varied balloon shapes and contours, including partially obscured or overlapping balloons, as well as balloons viewed from multiple angles, LEADING TO: recognise and direct ownship aircraft to avoid hot-air balloon visual complexity.	ML Safety-Training Requirement 4: The AVP ML component shall be trained to recognise a Hot Air Balloon in the context of varied balloon shapes and contours, including partially obscured or overlapping balloons, as well as balloons viewed from multiple angles.
Safety Requirement 5: AVP System shall avoid Hot Air balloons at various distances • IF: C3: Hot air balloon visual complexity depends on size and scale, • THEN: The AVP dataset shall include balloons at various distances and sizes, emulating both close-range and far-range balloon appearances, LEADING TO: recognise and direct ownship aircraft to avoid hot-air balloon visual complexity.	ML Safety-Training Requirement 5: The AVP ML component shall be trained to recognise Hot Air balloons at various distances and sizes, emulating both close-range and far-range balloon appearances.

Safety Requirement 6: AVP system shall avoid Hot Air balloons with varying textures • IF: C4: Hot air balloon visual complexity depends on texture and surface features, • THEN: The AVP dataset shall contain balloon images with varying textures, surface features, and common markings. LEADING TO: allow the AVP system to differentiate balloons based on subtle texture and surface features, such as logos or text.	ML Safety-Training Requirement 6: The AVP ML component shall be trained to recognise a Hot Air Balloon in the context of varying textures, surface features, and common markings.
System Requirement 1: The AVP system shall adjust confidence thresholds dynamically • IF: A3: Active AVP computes detection confidence score, • THEN: The AVP system shall implement a detection confidence scoring mechanism, adjusting thresholds based on scenario-specific complexities such as high object density and overlapping visuals. • LEADING TO: enable the AVP to flag uncertain detections for further processing or alerting when confidence is low, enhancing overall safety (I1).	ML Training Requirement 1 (non-safety): The AVP ML component shall be trained to adjust confidence thresholds dynamically and perform under various confidence scoring mechanisms. The thresholds shall be adjusted based on varying environmental conditions, high TOI density and pictorial distances.
System Requirement 2: The AVP system shall implement feature extraction	Not applicable

• IF: C5: Active AVP implements feature extraction, • THEN: The AVP system shall use feature extraction techniques, including attention mechanisms to prioritise relevant balloon features, • LEADING TO: detect hot air balloons.	
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I.6.4.6 ML dataset requirements derivation

A dataset training requirement can be derived from the training concept. We use the following structure to define requirements over actual datasets:

Dataset requirement structure:

The AS ML component [Training/Testing/Black Swan Validation] Dataset shall provide the trainee model with a valuable minimum variety of ...

In this case:

ML Development Dataset Requirement 1: The AVP's ML component Training Dataset shall provide the trainee model with a valuable minimum variety of various types of hot air balloons under various cloud cover scenarios (e.g., clear, partial cloud cover, heavy cloud cover).

I.7 Stage 3B: Comprehensive Operational Environment Definition⁵

We reduced the comprehensive ODD classification into the following categories for this application and considered Grade A uncertainty. For a detailed description of how to define ODD, refer to sections E. 9 and E. 10.

⁵ For a summary see section M.5

Table I.27 Operational Design Definition for AVP

System of interest		Computerised perception-based mid-air collision avoidance system (AVP)
Solution Operational Space		Open Airspace [Palo Alto City, USA]
ODD Uncertainty Grade		Grade A
Development purpose		Training, validation and testing
Environment Characteristics		Ideal natural environment system for perception
Natural Lighting Conditions		Sunny
Weather Conditions	Precipitation mm/h	0
	Wind km/h	<5
	Humidity %	50
	Visibility km	>8
	Cloud Cover	Clear or Blue Sky (0/8 oktas)
	Snow mm/12 hrs	0
	Pollen	No pollen
	Sand	Neither sand nor dust storms
	Temperature	15-25 degrees Celsius
	Sunshine Duration	12 hours
Time of the Year: Seasons-specific environmental characteristics		1 type of season
Landscapes type variety definition		1 type of Landscapes
Geographical region-specific natural phenomena		0 or 1 type of Infrastructure
Time of the Day		Midday (Noon)
Perceived Horizon Attitude		1 type
Sun sphere positioning		1

Moon sphere positioning	0
Specialised zones features	1 feature

Table I.30 defines the operational design domain (ODD) and environmental characteristics for deploying the AVP system in Palo Alto city open airspace under Grade A uncertainty. It specifies ideal natural conditions for perception, including sunny weather, minimal wind (<5 km/h), high visibility (>8 km), and clear skies. Additional parameters include a temperature range of 15–25°C, 12 hours of sunshine, and no pollen, sand, or dust interference. The operation is limited to a single type of season, midday timing, and one specific landscape type with minimal infrastructure variation. The table ensures a clearly defined, controlled environment for reliable system performance and accurate perception during deployment. We may choose different sets for ODD when developing testing datasets. However, we will keep the same ODD for generating the datasets in this application.

I.8 Stage 4: atypical AIC-Driven Black Swan scenarios Prediction⁶

In this stage, you may consider atypical AIC timing in section E4. Some of the output of this stage is to generate Black Swan validation datasets for validating ML components to handle Black Swan scenarios. However, nothing may stop the architect from dedicating some of the Black Swan scenarios to be part of the training and testing process of the ML component.

At first, we choose the complexity field we intend to perform the AIC perspective shift. For this we will select the last complexity field in Figure I.10:

⁶ For a summary see section M.6

The expected interaction between the hot-air balloon and the AVP dataset development is that the dataset should respond to the balloon's behaviour. The dataset should never influence the balloon's behaviour, regardless of how the complexity of the future field may evolve. The datasets are claimed to be supportive of balloons' purpose (based on the assumption that recognising hot-air balloons and including them in the datasets would contribute to their safety). At the same time, the unexpected variability of hot-air balloons would obstruct the reliability of AVP perception. This is what the architect is prophesising: what would happen in such a scenario. Any deviation from this established perspective norm would be considered unexpected or surprising, characteristics typical of Black Swan situations.

I.8.2 Step 2) Define the ArcMatrix

Define the interaction's current AIC contributing situations using the ArcMatrix (section F.2.1). We will give the hot air balloon a descriptive situation of “visually_variable_hot_air_balloon”.

Table I.28 Deep AIC factorisation for I.16 interaction

	Visually_variable_hot_air_objects	avp_development_datasets
Visually_variable_hot_air_objects		Supra Source: other_flying_objects. PrimeP: perform uninterrupted missions. Goal: avoid any obstacle to mission execution. Goal type: Influence. Action: visually complicate AVP datasets. Action type: Influence. Effect: Obstructive.
avp_development_datasets	Supra Source: Ownship aircraft.	

	PrimeP: Maintain safe operational flight and avoid mid-air collisions. Goal: complete coverage of potential operational scenarios. Goal type: Appreciation. Action: capture scenarios. Action type: appreciation. Effect: Obstructive.	
--	--	--

I.8.3 Step 3) Perform the Perspective Shift

Perform the perspective shift using AIC perspective shift STCoT. In this step, we choose an appropriate perspective shift that we believe will be a potential black-swan event not foreseen during modelling the interaction between `adv_drone_shapes` and `flying_eagle_drone`. We will select the following shift:

1. AIC type shift.

For this, we will apply the following Thought Step from F.7 approach (Appendix F):

General Systems rule: given a source or sink, it is possible that over time and with the change of complexity, the AIC goals are altered, leading to a new situation of complicatedness.

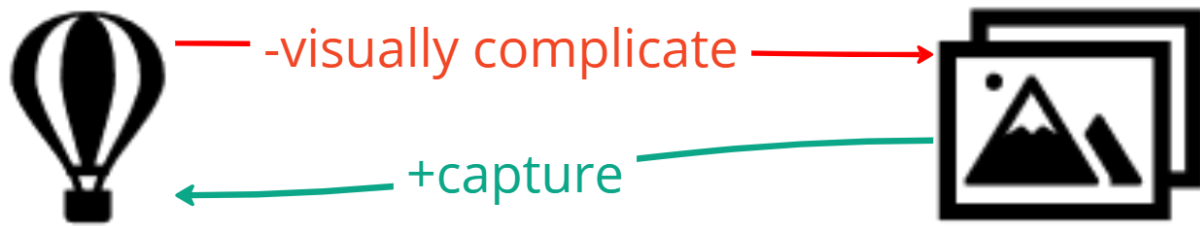
Predictive query: What would happen if the interaction flow and effect type remained the same, but the goal's AIC type and action type were altered in the future?

Guiding Prompt: Review the AIC dynamics of the observed complexes and alter the nature of goal and action. Then, define an appropriate action to bear alternative AIC types and describe a scenario in the shifted context.

Completion criteria: The step is considered complete when a scenario demonstrating an alternative AIC goal is detailed.

In this thought process, we will shift the AIC-type perspective as follows As per the following schema:

Pre-shift: Expected (normal event)



Post-shift: Unexpected (black swan event)

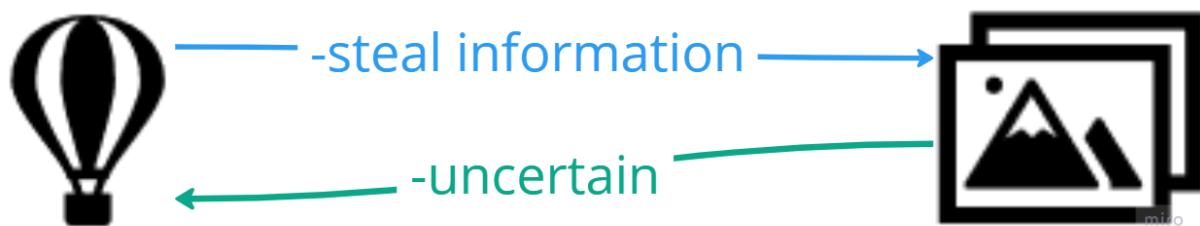


Figure I.18 Shifted perspective from influence to control

The above steps can be captured in the following table:

Table I.29 AIC-type perspective shift

Interaction	A1: [{ avp_development_datasets }_ [+capture]_ { other_flying_objects } I3: [{ other_flying_objects }_ [-visually complicate]_ { avp_development_datasets }]	
AIC contributing situations	Pre-shift perspective	Post-shift Perspective
Source	Visually_variable_hot_air_obje cts	Visually_variable_hot_air_objects
Sink	avp_development_datasets	avp_development_datasets
Supra Source	other_flying_objects	other_flying_objects
PrimeP	perform uninterrupted missions	perform uninterrupted missions

Source's Goal	avoid any obstacle to mission execution	Spoof the AVP system by using spoofing visual patterns on the hot-air balloon
Source's Goal type	Influence	Control
Source's Action	Visually complicate	Steal information of the type of data used for the AVP system
Source's Action type	Influence	Control
Source action effect on sink	Obstructive	Obstructive

How do we implement the F.7 approach (Appendix F) thought step above in this context?

The perspective shift captured in Table I.29 was derived using the AIC Type Shift Process, a structured methodology under F.7 approach (Appendix F) that anticipates how complex interactions evolve over time, leading to new situations of complicatedness. The process involves systematically analysing how source-sink relationships change, particularly in how they influence or control their interactions within an operational environment. The steps used to derive the changes in Table I.29 are explained below.

1. Applying AIC Type Shift (General Systems Rule)

The General Systems Rule situations that the goals of a source or sink can change over time due to increased complexity. In this case, the initial interaction was:

A1: [{ avp_development_datasets }_ [+capture]_ { other_flying_objects }]

Here, the AVP dataset captures and processes information from other flying objects, including visually variable hot air balloons. The pre-shift perspective assumes that hot air balloons only introduce passive visual complexity into AVP systems (influence-type interaction).

However, by applying the AIC Type Shift, the interaction was reconsidered under future adversarial conditions, leading to an alternative scenario:

I3: [{ other_flying_objects }_ [-visually complicate]_ { avp_development_datasets }]

Now, hot air balloons are no longer just visually complex objects; instead, they intentionally spoof the AVP system, shifting their role from passive complexity to an active adversarial control-type goal.

2. Predictive query Application

The Predictive query guides the shift:

What would happen if the interaction flow and effect type remained the same, but the goal's AIC type and action type were altered in the future?

Pre-Shift Perspective (Influence)

- The source (visually variable hot air objects) had the goal to avoid obstacles to mission execution, passively complicating the AVP dataset's processing.
- The action type was influence, meaning hot air balloons passively interfered with AVP perception without intent to deceive.
- The effect on the sink (AVP development datasets) was obstructive but unintended.

Post-Shift Perspective (Control)

- The source's goal changed from "avoiding obstacles" to "spoofing the AVP system."
- The source now actively manipulates visual complexity, deploying intentionally deceptive visual patterns to interfere with AVP decision-making.
- The source's action type shifted from influence to control, meaning that hot air balloons are no longer passive obstructions but actively disrupt AVP training and operation.

3. Guiding Prompt Application: Redefining the Nature of Goal and Action

To fully apply the perspective shift, the AIC interactions were re-examined with the following considerations:

- Reviewing AIC Dynamics:
 - Instead of random visual complexity, hot air balloons are now intentionally deceptive, tricking AVP into capturing misleading data.
 - The AVP system, in response, may begin learning incorrect features, weakening its performance.
- Altering Goal and Action:
 - The goal of the balloons changes from passively existing in the environment to actively misleading AVP models.
 - The action shifts from "visually complicate" to "steal information," implying an intelligence-gathering function against AVP datasets.

- Scenario in the Shifted Context:
 - In this new adversarial scenario, hot air balloons are designed with specific patterns to confuse AVP recognition algorithms.
 - AVP dataset integrity is compromised because it learns from misleading visual information, reducing its effectiveness in recognising actual airborne threats.

I.8.4 Step 4) Predict Black Swan scenario

In this step, we elaborate on the scenario further to define the sequence of potential actions. The architect may want to use 5HnWs, or they may use the 5-whys analysis for deeper analysis. To predict Black Swan scenarios, use the AIC perspective shift as clues and imagine situations where such a scenario may arise. Table I.30 captures the Black Swan situations.

Table I.30 Atypical Emergent Scenario (Black Swan situations)

Black Swan scenario	Rationale for prediction: it is possible that ...
Black Swan 1: In this unforeseen scenario, visually variable hot-air balloons appear in regulated airspace equipped with intentionally complex visual patterns that are difficult for the AVP system to classify accurately. These balloons deploy different visual spoofing strategies, such as: • Depicting bird-like images or shapes that are not traditionally detected as hazards within the AVP datasets. • Displaying visual cues resembling known objects (e.g., the shape of a common drone or aircraft) but embedded with confusing visual elements, such as unexpected colour schemes or reflective surfaces that alter appearance based on lighting and angle.	Impact on AVP System Resilience: The AVP system, trained in an typical-AIC environment without these spoofing scenarios, lacks resilience against such complex, variable stimuli. When faced with visually complex patterns intended to exploit gaps in its training data, the AVP system's collision avoidance capability is degraded. Resulting Vulnerabilities: This scenario reveals a critical vulnerability: the AVP system's dependency on highly structured datasets makes it susceptible to exploitation by novel or unforeseen patterns, challenging its adaptability and decision-making reliability.

The hot-air balloons' purpose is to confuse and mislead the AVP system's object detection and classification algorithms through repeated, intentional disruption of the AVP's vision processing model. This confusion introduces obstruction to the AVP's primary purpose, potentially causing the AVP system to either dismiss real hazards or mistake non-hazardous objects as threats.	
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Table I.30 presents a Black Swan situations that explores an unforeseen adversarial manipulation of the AVP system's object detection and classification algorithms. The scenario hypothesizes that visually variable hot air balloons appear in regulated airspace, intentionally designed with complex visual patterns aimed at confusing the AVP system. These balloons introduce visual spoofing tactics, such as bird-like imagery, deceptive object shapes, and reflective surfaces that distort their appearance under varying lighting conditions. This strategy exploits the AVP's dependence on structured training datasets, causing misclassification errors, potentially leading the AVP system to dismiss actual hazards or incorrectly classify non-threats as dangers.

I.8.5 Step 5) Define mitigating ML Development and Safety Requirements.

Each safety requirement outlined for Black Swan scenarios necessitates corresponding ML component safety requirements to ensure the AVP can handle unforeseen threats. Below, we define the ML component safety requirements in alignment with each system safety requirement.

Table I.31a AVP Training Requirements for Black Swan scenarios

Black Swan situations	Safety or systems requirements (Safe Operating Concept)	ML Safety-Training Requirement (Training Concept)
Black Swan 1	Safety Requirement 2: Enhanced Dataset Diversity IF: The AVP system may misinterpret or fail to detect spoofed objects in its environment. THEN: Include in the AVP training dataset a diverse set of complex	ML Safety-Training Requirement 2: The AVP ML component shall be trained to detect, classify, and track diverse set of complex and deceptive visual patterns (e.g., various colours, lighting angles, reflective surfaces)

	and deceptive visual patterns (e.g., various colours, lighting angles, reflective surfaces) resembling bird-like objects and drones in controlled airspace. LEADING TO: increase the AVP system's resilience against misidentifying spoofed objects and reduce the likelihood of collision risks.	resembling bird-like objects and drones in controlled airspace.
Black Swan 1:	Safety Requirement 3: Dynamic Perception Algorithms. IF: Standard object classification approaches may fail to recognise atypical or complex spoofing strategies in real time, THEN: implement adaptive anomaly detection algorithms that detect deviations from typical visual signatures. LEADING TO: improve the AVP's ability to recognise unknown or spoofed objects and prompt further verification.	ML Safety-Training Requirement 3: The AVP ML component shall be trained to detect deviations from typical visual signatures.
Black Swan 1:	Safety Requirement 4: Real-Time Adaptive Validation IF: The AVP system may be susceptible to spoofing by visually complex objects, THEN: integrate real-time cross-checks with air traffic control (ATC) data or other live data sources when confidence is low. LEADING TO: validate detected objects against live external information for more accurate decision-making.	No training required

Black Swan 1:	Safety Requirement 4: Security Measures to Safeguard AVP Datasets [non-functional] IF: The AVP datasets may be vulnerable to spoofed data exposure, THEN: limit access to AVP development datasets and implement strict security protocols for dataset updates. LEADING TO: reduce the risk of unauthorised access to AVP data and ensure data integrity.	No training required
Black Swan 1:	Safety Requirement 5: Security Measures to Safeguard AVP Datasets [non-functional] IF: The AVP's data processing could be compromised by intentional data corruption, THEN: conduct regular security audits and integrity checks on AVP datasets and systems. LEADING TO: identify any unauthorised changes to data or system parameters that could impact perception accuracy.	No training required

In Table I.31, we structured our process for defining safety and machine learning (ML) training requirements derived from the Black Swan 1 scenario. This scenario highlights the risk of visually variable hot air balloons equipped with deceptive patterns intentionally designed to mislead the AVP system's object detection and classification algorithms. The identified safety requirements aim to enhance AVP resilience against such adversarial visual complexity, ensuring robust classification and decision-making in high-risk operational environments.

The safety requirements were formulated by analysing how spoofed objects could degrade AVP performance and obstruct its ability to distinguish real hazards from non-threatening entities. Safety Requirement 2: Enhanced Dataset Diversity ensures that the AVP system is exposed to various deceptive visual patterns in training, including varying colours, lighting angles, and reflective surfaces resembling birds and drones. The corresponding ML training requirement mandates that

the AVP’s machine learning component must be explicitly trained to classify and track such complex visual stimuli.

To address the AVP’s potential failure to recognise atypical spoofing strategies, we define Safety Requirement 3: Dynamic Perception Algorithms. This requirement requires the implementation of adaptive anomaly detection mechanisms that identify deviations from standard visual signatures. The ML training requirement ensures that the AVP’s perception model can detect outliers and unknown spoofing techniques, improving its situational awareness and threat recognition.

Furthermore, we introduce Safety Requirement 4: Real-Time Adaptive Validation to counteract real-time decision-making vulnerabilities, which mandates cross-verification with external air traffic control (ATC) data when the AVP system encounters uncertain detections. This measure does not require additional ML training but instead leverages live data sources to enhance validation accuracy.

Recognising potential security risks, we establish Safety Requirements 4 and 5 to protect AVP datasets from adversarial manipulation. These requirements enforce strict access control measures and mandate regular security audits to prevent intentional data corruption that could compromise the AVP’s ability to distinguish real threats from spoofed objects.

Then, we derive an appropriate set of **ML-component development dataset requirements**:

Dataset requirement structure:

The [system of interest] ML component [Training/Testing/Black Swan Validation] Dataset shall provide the trainee model with a valuable minimum variety of ...

Table I.31b ML Safety-Training Requirements and Perception Dataset Specifications for AVP

ML Safety-Training Requirements (Training Concept)	ML Perception development datasets requirements
ML Safety-Training Requirement 2: The AVP ML component shall be trained to detect, classify, and track a diverse set of complex and deceptive visual patterns (e.g., various colours, lighting angles, reflective	Datasets Req 2: Bird-Like Shapes: The AVP perception ML component Training Dataset shall provide the trainee model with a valuable minimum variety of Images and should include bird-shaped objects in various sizes, colours, and poses to represent different flight orientations.

surfaces) resembling bird-like objects and drones in controlled airspace. ML Safety-Training Requirement 3: The AVP ML component shall be trained to detect deviations from typical visual signatures.	Examples: Hot-air balloons shaped like large birds (e.g., eagles, hawks, pigeons) with realistic and exaggerated colour schemes. Datasets Req 3: Drone and Aircraft Replicas: The AVP perception ML component Training Dataset shall provide the trainee model with a valuable minimum variety of balloon shapes or objects that resemble drones or small aircraft. Examples: Objects with quadcopter shapes, fixed-wing outlines, or fuselage designs similar to standard small aircraft. Pattern deviations: Objects should feature both matte and reflective surfaces that mimic metallic and plastic finishes to emulate typical drones or aircraft. Datasets Req 4: Complicated Visual Patterns and Camouflage The AVP perception ML component Training Dataset shall provide the trainee model with a valuable minimum variety of Mixed Patterns with Reflective Surfaces: Images should include objects with highly reflective and mixed-colour patterns that can shift appearance based on angle and lighting. Examples: Balloons with alternating dark and light patches, stripes, or polka dots that distort under different lighting angles. Reflective Materials: Objects with partially reflective or metallic surfaces that produce glare, mimicking potential interference with sensor recognition.
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	Datasets Req 5: complicated colour schemes and contrasts: The AVP perception ML component Training Dataset shall provide the trainee model with a valuable minimum variety of High-contrast images where objects use non-standard, bright colour schemes (e.g., neon, fluorescent) that deviate from typical environmental colours. Examples: Brightly coloured balloons with contrasting colours (e.g., bright yellow/black stripes, red/green patches) for the perception algorithms to introduce atypical colour contrasts. Datasets Req 6: Environmental Adaptation and Disguise The AVP perception ML component Training Dataset shall provide the trainee model with a valuable minimum variety of Tree and Cloud Camouflage: Images should feature objects with visual patterns or textures that could mimic common background elements, such as clouds, trees, or distant landscape elements. Examples: Balloons with camouflage patterns in grey or green tones that blend with cloudy skies or forested backgrounds. It also depicts a cloud on the balloon itself. Object-Background Blending: Ensure variations of these objects within varied natural settings (forests, urban, and rural areas) to train the system on distinguishing between foreground and background.
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I.9 Stage 5: CuneiForm-based Syllabus for Safety-Driven ML Development⁷

Outcomes from stages 3 and 4 reveal the epistemic uncertainties faced by the architect and their trainee machine regarding the problem domain. The architect must define the real world and outline potential unexpected scenarios for the trainee machine to meet objectives. Minimising these uncertainties to a reasonable level (ALARP) is crucial, achieved by creating CuneiForms that capture required training regimes. Thus, this process should be termed ML Epistemic Uncertainty Reduction Training, focusing on enhancing the machine's robustness by increasing the variety of real-world scenarios instead of increasing the variety of image augmentations. Dataset augmentation techniques (e.g., colour modifications) will follow, seen as “Aleatoric Uncertainty Reduction” since they alter pixel randomness rather than actual scenarios.

I.9.1 Step A) Articulate the pictorial problem context:

Datasets Req 6: The AVP perception ML component Training Dataset shall provide the trainee model with a valuable minimum variety of Environmental Adaptation and Disguise

Tree and Cloud Camouflage: Images should feature objects with visual patterns or textures that could mimic common background elements, such as clouds, trees, or distant landscape elements.

Examples: Balloons with camouflage patterns in grey or green tones that blend with cloudy skies or forested backgrounds. It also depicts a cloud on the balloon itself.

Object-Background Blending: Ensure variations of these objects within varied natural settings (forests, urban, and rural areas) to train the system on distinguishing between foreground and background.

⁷ Associated to section M.7



Figure I.19 We used DALL-E to generate the this Black Swan situations for validation.

Table I.32 CuneiForm Pictorial situation articulation

Pictorial Situation CoT step	Definition
Step 1) Define a minimum variety of TOIs and their pictorial appearances	architect prediction: the architect asserts that the perception system may face a pictorial situation with: A hot-air balloon with sky and clouds painted on its fabric.
Step 2) Consider a minimum variety of other objects that TOIs aim to influence	architect prediction: the architect asserts that: the hot-air balloon aims to influence the discoverability of itself by AVP.
Step 3) Consider a minimum variety of objects that TOIs must appreciate	architect prediction: the architect asserts that: the hot-air balloon may have to appreciate the following environmental scenery aspects: Cloud Cover Variability: The balloon should account for various cloud formations and densities, as clouds could enhance or diminish its camouflage.

	Sky Colour and Light Conditions: Different times of day or weather conditions will affect how the sky appears. The balloon's camouflage pattern may need to appreciate transitions from sunrise to sunset, clear skies, and changing hues.
Step 4) Consider a minimum variety of what other objects TOIs must control the correctness of predicting their shapes.	Architect prediction: The architect asserts that the hot-air balloon must carefully control its visual presentation to maintain inconsistent shape predictability, reducing the likelihood of correct classification or shape clarity within the AVP system's detection model. This includes: Varying the rotation of the balloon in order to increase the randomness of its reflective glossy surfaces, thus causing a maximum likelihood of AVP perception failure.
Step 5) Produce pictorial problem context.	Pictorial problem context: The architect asserts that the AVP perception system may face a pictorial situation with: A hot-air balloon{1} with a sky and clouds painted on its fabric{2}. A background of varying cloud formations and densities{3}. The balloon must adjust for changes in sky appearance based on different times of day and, including transitions from sunrise {4} to sunset{5}, clear skies{6}, and variable light hues{7}. The balloon varies its rotation {8} to increase randomness in its reflective glossy surfaces {9}, maximising the likelihood of AVP perception failure.

I.9.2 Step B) Characterise the Training Classes for CuneiForms:

In this step, we will define one CuneiForm that satisfies the pictorial situation context. We will consider a maximum of 5 objects in this CuneiForm. In this case study, we will need to define the range of general aircraft 3D orientation specification. We choose to define 18 possible variations:

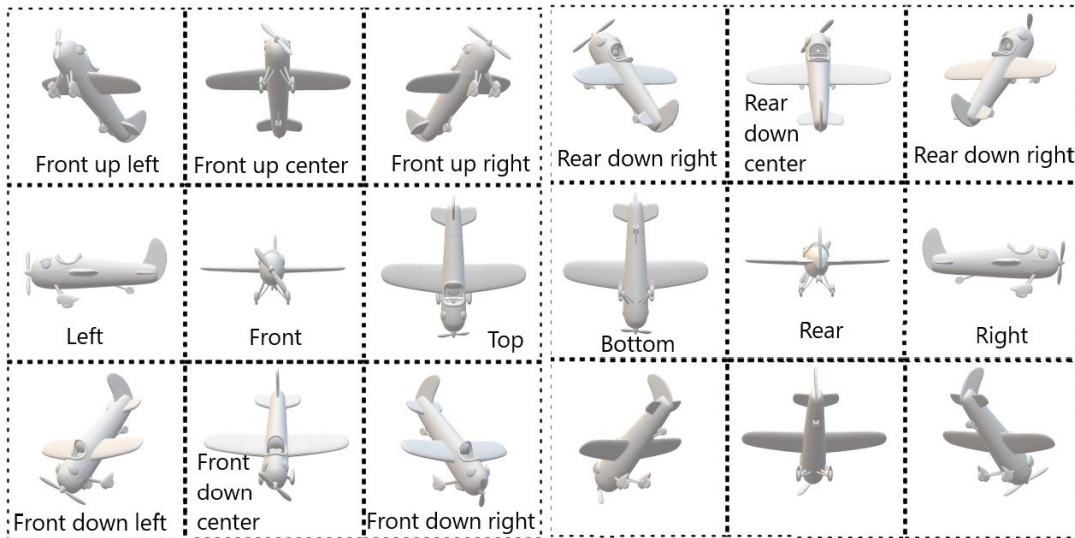


Figure I.20 general aircraft 3D orientation specification

In this step, we will define one CuneiForm that satisfies the context of the pictorial situation. We will consider a maximum of 5 objects in this CuneiForm.

Table I.33 Characteristic Training Classes definitions for a CuneiForm abstract image

CuneiForm characteristic	Definition
Step 1) Define visible horizon attitude.	No Horizon
Step 2) Define all the TOIs and their aesthetic complexity. Then, generate abstract representative icons for the CuneiForm abstract image.	Several hot-air balloons {1.1}, sky and clouds painted on its fabric {2.1}
Step 3) Define TOI's motion trajectory and dynamic optical situations. Then, update the generated abstract representative icons for the CuneiForm abstract image.	Motion trajectory: static, no motion {1.2}. Dynamic optical situation: captured without optical blur {1.3}.

Step 4) Define the background objects associated with TOIs and environmental scenery in the background of the CuneiForm. Then, generate abstract representative icons for the CuneiForm abstract image.	Clouds, Mostly Clear or Sunny (1/8 to 2/8 oktas) {3.1}, sphere of the sun during day {4.1}
Step 5) Define the background Objects' Motion situations and dynamic optical situations. Then, update the generated abstract representative icons for the CuneiForm abstract image.	Background objects' motion trajectory: static {3.2, 4.2} Dynamic optical situation: no motion blur {3.3, 4.3}
Step 6) Define TOI's positioning in the CuneiForm. Then, generate abstract representative icons for the CuneiForm abstract image.	Any {1.4}
Step 7) Define TOI's 3D orientation. Then, update the generated abstract representative icons for the CuneiForm abstract image.	Front {1.5}
Step 8) Define the optical distance for each TOI in nindans. Then, update the generated abstract representative icons for the CuneiForm abstract image.	A balloon is represented in a pictorial distance of 3 nindan (equivalent to 3/9 of the total area of the pictorial frame) {1.6} A balloon is represented in a pictorial distance of 1 nindan (equivalent to 1/9 of the total area of the pictorial frame) {1.7}
Step 9) Design the relevant icons to produce the CuneiForm and give an example of an instantiating image	Hot-air Balloon:  , Clouds:  , Sun: 

I.9.3 Final CuneiForm

An output cuneiform would be:

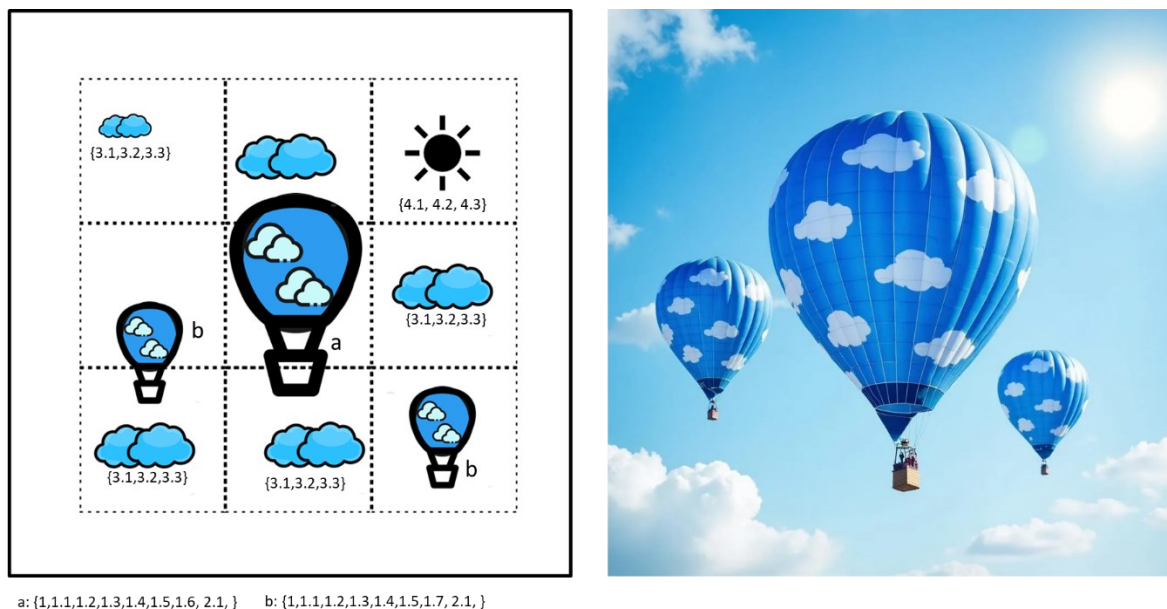


Figure I.21 Example CuneiForm and instantiated image

An example of CuneiForm non-compliant images may be:

Table I.34 Examples of non-compliant instantiations

CuneiForm non-compliant Image	Reasons	Decision
	Such an image would not pass the CuneiForm-I.20 characterisation, as it inadequately captured the presence of clouds and the sun's position on the opposite side. Nonetheless, despite this shortcoming, the image illustrates a rare event. Consequently, it may be considered for inclusion in validating a new variation of CuneiForm.	To be added to the Black Swan validation datasets.
	Such an image would not pass the CuneiForm-I.20 characterisation, as it inadequately captured the presence of cloud paintings on the balloon. Nonetheless, despite this shortcoming, the picture	To be added to the Black Swan training dataset, to instantiate another Black Swan CuneiForm training class.

	illustrates a rare event. Consequently, it may be considered for inclusion in validating a new variation of CuneiForm.	
	Such an image would not pass the CuneiForm-I.20 characterisation, as it inadequately captured the presence of cloud paintings on the balloon. Nonetheless, despite this shortcoming, the image illustrates a rare event. Consequently, it may be considered for inclusion in validating a new variation of CuneiForm.	To be added to the Black Swan training dataset, to instantiate another Black Swan CuneiForm training class.
 Link	Such an image would not succeed in passing the CuneiForm-I.20 characterization, as it does not capture the presence of cloud paintings on the balloon. However, it certainly captures a rare or seasonal event.	To be added to the Black Swan validation datasets, to instantiate another Black Swan CuneiForm validation class.
 Link	Such an image would not succeed in passing the CuneiForm-I.20 characterisation, as it does not capture the presence of cloud paintings on the balloon. Also, the situation in the image is clear of any cloud.	To be discarded as it does not necessarily add valuable information in comparison to the above.

I.9.4 Develop the Training, Testing and Black Swan Validation Datasets

The final step of this process would be gathering various images that conform to the CuneiForm. In this exercise, we will assume that the architect has done the due diligence per their data generation methods. Instead of gathering and generating a dataset, we will validate the AVOIDDS training strategy against the CuneiForm framework and training classes coverage criteria.

I.10 Stage 6: Black Swan-driven ML Development and Testing⁸

Usually, there would be the model training process begins. We will assume that a model was developed, and the client needs to validate the coverage of the training dataset. So, in this stage, we will use the CuneiForm method to validate the quality of a pre-existing dataset named AVOIDDS. This repository contains datasets, models, and simulators for the AVOIDDS (Aircraft Vision-based Intruder Detection Dataset and Simulator) benchmark, which centres around the vision-based aircraft detect-and-avoid (DAA) problem. The full AVOID dataset of 72,000 samples is available here: purl.stanford.edu/hj293cv5980.

Our validation methodology was based on covering the CuneiForm framework for Epistemic Training Strategy, which allows for structured epistemic analysis of dataset completeness by categorising data based on key dimensions, including:

- Time-of-day distribution.
- Cloud type.
- Horizon orientation.
- TOI's pictorial positioning zones.
- TOI's pictorial distance.

Key Assumptions:

We will use the following definitions:

Concept	Definition
ALARP	As Low As Reasonably Practicable
ArcUC	Architect Epistemic Uncertainty Curve for a given dataset
PHI	Pictorial Visible Horizon Attitude Indicator
TOI	Target Object of Interest

- The sample_small folder of training and validation datasets is statistically representative of AVOID dataset (73,000 instants).

⁸ Associated with section M.8

- 30 images, training sample.
- 30 images validation sample.
- Time of day, Clouds types, file names all taken directly from the provided situation_data.xlsx in the repository.
- The CuneiForm coverage results on the representative sample are accepted as a quality assessment of the entire AVOID dataset.
- For this exercise, we needed to define TOI's 3D orientation categories for completeness to achieve coverage or 3D orientation and minimise epistemic uncertainty ALARP. To such end, we define the following categorical system of airplane orientation training classes:

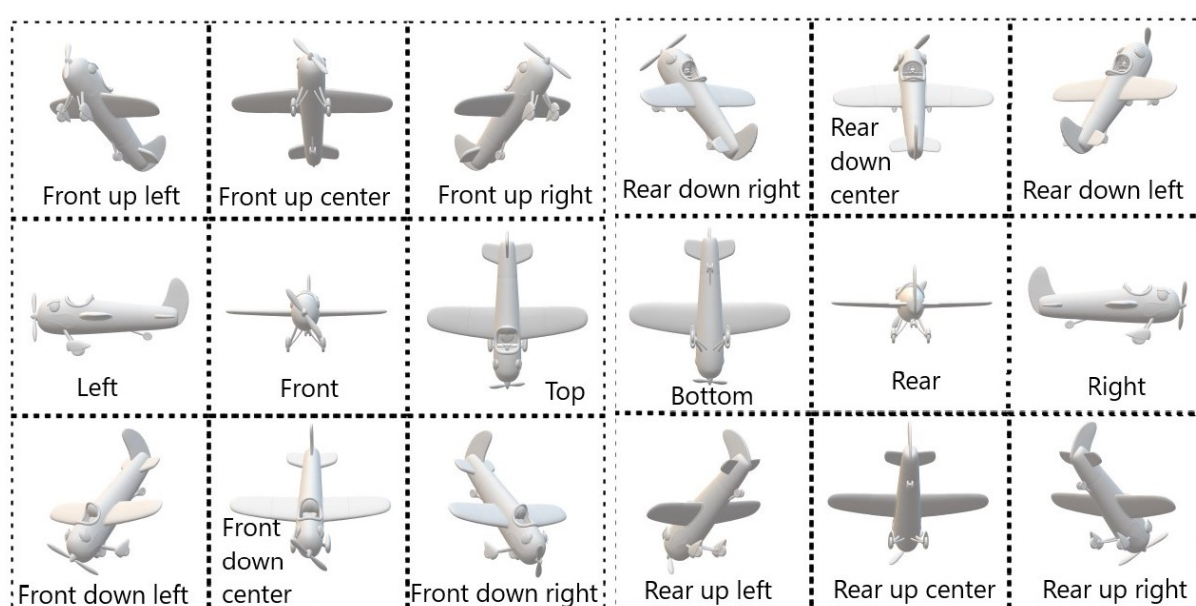


Figure I.22 aeroplanes 3D orientations categories (training classes)

Figure I.21 presents a structured classification system for defining aircraft's three-dimensional (3D) orientation categories within a training dataset. This classification aims to cover various orientations comprehensively, ensuring minimisation of epistemic uncertainty to As Low as Reasonably Practicable (ALARP) levels. This structured approach improves dataset completeness in training AI-based perception models in airborne object detection and classification tasks.

The figure systematically categorises airplane orientations by organising them into distinct viewpoints based on azimuthal and elevational perspectives. These orientations are essential for ensuring that trained AI models generalise effectively across various real-world aircraft configurations. The classification system is divided into four major reference perspectives:

1. Frontal Perspective (Front, Front-Up, Front-Down)

- Captures variations in viewing the aircraft from the front-facing direction.
 - Includes standard frontal orientation and elevated or depressed views at different angles (e.g., Front Up Left, Front Up Right, Front Down Center).
2. Rear Perspective (Rear, Rear-Up, Rear-Down)
- Represents aircraft as viewed from the rear, covering both standard rear-facing angles and various elevations.
 - Important for detecting aircraft retreating or in trailing scenarios that challenge detection algorithms.
3. Lateral Perspective (Left, Right)
- Includes aircraft viewed from the side, either left or right.
 - Ensures the model can recognise aircraft from side angles, which is critical in multi-aircraft detect-and-avoid scenarios.
4. Top-Down and Bottom-Up Perspectives
- Captures aircraft directly from above or below (e.g., Top, Bottom).
 - These perspectives are particularly challenging for vision-based models as they often resemble ground projections or silhouettes against the sky.

By structuring the training dataset around these 16 orientation categories, the approach ensures that the AI system is exposed to a complete representation of possible aircraft views in 3D space. This methodology is essential for reducing epistemic uncertainty and enhancing generalisation in vision-based models for airborne object detection, tracking, and avoidance in dynamic operational environments.

The inclusion of elevational shifts (up, down) and lateral tilts (Left, Right) provides additional robustness against biases that may arise from incomplete viewpoint exposure in training datasets. Failure to account for these diverse orientations could lead to inconsistent model performance in real-world scenarios, where aircraft positions vary significantly due to dynamic flight manoeuvres.

I.10.1 CuneiForm Training Strategy as a Validation Process for Datasets

We will use some of the training classes we defined in section **E.7: CuneiForm ML Epistemic Training Strategy** to reference what needs to be covered in the AVOID training dataset. The following is a snapshot of how the training sample looks:

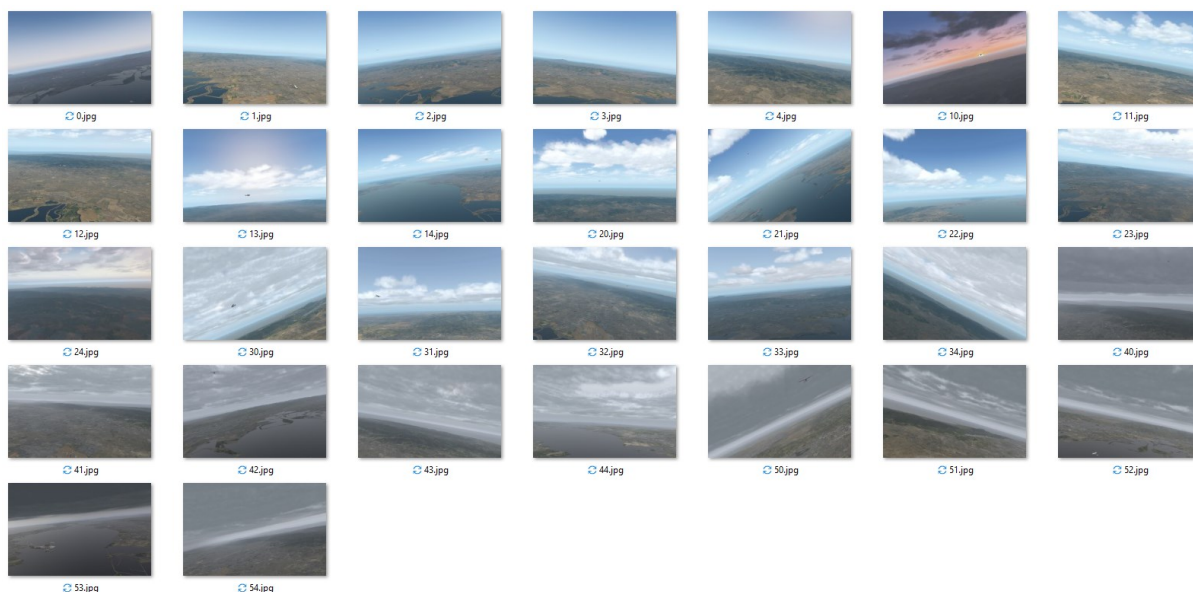


Figure I.23 AVOID training dataset random sample

Figure I.22 shows the set of images we used to evaluate the quality of the AVOID training dataset in terms of coverage using CuneiForm epistemic dimensions coverage. In other words, we look at the dataset and examine any imbalance of coverage against the categories for some of the CuneiForm dimensions. In this exercise, we included three CuneiForm Training classes in our examination and produced an examination report, which can be found in here:

- Pictorial Horizon orientation.
- TOI's pictorial positioning zones.
- TOI's pictorial distance.

We also included one aspect of our ODD, the Time-of-Day categorisation. AVOIDDS provides a balanced (uniformly distributed) distribution of information about the types of clouds.

The validation process was structured into the following key steps:

I.10.1.1 Examining AVOIDDS Training Strategy in Covering Time-of-Day Training Classes

Our examination discovered that there exists an imbalance in coverage, with 47% of instances recorded at night and midday (noon) demonstrating strong bias. The lack of balanced exposure constitutes a potential for exponentially unpredictable, high-risk emergent Black Swan behaviour performed by the perception. The dataset does not satisfactorily meet the ALARP requirement. Therefore, we have sufficient reason to believe that this dataset does not pass the ALARP criteria for covering time-of-day categories. The unbalanced coverage leads to hazardous uncertainty in the

intelligent system's behaviour during Black Swan scenarios in a respective category. The figure below captures the results of examining AVOIDDS time-of-day coverage based on their sample data.

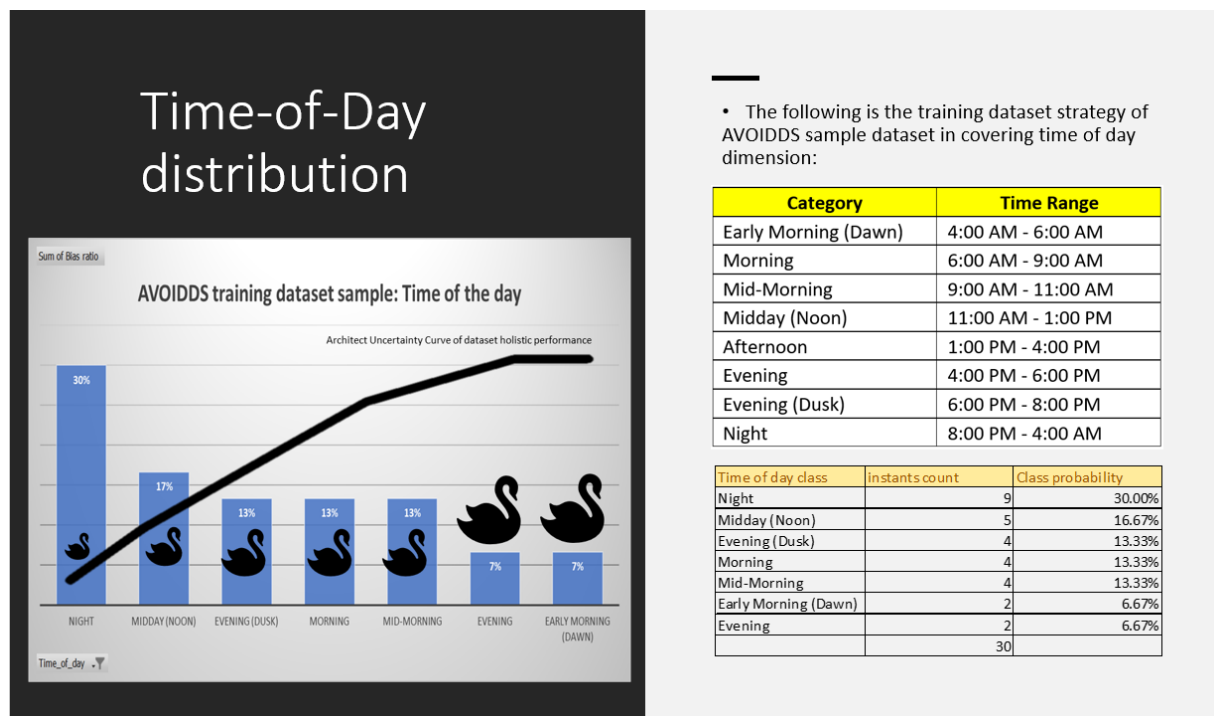


Figure I.24 AVOIDDS Time-of-Day coverage validation report based on the training sample folder

Figure I.23 presents our examination of the time-of-day coverage within the AVOID training dataset, highlighting critical imbalances in its distribution. The dataset exhibits a strong bias towards night and midday (noon) instances, which account for 47% of the recorded samples. This skewed representation results in an incomplete epistemic coverage, introducing potential hazardous uncertainty when the model encounters underrepresented time-of-day conditions in real-world applications.

The figure employs a Black Swan icon size to visually depict the degree of epistemic uncertainty across different time-of-day categories. Larger swan icons indicate higher uncertainty, suggesting that the dataset is less reliable in those categories and that the trainee model is likely to exhibit erratic or unpredictable behaviour in those underrepresented conditions. Specifically, early morning (dawn), evening and mid-morning classes exhibit a notably low representation, increasing the likelihood of exponential performance degradation in these unseen scenarios.

From a safety-critical AI validation perspective, the dataset's coverage fails to meet the ALARP (As Low As Reasonably Practicable) criterion, as the uneven distribution introduces a significant risk of Black Swan emergent behaviours. The absence of balanced exposure across all time-of-day categories weakens the model's ability to generalise and respond effectively in diverse lighting and environmental conditions, which is crucial for robust aircraft detection-and-avoid applications.

Therefore, the findings illustrated in Figure I.22 underscore the importance of a more balanced dataset composition. Without further rebalancing or augmentation of underrepresented time-of-day classes, the AVOID dataset remains susceptible to uncertainty, reducing its reliability and safety for real-world deployment in dynamic aviation environments.

I.10.1.2 Examining AVOIDDS Training Strategy in Covering Clouds type

The dataset was evaluated for **balanced exposure to different cloud formations**. While it passed the ALARP (As Low As Reasonably Practicable) requirement for general cloud-type coverage (Figure I.24), if we consider the Time-of-Day per cloud-type distribution, we discover potential biases in the dataset (Figure I.24). The figure below reflects the training sample cloud-type alone coverage:

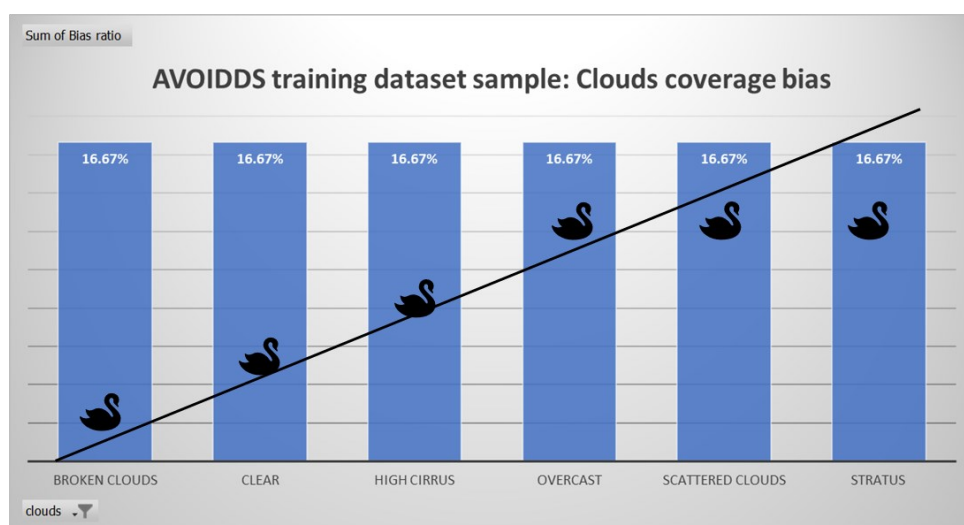


Figure I.25 types of cloud coverages examination

The AVOID training dataset has a balanced coverage of all types of clouds. Figure I.24 presents a high-level overview of the dataset's cloud-type distribution, indicating that each cloud category, broken clouds, clear sky, high cirrus, overcast, scattered clouds, and stratus, is covered equally at 16.67%. This uniformity suggests that, at first glance, the dataset satisfies ALARP (As Low As Reasonably Practicable) requirements regarding cloud-type representation.

We then examined a more refined approach that examines the type of clouds and type of day. Below is the histogram that showcases the types of missing scenarios in the training dataset sample:

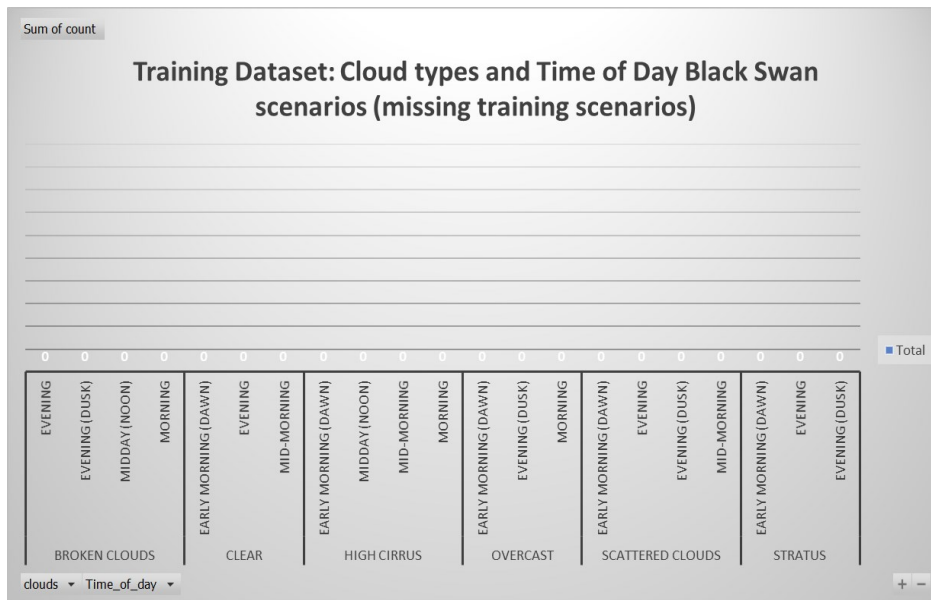


Figure I.26 Potential epistemic confusion in Black Swan scenarios

Figure I.25 captures a more refined description of potential real-world scenarios in which we have sufficient reason to lose trust that any ML component trained on the AVOIDDS would detect objects' reliability if faced with the Black Swan situations in those conditions. In other words, Figure I.25 extends this analysis by examining the intersection of cloud types with time-of-day variations, highlighting the absence of several crucial Black Swan scenarios. The missing instances indicate conditions in which the dataset does not train the model adequately, increasing epistemic uncertainty. If an ML perception system is trained on AVOIDDS but deployed in a scenario where, for instance, a broken-cloud sky occurs at dusk, an untrained combination, there is an increased risk of unpredictable model behaviour, potentially leading to failures in object detection.

In contrast, the training dataset sample reveals the following combined distribution between cloud and Time-of-Day in Figure I.25. It shows that AVOIDDS has a training strategy that covers scenarios such as:

- Broken cloud sky during dawn.
- Clear sky during morning.
- High cirrus clouds during the night.
- Overcast clouds during the evening.
- Scattered clouds during midday.
- Stratus cloud during morning...etc.

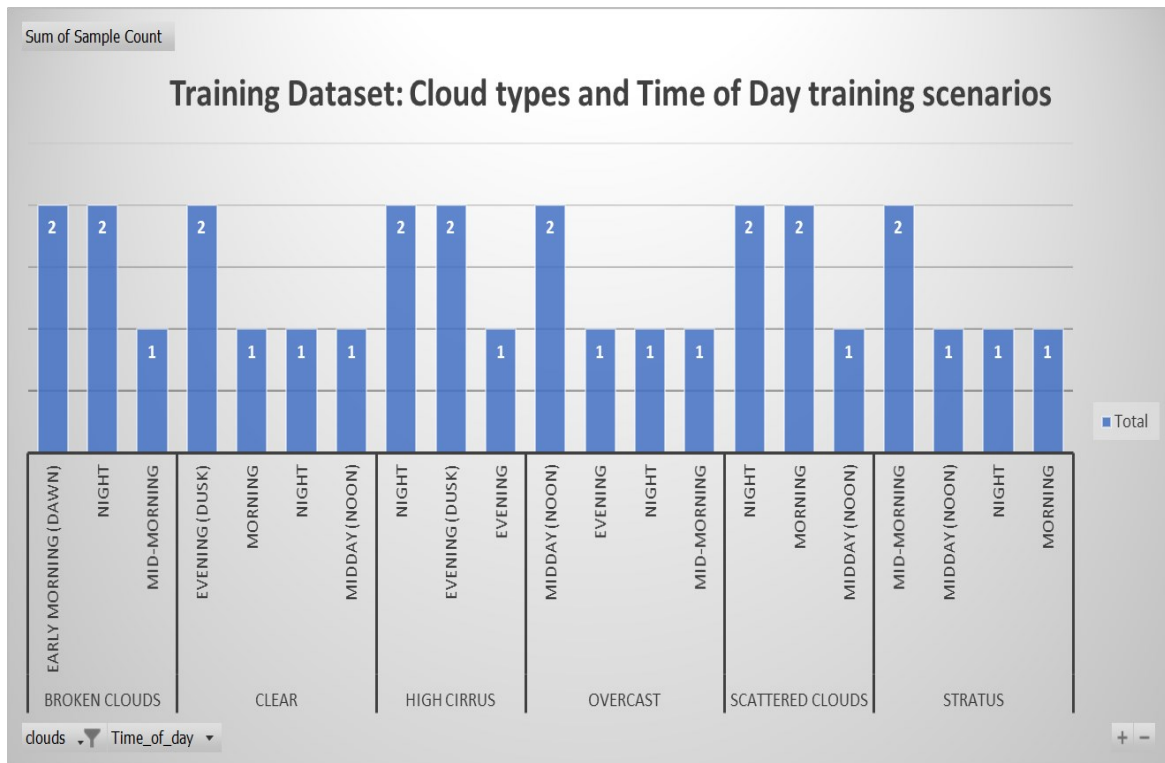


Figure I.27 Example scenarios covered by AVOIDDS training strategy

Figure I.26 further refines this evaluation by showcasing the specific cloud-time-of-day pairings in the dataset. It demonstrates that while the dataset covers some expected scenarios, such as high cirrus clouds at night and scattered clouds at midday, it remains incomplete for ensuring robust real-world performance. The analysis confirms that while the dataset's cloud coverage is theoretically balanced, its intersection with time-of-day conditions introduces significant epistemic uncertainty, limiting the reliability of any trained ML model under novel operational conditions.

I.10.1.3 Examining AVOIDDS Training Strategy in Covering Pictorial Distance Training Classes

See sections 4.5.5.5 and 5.9.5 for a background definition of this training class. The dataset was tested for coverage across different object distances (close, medium, far, and extremely far). A major imbalance was found, with 50% of instances covering only 20% of the required categories. Additionally, no instances were classified as "dangerously close distances," which could cause critical failures in AI-based collision avoidance models. See Figure below:

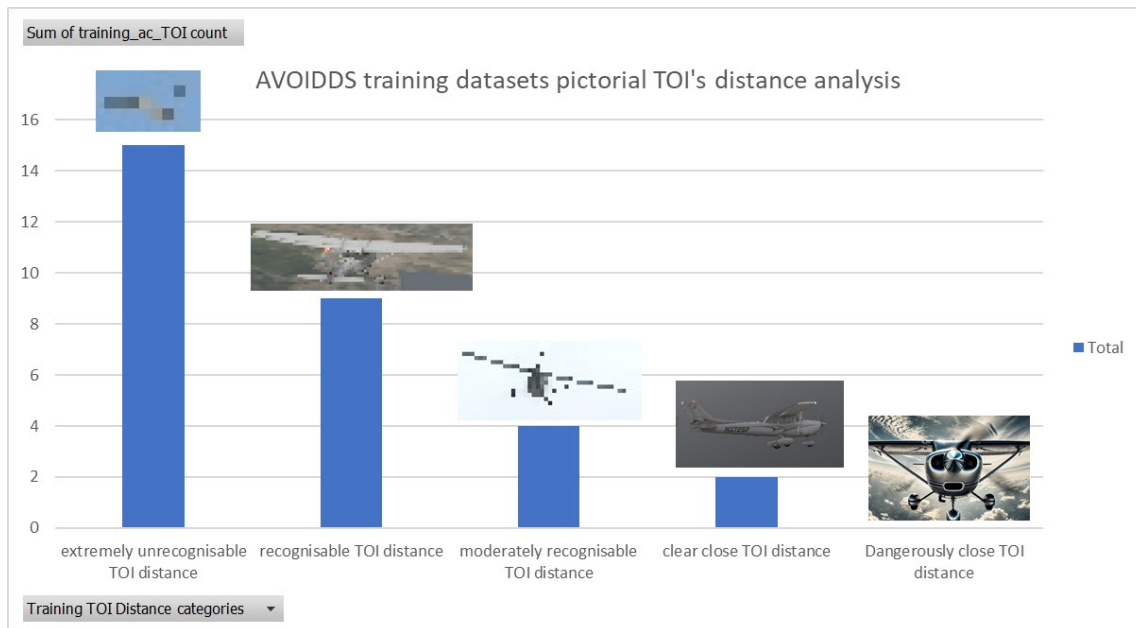


Figure I.28 TOI distance categories distribution

Figure I.27 presents our examination of the AVOID training dataset's coverage of TOI distance training classes, revealing a significant imbalance in the representation of different distance classes.

Regarding trustworthiness in facing Black Swan scenarios, the distribution indicates more uncertainty (less trust) about the “dangerously close distance” category. Signified by the size of the Black Swan icon in the figure below:

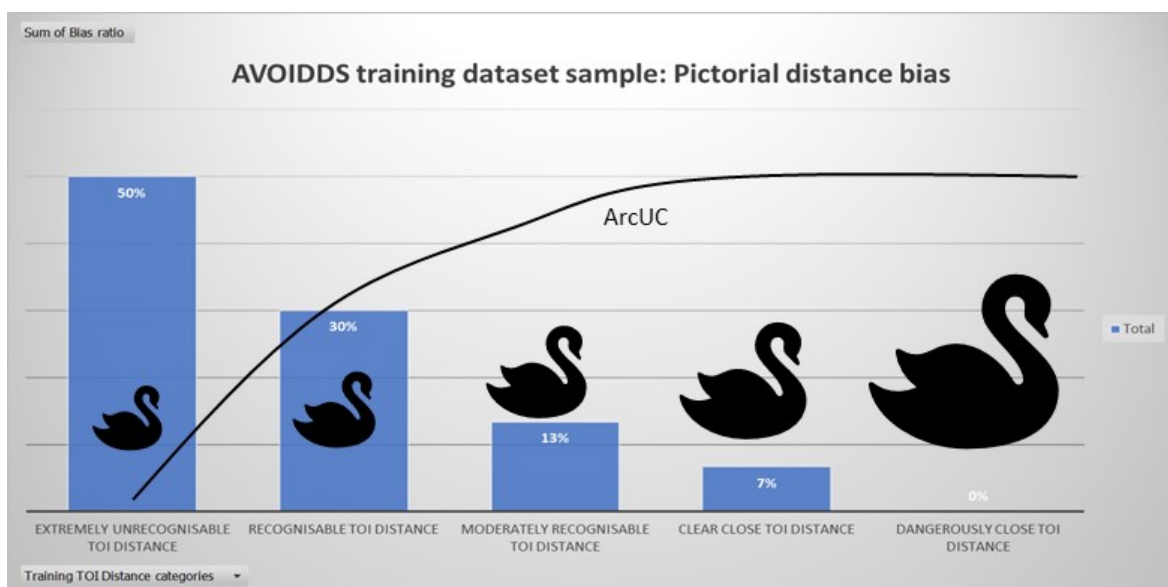


Figure I.29 Trust in performance in facing Black Swan scenarios related to a respective category.

Figure I.287 shows how the architect's uncertainty increases as the likelihood of unreliable performance during a Black Swan situation in a given category increases. It is interesting to notice

that AVOIDDS did not consider a dangerously close aeroplane for a safety-critical application such as avoiding mid-air collisions. Considering that some incidents happened because two aeroplanes came dangerously close to each other without the pilots noticing, see the following [2]. Consequently, the dataset does not satisfactorily fulfil the ALARP requirement for pictorial distance coverage.

I.10.1.4 Examining AVOIDDS Training Strategy in Covering TOI's Positioning Training Classes

See section E.7.4.4 for a background definition of this training class. The dataset's representation of **object positioning in the visual frame** was analysed. We found that the dataset exhibits a significant imbalance, with 60% of perception experiencing TOIs in only 20% of the possible positioning quadrants. Notably, there is a complete absence of experience in the following possible positions:

1. up center/center
2. up left/center left
3. center left/center
4. up left/up center
5. down left/down centre

The lack of exposure constitutes a potential for exponentially unpredictable, high-risk emergent Black Swan behaviour performed by the perception. Figure I.28 demonstrates the lack of coverage in the listed positions, indicating potential sources of sufficient concerns in facing Black Swan scenarios in such categories. The histogram shows a heavy concentration of training instances in central and slightly off-centre positions, with the highest frequency recorded at "centre" (17%) and "downright" (13%). Other positioning categories, such as "center left," "center right," "down left," and "down right," receive moderate representation (7%), while numerous other positional categories have significantly lower representation (3%). The Black Swan icons in the figure depict the increasing uncertainty in model reliability as the training dataset lacks exposure to key TOI positional scenarios. Most notably, the dataset completely lacks experience in five critical positions: "up center/center," "up left/center left," "center left/center," "up left/up center," and "down left/down center". The absence of these positioning categories presents a risk in real-world object detection, particularly in aviation safety applications, where TOIs may appear in any arbitrary position relative to the observer. The dataset's failure to comprehensively train AI models across the full spectrum of TOI positions indicates a significant epistemic gap, increasing the likelihood of unpredictable Black Swan failures in operational environments.

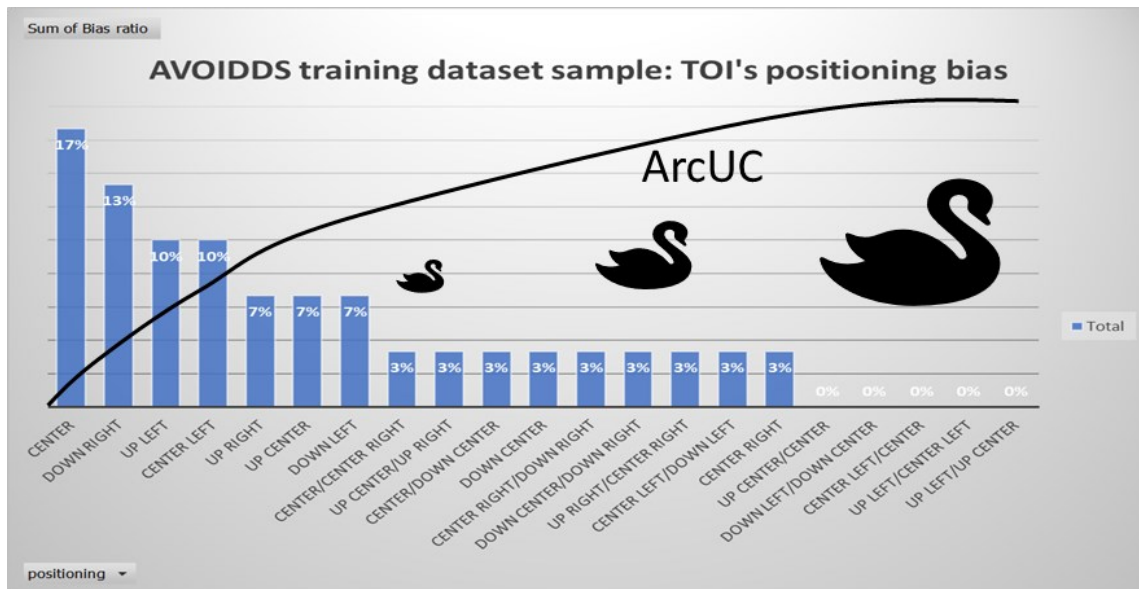


Figure I.30 TOI's positioning training classes coverage.

Figure I.30 shows which training scenarios the AVOIDDS epistemic training strategy missed in its training dataset to illustrate the missing training classes clearly. Red zones are the areas in images that the training dataset is missing, which gives us sufficient reason to suspect that the trained ML may struggle with scenarios where TOIs appear to be in those regions.

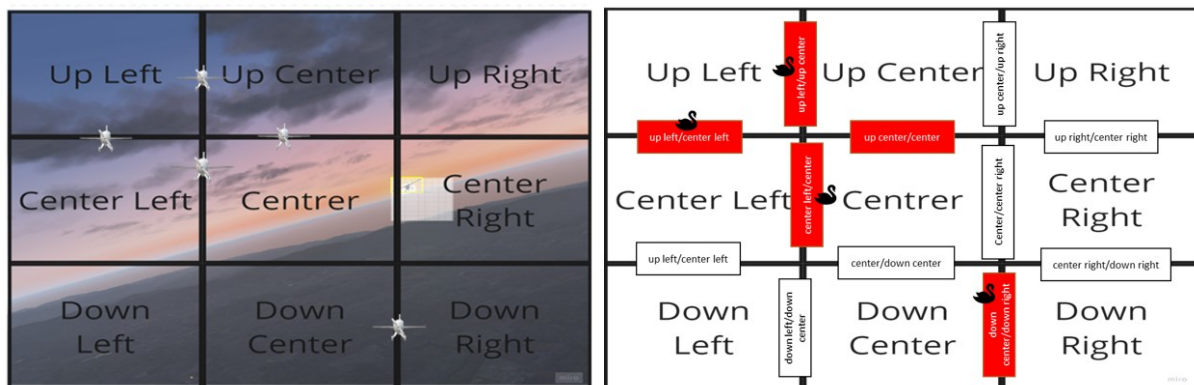


Figure I.31 visualising the areas of missing scenarios in AVOIDDS training strategy

Consequently, the dataset does not satisfactorily fulfil the ALARP (As Low As Reasonably Practicable) requirement for TOI's positioning coverage.

I.10.1.5 Examining AVOIDDS Training Strategy in Covering Pictorial Horizon Attitude Training Classes

See sections E.7.5.3, 4, 5 for a background definition of this training class. Our examination of coverage against Visible Horizon Attitudes training classes showed that AVOIDDS training strategy exhibits significant imbalance, with 60% of perception experiencing TOIs in only 23% of the possible

horizon attitudes. Notably, there is a complete absence of experience in the following possible horizon attitudes:

- Level Horizon
- Negatively Tilted Elevated Horizon
- Acute Angled Bird's Eye Ground View
- Bird's Eye Ground View
- Ascending Rocket Sky View
- Negatively Tilted level Horizon
- Acute Angled Rocket Sky View

The lack of exposure constitutes a potential for an exponentially unpredictable high-risk emergent Black Swan behaviour performed by the perception.

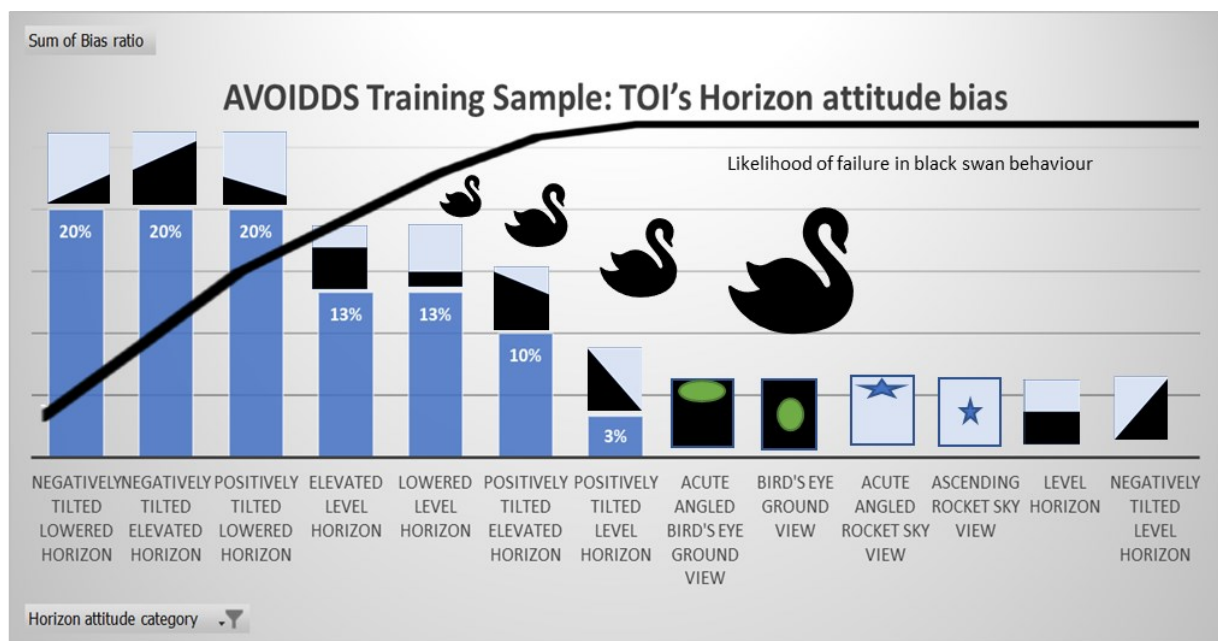


Figure I.32 AVOIDDS training strategy coverage of Horizon Attitudes training classes. The icons on top of the pins depict the nature of the horizon attitude and how the TOI would look.

Figure I.31 presents the analysis of the AVOID training dataset's coverage of TOI horizon attitudes, revealing substantial biases in exposure to different orientation perspectives. This distribution results in a considerable epistemic gap, decreasing our trust in the dataset to enable the trainee model handling Black Swan scenarios where TOIs appear in underrepresented attitudes. The Black Swan markers indicate the rising architect uncertainty about AVOIDDS quality, allowing any trained model to generalise to unseen cases. Consequently, the dataset needs to satisfactorily fulfil the ALARP (As Low As Reasonably Practicable) requirement in terms of horizon attitude coverage.

I.10.2 Possible Missing Pictorial Hazards Identified in AVOIDDS

Our examination of the AVOID dataset has revealed limitations that affect its suitability for real-world safety-critical applications, particularly in mid-air collision avoidance perception systems. While AVOIDDS is a useful benchmarking tool, it lacks the robustness to ensure reliability under complex operational conditions. The identified hazards highlight areas where the dataset introduces epistemic uncertainty, increasing the likelihood of perception failures. Table I.35 outlines the key hazards and associated mitigations necessary to enhance the safety and reliability of AVOIDDS to enable a reliable ML model.

One of the primary failure modes identified is ambiguity in object identification, where the perception system may struggle to correctly recognise aircraft that are either at long distances or appear from unconventional orientations. This ambiguity poses a critical risk, as misinterpretation of another aircraft's position and movement can lead to incorrect avoidance actions. To mitigate this risk, the perception system must be trained on various aircraft orientations, including atypical perspectives such as inverted or tilted positions, which may arise due to dynamic flight manoeuvres or environmental contributing situations.

From an assurance perspective for real-world application, to ensure that the datasets accurately reflect the operational domain, the architect must provide objective evidence that can be quality assured by third parties. The AVOIDDS sample indicates the presence of images that human validators cannot easily verify to determine whether those images cover a particular risk. For example, image 24.jpg part of the training sample, includes a TOI where it is unclear which 3D orientation it is captured in. See the Figure below:

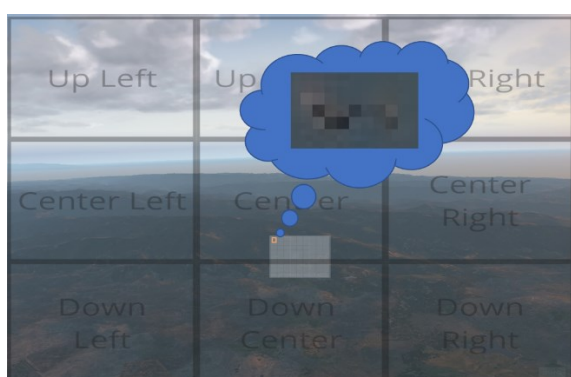


Figure I.33 a sample training image 24.jpg in AVOIDDS dataset where the TOI's 3D orientation is unrecognisable

The figure above showcases a weakness in the dataset, in that there are TOIs that are too far, so much so they may have a distorted visual appearance. This unrecognisable visual appearance may lead the model to an increased number of recall (false positives).

Another critical issue (can be claimed to have been attempted to be addressed by the AVOIDDS) is difficulty recognising distant or low-visibility objects, particularly in adverse conditions such as night-time, poor lighting, or heavy cloud cover. Aircraft appearing in such situations may not be effectively detected by a model trained on AVOIDDS, which lacks sufficient representation of these challenging scenarios. To address this limitation, the system should be enhanced to recognise objects at extreme distances and under varying atmospheric conditions. This includes handling low-contrast objects and those partially obscured by environmental contributing situations such as clouds or atmospheric scattering, ensuring reliable detection even in suboptimal conditions.

A further limitation arises from uncertainty in aircraft orientation and behaviour, where the system may fail to interpret an aircraft’s attitude and trajectory correctly. This issue is particularly concerning in high-risk scenarios, such as two aircraft crossing paths or one executing an unexpected manoeuvre. If the perception system cannot distinguish between a stable flight path and an erratic movement, it may take inappropriate or delayed actions, leading to an increased risk of collision. To mitigate this, the model must be trained on a comprehensive range of aircraft orientations and dynamic flight behaviours, ensuring it can accurately predict trajectories and distinguish between normal and anomalous flight patterns.

Table I.35 Hazards and mitigations associated with AVOIDDS

Missing Perception Failure mode	ML Safety-Training Requirement (Training concept):
Ambiguity in Object Identification: Risk: The system might fail to recognise other aircraft when they are at a very close distances or have an orientation that makes them difficult to identify. For example, if an aircraft is coming from an unconventional angle (e.g., upside down or tilted), the system might misinterpret its position and distance, leading to incorrect avoidance actions.	Mitigation: The perception system shall be trained to handle diverse orientations, particularly those that might appear unconventional due to the aircraft's flight dynamics or environmental conditions.

Difficulties in Recognizing Distant or Low-Visibility Objects: Risk: In low-light (e.g., night), weather, or long-range scenarios, the system might fail to detect aircraft that are at extreme distances or partially obscured by clouds, smoke, or other atmospheric contributing situations. The system’s performance could degrade, especially in critical situations where detection and timely reaction are necessary to avoid collisions.	Mitigation: The system should be trained to handle ambiguous or partial visual information by recognising objects at varying distances and through weather conditions like broken clouds or low visibility. Proper handling of low-contrast or poorly illuminated objects will ensure that the system remains effective even in suboptimal conditions.
Uncertainty in Aircraft Orientation and Behavior: Risk: Due to their flight path (e.g., ascending, descending, tilted), aircraft can appear in various orientations. If the perception system cannot accurately determine the orientation or trajectory of another aircraft, it might misinterpret its movement or predict the wrong course of action. This is particularly dangerous when aircraft are near each other or crossing paths.	Mitigation: The system must be trained to interpret diverse aircraft orientations, especially in scenarios where they might not be level, and to account for dynamic flight behaviour. This would ensure that it can differentiate between aircraft in a stable flight path versus those in unusual orientations, thus avoiding collision.

I.10.3 Retrospective Production of CuneiForm Abstract Images

In our examination of the AVOID dataset, we retrospectively reconstructed potential CuneiForms that may have been implicitly guiding the dataset’s training strategy. This process involved analysing the dataset’s sample characteristics regarding object positioning, motion, environmental context, and horizon attitude and mapping them onto an abstract CuneiForm framework. The objective was to determine what the dataset’s composition would be like if we were to generate a CuneiForm abstraction.

To reconstruct the CuneiForm representations, we extracted their key dimensions by examining a selection of instantiated training images, such as cessna_ac_training50 through cessna_ac_training54. The reconstruction involved segmenting each image according to the CuneiForm dimensional framework, which captures essential training characteristics. These dimensions included TOI (Target of Interest) pictorial positioning, TOI’s 3D orientation, TOI pictorial

distance, and visible horizon attitude. The following is the example CuneiForm Training Strategy framework that covers training images 50 to 54:

Table I.36 Example CuneiForm Training Strategy

Abstract CuneiForm Characteristics (dimensions)	Abstract CuneiForm Characteristics definitions
TOIs definition and their aesthetic complexity	Single-engine propeller aeroplane {1}
TOI Motion and Dynamic optical situations	Motion trajectory: Linear motion captured in consecutive images where the aeroplane appears to move in a straight line at a constant speed (no acceleration) {1.2}. Dynamic optical situation: captured without optical blur {1.3}.
Background Objects associated with TOIs	Stratus Clouds{2} green-terrain {3} water surface{4}
Background Objects Motion and Dynamic optical situations	Background objects' motion trajectory is static {2.1,3.1,4.1} Dynamic optical situation: no motion blur{2.2,3.2,4.2}
Visible horizon attitude	Negatively Tilted Lowered Horizon{5} Positively Tilted Lowered Horizon{6} Negatively Tilted Elevated Horizon{7}
TOI's Pictorial Positioning	center left/down left{1.4} center/down center{1.5} down left{1.6} up center/up right{1.7}
TOI's Pictorial Distance	recognisable TOI distance{1.9}, clear close TOI distance{1.10}, extremely unrecognisable TOI distance{1.11},
TOI's 3D Orientation	Front{1.12} rear down right{1.13} Right{1.14}

	Unknown{1.15}
--	---------------

Table I.37 presents an Example CuneiForm Training Strategy, which defines a structured framework for training an AI perception model using a well-defined dataset. This framework ensures comprehensive epistemic coverage by categorising essential visual and spatial characteristics of the TOI and its surrounding environment. Each row in the table corresponds to a specific CuneiForm dimension, encapsulating key elements that define an object's training exposure within the dataset.

The TOI's definition and aesthetic complexity establish the primary subject of interest in the dataset, identified here as a single-engine propeller aeroplane. This ensures that all training instances adhere to a consistent object classification for the AI model. The TOI Motion and Dynamic Optical Situations section specifies the expected movement behaviour of the aeroplane. The TOI follows a linear motion trajectory with no acceleration, ensuring that each captured frame maintains a structured and predictable optical flow. Additionally, the dynamic optical situation is carefully controlled to prevent artificial motion blur, ensuring clarity in feature extraction.

The Background Objects Associated with TOIs dimension details the environmental elements that accompany the aeroplane in the dataset. This includes stratus clouds, green terrain, and water surfaces, providing a variety of natural contexts. Correspondingly, background objects' motion and dynamic optical situations confirm that these elements remain static with no observable motion blur, ensuring that the TOI remains the focal point of training data.

The Visible Horizon Attitude dimension introduces tilted horizon categories, including negatively tilted lowered, positively tilted lowered, and negatively tilted elevated horizons. These variations reflect real-world flight conditions, simulating different aerial perspectives that an AI-based vision system may encounter.

The TOI's Pictorial Positioning dimension defines where the aeroplane appears in the frame. Positions such as center-left/down-left, center/down-center, down-left, and up-center/up-right ensure the AI model is trained to recognise TOIs across various visual placements, reducing spatial bias.

The TOI's Pictorial Distance captures different levels of visibility, ranging from recognisable distances to extremely unrecognisable instances. This diversity in the distance ensures that the AI system learns to detect objects at varying proximities, addressing key challenges in perception robustness.

Finally, the TOI's 3D Orientation specifies the possible viewpoints from which the aeroplane can be observed, including front, rear down-right, right, and unknown orientations. By covering multiple

perspectives, this strategy enhances the AI model's ability to generalise across different real-world viewing conditions.

The output CuneiForm Abstract Image is as below:

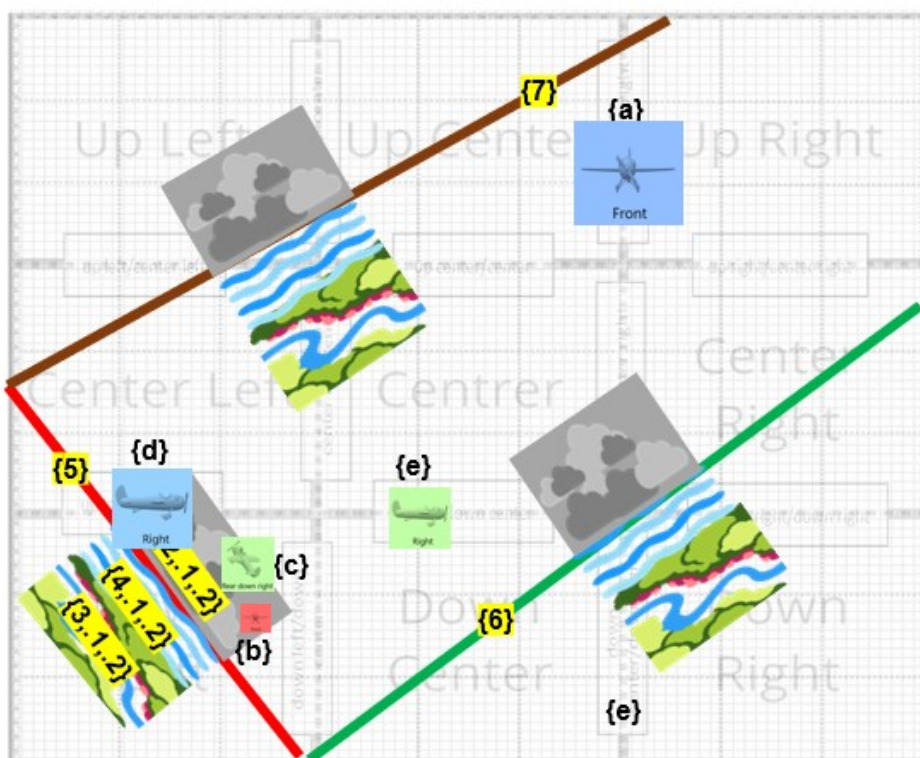


Figure I.34 Example CuneiForm for AVOIDDS. Coloured straight lines represent the horizon attitudes covered. Colouring of the TOIs orientation represents the pictorial distance.

Figure I.33 illustrates an Example CuneiForm representation for the AVOID dataset, showcasing a structured approach to characterising training data using pictorial positioning, TOI orientations, horizon attitudes, and environmental context. The figure employs coloured straight lines to indicate the visible horizon attitudes covered within the dataset. At the same time, the TOIs (Targets of Interest) are colour-coded to reflect their respective pictorial distance categories.

The horizon attitudes, marked by numbered annotations, provide insight into the spatial inclination of the observed environment. The red line corresponds to the Negatively Tilted Lowered Horizon {5}, the green line to the Positively Tilted Lowered Horizon {6}, and the brown line to the Negatively Tilted Elevated Horizon {7}. These horizon attitudes indicate that the dataset includes diverse viewing conditions that simulate real-world variations in aerial perspectives.

The TOIs (Targets of Interest) in the figure are represented by aircraft instances, with distinct spatial placements and orientations. For instance, the TOI labelled {a} is a single-engine propeller aeroplane {1} positioned at the up centre/up right {1.7} location, captured from a Front {1.12} perspective. Meanwhile, the TOI marked {d} is positioned at Right {1.14}, providing additional viewpoint diversity.

These variations in 3D orientations contribute to an enriched training set, enabling a perception model to generalise across multiple viewing conditions.

Additionally, the pictorial distances of the TOIs are indicated by their colour coding. The dataset incorporates instances of recognisable TOI distances {1.9}, clear close TOI distances {1.10}, and extremely unrecognisable TOI distances {1.11}, ensuring variability in how objects appear within different observational contexts.

Furthermore, the background objects within the dataset include stratus clouds {2}, green terrain {3}, and water surfaces {4}, reinforcing the need for AI models to operate reliably under different environmental conditions. These background elements maintain static motion {2.1, 3.1, 4.1}, and the dataset ensures that there is no motion blur {2.2, 3.2, 4.2}, thus preserving the clarity and stability of the TOI within the training images.

The figure below shows how the Cuneiform is abstracted. We associated the instantiated images with the part of the cuneiform that abstracts it. For example, image 50.jpg initiates the green horizon line.

AI Training CuneiForm 5

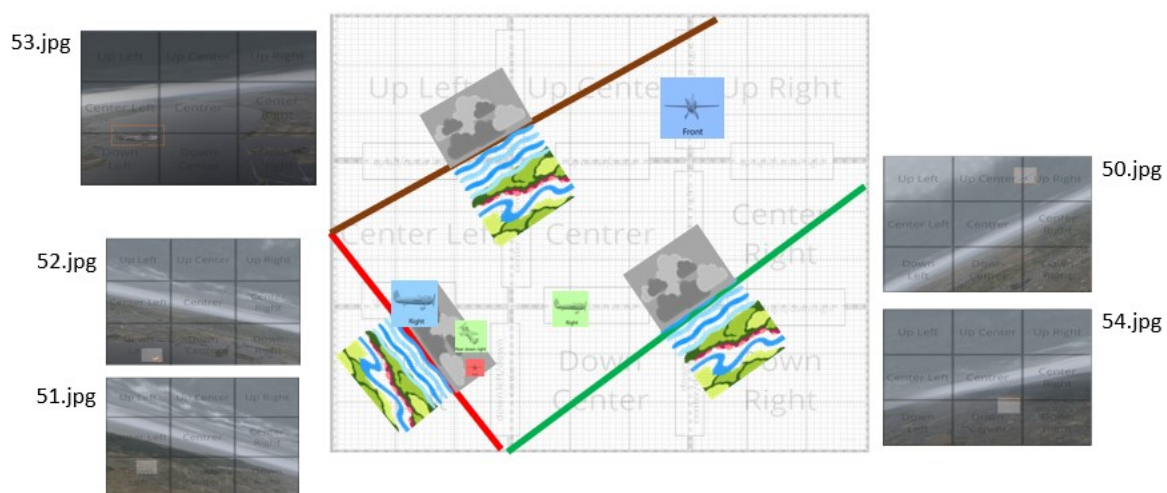


Figure I.35 How cuneiform 5 is instantiated in the AVOID training dataset sample

Figure I.34 suggests that there should be other similar training examples in the training dataset itself that they would have used. However, the team did not provide the actual 72,000 images they generated, and it appears as if the team had done a `generate_traffic_data.py` run and generated the sample images in the repository. For the sake of this case study, as we said at the beginning, we are assuming that the dataset is a sound statistical representation of a dataset generated for the purpose of the AVOID system.

I.10.4 CuneiForm Validation Artifact

The CuneiForm Validation Artifact serves as a structured mechanism to assess the epistemic coverage and trustworthiness of datasets used in AI-based safety-critical applications. This artefact provides concrete evidence supporting the dataset's adequacy in training machine learning models by mapping various perceptual dimensions, such as pictorial distance, 3D orientation, time of day, and horizon attitude, within the Operational Design Domain (ODD). By incorporating detailed specification tables, annotated images, and computational analysis tools, the CuneiForm artefact enables a rigorous validation framework that can be used to support safety case claims regarding the dataset's sufficiency for real-world operational scenarios. In this section, we will describe the artefact using an example taken from AVOIDDS validation report for one of the training images samples in Figure I.32:

50.jpg

ODD Dimension	Specification
Weather Conditions	stratus
Time of Day	Mid-Morning, 10:07:30
CuneiForm Dimension	Specification
TOI's pictorial distance (nindans)	$6561/72 = 91.1$
TOI's Pictorial positioning	up center/up right
TOI's 3D orientation	Front
Horizon attitude	Roll: -18, Pitch: -10

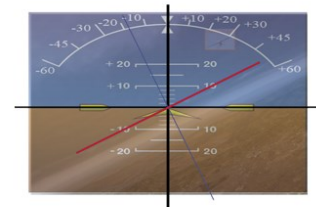
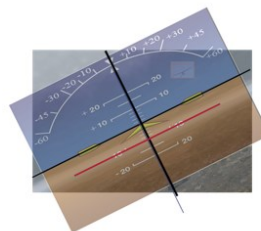
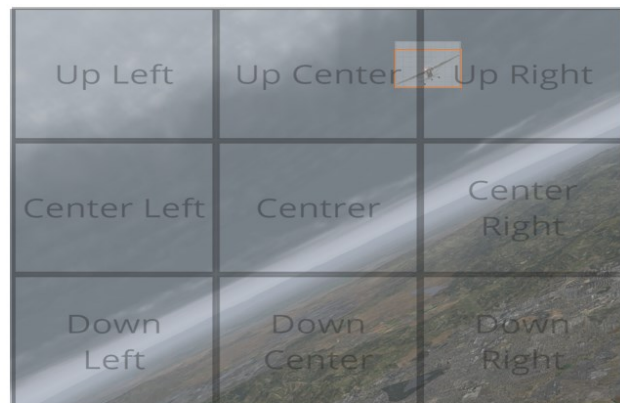


Figure I.36 Example CuneiForm validation artefact for 50.jpg sample image

I.10.4.1 Structure of the CuneiForm Validation Artifact

Each CuneiForm Validation Artifact consists of multiple structured components designed to capture and evaluate the dataset's coverage over essential perception dimensions. The artifact includes:

1. CuneiForm and ODD Specification Table

- This table systematically classifies the dataset's attributes concerning its Operational Design Domain (ODD) and CuneiForm dimensions.

- It includes information on weather conditions, time of day, pictorial distance, pictorial positioning, 3D orientation, and horizon attitude of the Target Object of Interest (TOI) in the dataset.
- An example is shown in the validation artifact for the training image "50.jpg", where the conditions are specified in the following training classes as:
 - Weather Condition: Stratus
 - Time of Day: Mid-Morning, 10:07:30
 - TOI Pictorial Distance: $6561/72 = 91.1$ (measured in nindans)
 - TOI Pictorial Positioning: Up Center / Up Right
 - TOI 3D Orientation: Front
 - Horizon Attitude: Roll: -18, Pitch: -10

For example:

Table I.37 ODD and CuneiForm training classes

ODD Dimension	Training class spec
Weather Conditions	clear
Time of Day	late afternoon, 19:20:53
CuneiForm Dimension	Training class spec
TOI's pictorial distance (nindans)	$6561/15 = 437.4$
TOI's Pictorial positioning	down right
TOI's 3D orientation	front down right
Horizon attitude	Roll: 1, Pitch: 7

2. Graphical Evidence for Validation

- The CuneiForm Canvas:
 - A 9x9 grid overlay on the image highlights the TOI's pictorial positioning within the image frame.
 - In the case of 50.jpg, the TOI is positioned in the upper center/right region, visually demonstrating its placement within the dataset.
- Pictorial Visible Horizon Attitude Indicator (PHI) Tool:
 - The PHI tool measures the horizon attitude by computing the scene's roll and pitch angles relative to a level horizon.

- For 50.jpg, the PHI tool indicates a roll angle of -18 degrees and a pitch of -10 degrees, which is crucial for evaluating model robustness under various horizon distortions.
- 3D Orientation Abstraction:
 - A visualised TOI extraction showing the aircraft's 3D orientation class is included to provide an abstract classification of the object's orientation.
 - In 50.jpg, the TOI's orientation is front-facing, meaning the model will need to recognise the aircraft's frontal view under the given environmental conditions.

3. Incorporation into the Safety Case

- The CuneiForm Validation Artefact is critical for safety case development, as it provides quantifiable and reproducible evidence regarding the dataset's epistemic coverage.
- The combination of structured tabular specifications and graphical evidence allows for traceable validation of the dataset's ability to generalise across different conditions, reducing epistemic uncertainty in machine learning-based object detection systems.
- The inclusion of diverse pictorial positioning, horizon attitudes, and environmental contexts ensures that AI-based safety-critical perception systems are evaluated against real-world operational challenges rather than just standard scenarios.

We use the CuneiForm Validation Artefact to evaluate how the dataset meets the ALARP standard for epistemic uncertainty reduction. By using structured CuneiForm dimensions, grid-based positioning, PHI horizon alignment, and 3D orientation abstractions, this methodology enhances the credibility of the dataset's trustworthiness in supporting AI-driven perception models. This structured validation process directly contributes to the safety case claims, reinforcing the dataset's reliability in detecting and classifying objects under real-world conditions, including Black Swan scenarios.

I.11 STCoT Validation Report for AVOIDDS case study

Below is a sample STCoT Validation Report drafted by the L.1 STCoT Validation Report Template. This example applies to the "Identification and Analysis of Unsafe Problematic Behaviours in Mid-Air Collision" STCoT, Table I.1, validating the correctness and usefulness of "Architect Assertion 1.1.4" about confusing and unsafe behaviours that may lead to collisions.

I.11.1 Purpose of the Report

This validation report evaluates the “Identification and Analysis of Unsafe Problematic Behaviours in Mid-Air Collision” STCoT, ensuring that its predictive thinking steps systematically uncover unforeseen or unsafe behaviours in the problem domain. It also confirms whether the STCoT product meets the expected completeness, consistency, and alignment with General Systems Model and organisational needs.

- **STCoT Under Review:** “Unsafe Problematic Behaviours List Identification”
- **Architect Assertion 1.1.4:** Outlines two major problematic behaviours in mid-air collision contexts:
 1. Aircraft deviating unexpectedly from assigned flight paths
 2. By-passing aircraft moving erratically at high altitudes near crowded airspace.

I.11.2 Scope of Validation

This validation covers:

- Examination of how the STCoT identifies and describes confusing or unsafe behaviours in the domain of mid-air collision avoidance.
- Authentication that the **Predictive Thinking** steps are properly followed, i.e., that they adhere to “General rule A” and “General rule B” from your General Systems Model.
- Assess alignment with the Step Completion Criteria (1.1.3) to confirm that the STCoT product sufficiently describes the problem context and indicates undesirable impacts without prematurely jumping to design solutions.

Out of Scope:

- Detailed assessment of proposed solutions or mitigations for any of the identified unsafe behaviours.
- Authentication of other STCoTs beyond the “Identification and Analysis of Unsafe Behaviours” focus.
- Full end-to-end system safety or certification processes outside the immediate STCoT’s boundaries.

I.11.3 STCoT Output Validation

I.11.3.1 Validation Criteria

We define the following **pass/fail** criteria to determine whether the “Identification and Analysis of Unsafe Problematic Behaviours in Mid-Air Collision” STCoT output is valid and valuable:

1. Alignment with General Systems Model

- The STCoT must illustrate that **General rule A** (unsafe behaviour is a type of confusing behaviour) and **General rule B** (confusing behaviours cause undesirable emergent outcomes) are correctly applied.

2. Relevance to the Stakeholder/Architect Sphere of Concern

- The problem statements must show that the described behaviours (e.g., erratic or unauthorised flight path deviations) fall within the domain’s sphere of concern (e.g., collision risk in congested airspace).

3. Identification of Realistic Undesirable Outcomes

- The STCoT must show how each unsafe or confusing behaviour leads to specific, undesirable consequences (e.g., near misses or actual collisions).

4. Exclusion of Solutions

- The deliverable should focus on analysing behaviours and not present solutions or mitigations (in line with the step’s instructions).

5. Step Completion Criteria (1.1.3) Preserved

- The STCoT must sufficiently describe the problem (the “Confusing Complex”) and the unsafe impact on some element B.

I.11.3.2 Validation Results Table

Below is an example table summarising the evaluation of each step or key aspect in the STCoT.

Validation Step	Evaluation Criteria	Pass/Fail	Notes
Step 1: Identify	Does the STCoT describe a mid-air collision scenario with enough clarity?	Pass	The problem domain is stated clearly: mid-air collisions. The scenario references “aircraft deviate from assigned path.”

Problem Domain			
Step 2: Apply General Rule A, B	Are unsafe/ confusing behaviours recognised as contradictory? Do they cause emergent risk?	Pass	Identified two primary behaviours: unexpected flight path deviation and erratic flight near crowded airspace, leading to collision risk.
Step 3: Problem-Focused, No Solutions	Does the text avoid introducing solutions?	Pass	The STCoT only defines problems: “Aircraft deviating” and “Moving erratically...” It does not mention new solutions or systems.
Step 4: Step Completion Criteria	Does an “undesirable outcome accompany each behaviour”?	Pass	The undesirable outcomes are clearly described as “Increased collision risk,” “ATC confusion,” and “reactive manoeuvres.”
Overall	Does the STCoT produce an acceptable description of the Confusing Complex?	Pass	The problem context is consistent, relevant, and no solutions overshadow the problem articulation.

I.11.4 Summary and Next Steps

I.11.4.1 Summary of Findings

- The STCoT effectively identifies critical unsafe behaviours (such as “unexpected flight path deviations” or “unauthorised, erratic flight in crowded space”).
- Each behaviour is framed as “confusing” and includes specific undesirable outcomes (e.g., collision risk, unpredictability for other pilots).
- The content adheres to General Rule A and General Rule B, explaining how each contradictory behaviour escalates collision likelihood and thus is within the “sphere of concern.”
- No solutions appear in the text, thus fulfilling the requirement to remain problem-focused.

All these observations indicate that the STCoT output meets the objectives set out in the instructions for “Identification and Analysis of Unsafe Problematic Behaviours.” Hence, the STCoT is validated as meeting the acceptance criteria for this stage.

I.11.4.2 Recommended Actions

To refine future STCoT steps or expansions, the following actions are suggested:

1. Feed into Next Phase

- Use these validated unsafe behaviours to inform the following process steps:
 - Step 1.2) Generate a descriptive image that visualises the unsafe behaviour

I.11.5 Appendices (Optional)

If needed, include:

- Minutes of problem articulation reviews with the customer that support this STCoT's finding.
- **Illustrative diagrams** or references to concept sketches of the flight paths, relevant logs of near-misses, or any additional data compiled during the problem articulation.

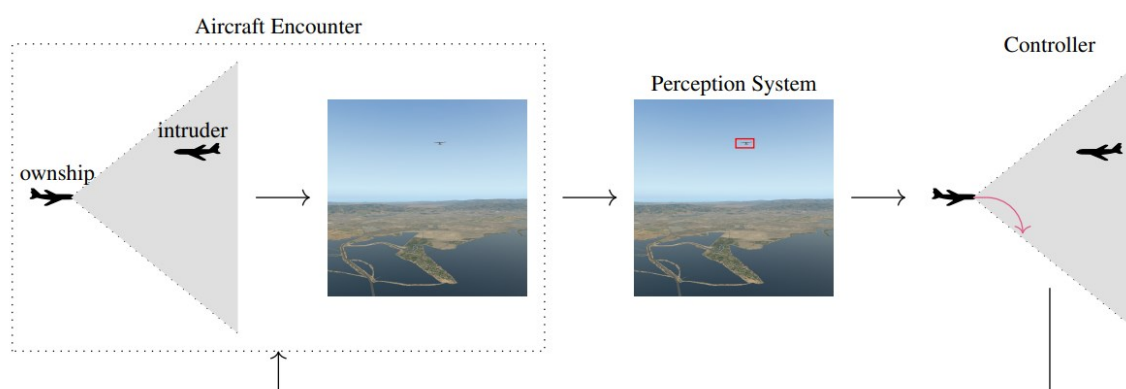


Figure I.37 Diagram taken from [1]

I.11.6 STCoT Validation Review Process

- **Report Author/Architect:** Haider Al-Shareefy [University of Southampton], responsible for generating the STCoT validation content.
- **Reviewer 1:** Prof Michael Butler [University of Southampton], validating alignment with General Rules D, E, F.
- **Reviewer 2:** Dr Son Hoang [University of Southampton], verifying coverage of elements and clarity of PrimePs.

- **Reviewer 3:** Prof Hamid Asgari [Thales], performs customer acceptance of validation report output.
- **Sign-off:** The above signatories affirm that the **STCoT validation** process was carried out thoroughly, and the results are valid as of this report date.

By signing below, the reviewers confirm that **I.4.1.1 Step 1.1) Identify a list of unsafe or confusing behaviours** of the STCoT is validated as meeting the defined criteria.

Signatures:

Signatures:

- Architect: _____ (Date: _____)
- Reviewer 1: _____ (Date: _____)
- Reviewer 2: _____ (Date: _____)
- Reviewer 3: _____ (Date: _____)

End of STCoT Validation Report